Wiki of features for Strawberry MC*

Arranged by Aadarsh Aggarwal, Eric Xia, Sai-Chatura Prakar, Zafar Fayzullev, Hrid Bandyopadhyay, Alex Hu, Caleb Zhang, Rishaan Agarwal, and Theodore Mai.

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Made possible by MCreator and NeoForged

Credits of inspiration and sources

Youtube:

- Youtube/Nekoma
- Youtube/Doctor4t
- Youtube/ZERB
- Youtube/Kenadian
- Youtube/Kenadian the Cat

r/minecraftsuggestions:

- u/Pasta-alla-Pixel
- U/Jackuhlf23
- U/CitrusMikasa2
- u/Super_Leadership_808
- u/Minute-Abies2173
- U/mjmanella
- u/cs386148,

Pixabay:

- Pixabay/TanwerAman

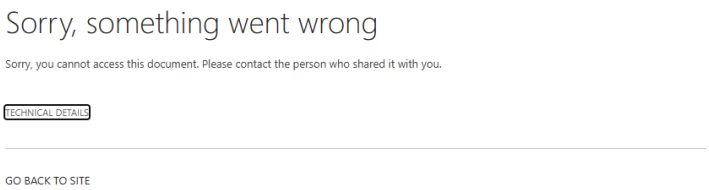
Minecraft wiki

Named assets credits:

Asset	Piece name	Author, Arrangement
Lapis golem fight, music disc	“Toccata”	By Aran Khachaturian, Arr. Eric Xia
Ambience overworld morning birds	n/a	By Pixabay/TanwerAman, Arr. Eric Xia
Music disc	“No escape”	By credit: T_en_M @spotify, Laudividni
Item texture	Plant fiber	Inspired by: RLCraft @NeoForged
Lapis golem fight, music disc	“Tre Pezzi Picolo” No. 3: Vivo	

Aadarsh music disc	Aadarsh's musical UH + WHY Australian accenteedition	Eric's podcast and Aggarwal Aadarsh, Arr. Eric Xia

Other links [dev]



https://drive.google.com/file/d/13n0_SEtUl5N6nWZGSUNYUiHeXGx89vOd/view?usp=sharing
WATCH THIS VID: HERE
WATCH THIS OTHER VID FOR PARABOLA MODE: HERE
Public suggestions for mod ideas

GO HERE FOR COMPILED LINKS

FOR CODING INFO!!!!
Github repository [private]
Strawberry folder

r/minecraftsuggestions
FUNGAL SURVEY AT PCMS: HERE
Population Statistics for the Federated States of Micronesia: Here
9/24 last updated on 9/25:

The github repository has been changed to uncraftable-strawberry-pickax from uncraftable-strawberry-pickaxe due to desync errors.

9/30:

Due to file corruption, we will be manually editing some files which may extend distribution date

10/6:

Due to multiple desync arrors, the corrupted elements have been remade. Waiting for a. aggarwal to finish neoforge elements

Introduction

Strawberry Minecraft is a new and innovative mod featuring many new features to enhance gameplay and the overall experience. This mod is currently under development and will be released before the end of 2028 (Because we will become 9th graders grinding homework). Note that some ideas for this document are inspired or contributed by creditors unless otherwise listed, which are only seen in the credits section. Other element[s] from this mod are copyrighted. Please read the Apache license in the prologue.

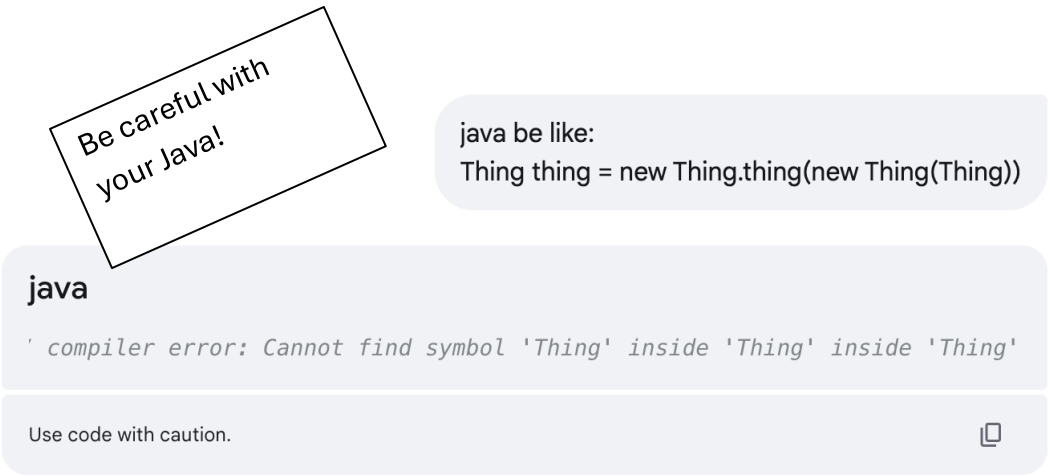
Textures and Coding progress bar:

10%{*_____}100%**

What we are working on right now:

Texture mapping, creating .jar files

TERMINAL



Stats for nerds (AKA all of us) [dev]

Coding list

Recovery Pearl

Procedure: Complete

Texture: Complete

Fletching Table and Arrow Modifications

General

Procedure: Complete

Textures: Incomplete

Iron nugget

Procedure: working on it

Covenant of Witches

Customize/Remix our mod

> *Note that credit must be noted and placed in an obvious location upon non-commercial distribution of remix. Remix commercial distribution is strictly prohibited. Read terms and conditions for more details; remixes are subject to the same terms.*

If you are unsatisfied with our mod or would like to add/remove features, you can use the following manual to customize our mod according to your needs. The first step will be installing MCreator and VScode if you haven't already. Be careful when installing Mcreator, as there are virus ads with fake download buttons that you would want to avoid. Select the 2026.2 version compatible with this mod, then click Download.

While these are installing, install git. Search up for example git for windows and click the first result and download git api.

After setting up MCreator and VScode, link your VScode to your github account. If you don't have a github account, you can make one at github.com.

Next step will be cloning our repository for our mod on github. Open github first and go to our repository which I can't link right now due to Securly. The repository should be called Uncraftable-strawberry-pickax (without the e), although there are 2 repositories named Uncraftable-strawberry-pickaxe, but copy the URL of the one by i-am-also-coder-314159 (not i-am-a-coder 314159). Make a new folder in the directory where you would like to store your remix. Open the folder on VScode by clicking on the open folder option on the homepage. Powershell or Terminal should show up at the bottom in a black box and should be labeled #zsh. If you don't see a terminal, go to the taskbar at the very top and click "terminal", then "new terminal".

Once you have Powershell/terminal open at the bottom, type git clone <repo URL> and files should appear if you did everything properly.

How to use git for commit [dev]

Committing your changes

If you have no staged changes, first use `git add .`

After staging you changes, merge your checkout to main with `git merge main`, or if you want to keep it a separate branch, use `git commit –m"commit message”`

This article is a stub. You can help by expanding it.

Claimed calls

If you haven’t already, install these apps:

Blockbench: https://www.blockbench.net/ for editing minecraft textures **DO NOT CHANGE SETTINGS OF BLOCK, JUST FOR TEXTURES**
Musescore: https://musescore.org/en/download for editing music or ambience or sounds
VSCode: Search up in apple app store or microsoft shop for editing code
NeoForge: https://neoforged.net/ For debugging and analyzing mmd, and coding
IntelIJ idea java: i dont have a url sorry but ask aadi i think he has
MCreator: Linking everything together

Idea	Claimed by	Type	Status
Crystal	Aadi	Feature	80% done
Recovery pearl	Aadi	Item/Projectile	Resolving git
Dendrite,Garnet	Ewic	Texture	Eeep
nx life			

Sai’s calls

GIVE ME SOME STOOF

Eric’s calls

1. End cauldron (working on rn)
2. Sulfur lantern (completed)
3. End lantern
4. Waypoints(completed)
5. Endillary(completed)
6. Spectral (completed by Sai {In Document})

7. Core of linkages(completed)
8. Dragon egg in core of linkages
9. Chorine smithing template(completed)
10. Sensegrass item and block model(completed)
11. Charred log, planks, and wood
12. Burn morel
13. Chiseled cauldron (should have Mojang-galactic runes chiseled in)
    Uncraftable potion
14. Ancient dyes **(IP)**

Link to MC controversy page

Strawberries for sharing files

Credits

Heads of procedural Java software and debugging

Aadarsh Aggarwal

Eric Xia

Heads of Creative Concepts

Sai-Chatura Prakar

Caleb Zhang

Alex Hu

Assistants Creative Concepts

Hrid Bandyopadhyay

Zafar Fayzullev

Playtesting and Document Review

Ruichen

Ansh Bansal

Documentation Arrangement and Editing

Eric Xia

Sai-Chatura Prakar

Aadarsh Aggarwal

Alex Hu

Head of Phonic Assets

Eric Xia

Navigating this document [dev]

There are 2 ways to navigate this document: the recommended way is to use ctrl/cmd + F to search in the sidebar for your destination within this document. The other way is to use the glossary provided below, though not recommended. To not get lost in this document, follow the headers and use the page numbers to navigate it. Sometimes the document may lag, in which you should stop typing, and wait for the document to stabilize again or just reload the page and hope your changes were saved.

Shader packs and mods for realism mode [dev]

Distant horizons mod is HIGHLY recommended for best experience, although make sure you know if you are using BLAZE rendering by Mojang, Vulkan, or any other rendering engine.

GET SHADERS AND MODS

Writing Your Section: formatting and organization [dev]

Forced DJABBAC style

If you are bad at naming things, use the forced DJABBAC style and give all your things [unnamed *n*] tags instead of naming yourself (to prevent rejection)

Headers [IMPORTANT]

When writing your own section, remember to use the correct header format and make sure your headers are not interfering with other people's sections. Another thing to keep in mind is to capitalize your heading correctly such as using More About French Fries instead of More about french fries. Note conjunctions in titles such as is, and, in, or of do not need to be capitalized.

Headers in a document should go like this (in order):

- Heading 1: Collection of things (You should not need to use this while writing unless you are starting an entire new section agreed upon by everyone).
- Heading 2: most of the time will be the main title or topic of what you're adding to this document. Make sure that you put your section in the correct super-category (heading 1's), such as {Rendering}, {Biome related}, and {Standalone features}.

- Heading 3: divisions of what you're adding to your document, for example you are adding french fries, then you would want to divide it into Obtaining, Effects, and more.
- Heading 4: subdivisions of your divisions
- *Heading 5*: even smaller sections of your subdivisions
- Heading 6: Although rare, sometimes you must use heading 6.
- Bullet points or number lists: these are on the same level of depth as heading 6 but are a more organized way to list lots of content.

Dividing sections

When dividing and naming your section, it is important to remember to use specific titles, for example if you are creating a section on french fries, do it like this:
French Fries [heading 1]
[description]
Potato French Fries [heading 2]
[description]
Sweet Potato French Fries [heading 2]
[description]
NOT LIKE THIS:
French fries [heading 1]
[description]
Potato [heading 2]
[description]
Sweet Potato [heading 2]
[description]
We do this so that when people are using ctrl+F to search this document, they are not getting confused about a section on sweet potatoes in the middle of nowhere.

Tense and Subject

This document officially uses past tense and uses “the player” to refer to the player.

Symbols and Abbreviation

On the document, the use of symbols like “&” is not allowed, as they are only used when conjoining 2 different married couples’ names.

Comments [dev]

When changing or adding anything, add a comment “clogpost: made changes” so that a simple scroll through the doc will tell you the most recent additions

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This condensed encyclopedia features a comprehensive list of features that will most likely be added to our upcoming mod:

Gameplay

Realism mode

First things the player must do

The first step for the player to survive in realism mode is to obtain flint tools. In order to obtain flint tools, the player must break gravel to obtain flint chips, which for every 4 flint chips, can be crafted into a flint shard. These flint shards as well as plant fiber[read also: plant fiber] with sticks that can be found by punching leaves are used to craft a flint hatchet.[read flint dagger for more information on crafting a flint dagger]

Using this hatchet, the player can proceed to use the hatchet to harvest logs (very slow). From these logs and a flint hatchet in the player's inventory crafting grid, these logs are crafted into wooden planks. Note that the flint hatchet takes 5 durability out of 15 durability to make the wooden planks. These planks along with flint shards from mining and crafting gravel, can be used to craft a basic crafting table. This basic crafting table can be used to directly craft logs into planks.

Assuming the player survives to this far into their journey, the player now must use the dagger to slowly damage and kill sheep. The sheep once killed will turn into a sheep corpse, which can be placed in the crafting table for raw mutton, animal hide, and wool. Along with more planks crafted in the crafting table and fiber, the player can make a basic bed, which allows the player to sleep from midnight to dawn. This basic bed can be upgraded to a comfortable bed that allows the player to sleep before dusk and wake up after dawn.

Once the player passes their first night, they must be cautious about splinters, tripping, and they also need a food source. Depending on the biome, different fruits, berries, vegetables, and grains can be found. Meat and animal products are not recommended as obtaining a furnace can be hard with starter items. For food, look near charred areas for roasted ember lily tubers and burn morsels, jungle areas for bananas and grains, taigas and colder biomes for berries, and more[read food mechanics for more suggestions]. A water filter is also recommended; water filters can be found in ruins around dry or hot biomes or be crafted later in the game. Once a water filter is obtained, the player can then get charcoal from charred forests and use sand, red sand, gravel, or suspicious mediums to filter water in bottles (obtained from ruins' chests).

When the player can finally support themselves and have time that is not used for gathering food and surviving, the player can finally build a base. Starter bases can be carved into hills and mountains with plentiful soil, or a log cabin made from surrounding logs. At this point, the player can use 2 flint trowels [read also: flint trowel crafting recipe in appendix] (as one will break when halfway through breaking the stone). After halfway through, the stone will drop rocks, and to completely mine the stone, another flint trowel is needed which at this point will drop another rock item. 3 rocks obtained from using the flint trowel can be crafted in a stone cutter from a village for 1 cobblestone slab, or 6 rocks can be placed in the way a trapdoor is crafted in a basic crafting table to result in 1 cobblestone slab.

Once the player obtains a cobblestone slab, the cobblestone slab can be placed at the bottom center slot of the crafting table with one plank of any wood type above; and 3 sticks surrounding the plank like so:

This will result in 1 unlit campfire, which the player can ignite using a flint knife and a sharp rock. Sharp rocks can be obtained by either inserting 2 rocks into a crafting table for 1 sharp rock or by inserting 1 rock into a mason villager's stone cutter to get 1 sharp rock. But before lighting the campfire, the player needs to find tin ore. Tin ore can be found on cliff faces and can be mined with a flint trowel. Raw tin ore can then be placed on the campfire which can be lit, to obtain pig tin nuggets for every two tin ore. One pig tin nugget can be crafted with1 any fibrous material such as plant fibers or rope along with one stick to obtain a tin hatchet.

	Tin hugget	
	 *	
		

*Credit: RLCraft textures

A tin hatchet has 15 durability, which must be used to mine iron ore, which now can directly generate under dirt on the surface. A flint trowel can break 4 dirt before breaking,

so 4 stone can be revealed to expose possible iron ore since dirt in realism mode is only 1 block thick. If any iron ore is revealed, the tin hatchet can be used to obtain iron ore nodules, which take 2 durability to mine. 3 Iron ore nodules can be combined to create 1 iron ore block. Only iron ore blocks can be smelted in basic furnaces.
A basic furnace can be made by lining the base of the crafting grid with cobblestone slabs followed by an upside-down U shape of clay balls above, leaving one empty space in the middle. The clay balls can be obtained by using a tin hatchet or trowel to excavate clay generating under snow on mountainsides and in freshwater rivers as well as sometimes in pockets under or in sand banks.
After the basic furnace is created, the player must leave the furnace to dry for 5 real life minutes.
Coal collected from riverbeds is used as a fuel source, and the iron ore block is inserted into the furnace. The furnace has enough durability for 3 iron blocks to be smelted. Each iron ore block outputs 2 iron ingots and 3-5 slag.

Food mechanics[realism mode only]

In realism mode of Strawberry Minecraft, the player must eat a diet consisting of balanced nutrition to survive. If the player does not eat nutritiously over 3 days, the player has a 70% chance for each day without eating the needed food source to gain the malnourished effect, which can have a wide variety of side effects depending on what the player is deficient in. The good in-game food sources are:

- Apples
- Potatoes
- Melon slices
- Coconut milk bottles
- Coconut flesh (the white inside of a coconut extracted using crafting table[read also: coconuts])
- Milk buckets
- Raw mushrooms
- Sweet berries
- Juniper berries
- Beetroots
- Raw Carrots
- Wild Oats [realism mode only]
- Lava / Hot Spring seared Meat (seared by placing meat item on stone or exposed stone next to lava)
- Roasted Mushrooms
- Boiled Eggs (use hot spring features above sulfur caves or in volcanic jagged peaks)

Malnourished

The Malnourished effect comes in 2 levels; Malnourished I and Malnourished II. This Malnourished effect can affect the player in the following ways:

- Blindness
- Darkness
- Hunger
- Infested
- Instant damage I throughout the day randomly
- Poison
- Slowness
- Weakness
- Mining fatigue

The Malnourished effect occurs when the player has not eaten all the required food categories in Realism mode for 3 days, with this effect having a 70% chance to occur for every day the player has not fulfilled their diet after the 3 days.

Water and stamina[realism mode only]

Several types of water replenish different amounts of hydration:

- Muddy Water Bottle or Muddy Water Canteen replenishes 1 thirst.
    - Has 90% chance of contracting Fatal Poison to the drinker[bedrock only]
    - Has 90% chance of contracting Poison to the drinker[java only]
- Water Bottle or Water Canteen (not purified) replenishes 2 thirst for every bottle.
    - Has 40% chance of contracting Poison to the drinker.
- Purified Water Bottle or purified water canteen replenishes 4 thirst for every bottle.
- Saline Water Bottle or Saline Water Canteen (salinity >1) replenishes 1 thirst for every bottle.
    - Has 100% chance of contracting Thirst I or II depending on salinity

If the player runs out of thirst, the player instantly dies (regardless of gear, totems, or potion effects). Thirst drains 2x faster than hunger, so the player should stay near water sources. Stamina depletes when running, swimming, jumping, crouching (repeatedly), and block mining. The only ways to replenish Stamina are to sleep, eat food, and stand still for a while or rest on the ground.

Parabola Mode

Parabola Mode is a gamemode that can be either switched into by using /difficulty parabola or by selecting the Parabola difficulty when creating a world using the Lobby [read also: Lobby]. Parabola Mode is a compilation of April fools features with funky mechanics that

are supposed to confuse the player. Parabola Mode is best used with survival mode or using Parabola Mode features for escape rooms.

Renaming

Examples:

Name	Parabola Mode Name
Iron ingot	White Butter
Nether bricks	Burnt Fake Butter
Gold ingot	Rare butter
Copper ingot	Medium butter
Copper (non ingot)	Waxed lightly weathered
Bricks	Fake butter
Shard	Slightly poky <shard material>
Shridnite	Oar
Pickaxe	Pic del <material>ii [read also: ore sticks]
Sword	<material butter name> Butter slicer supreme
Shovel	Potatoic device
Hoe	§k*Redacted* §r [reference to the word “Hoe” not allowed on chat in servers in the UK]
Axe	<material> Deforestation on a stick
Uncraftable Potion	Hunk
Bow	Charles’s dream
Crossbow	Charles’s reality
Melon	Extremely watered melon
Bread	Watered wheat powder
Apple	Apple but its not bad for once
Firework Rocket	Grayish powder in a paper
Enchanted	Glinty <equipment name>
Carrot	Thick root
Beetroot	Seedy rut
Golden dandelion	DANGER: OXIDIZER
Netherite	ite
Nephrite	Dyslexic
Sand	graviton
Ancient debris	debris of the OLDEN DAYS
Helmet	safety 3rd

Chestplate	<attribute> Shirt
Leggings	<attribute> Pants
Boots	LOOK AT MY COOL <attribute> SHOES
Prismarine	Unknown blue hunker
Fire	NFPA704 says “danger in the box”
Blaze	NFPA704 says “VERY DANGER THINGY IN THE BOX”
Wind	Volatile
Breeze	Pokably Volatile
Charge (projectile)	greased greece
Lapis	La peace
Diamond	Lapisn't
Emerald	Green barrels
Stone	Chimpanzees
Sulfur	Sulphur <item name>
Golden Apple	Good Bad Good Appel
Enchanted Golden Apple	Good Good Good Appel HOW IS IT SO SHINY
Gunpowder	OL CANON POWDEROUS
Sandstone ore	Chimpanzeenean flour
Chicken	chIkennugeat
Fox	thing
Snow Fox	COLD NEOFORGE
Bottle o’enchanting	Bottle o’ no longer IQ
Spawner minecart	useless…
Spawner	COOL LOOKIN BOX
Spawn egg…	Egg of <mob>
Brown mushroom	boletus lookalike
Wooden bowl	yuuuuuuuummy SOUP
Mushroom stew	TEMU REFRIED BEANS
TNT	NFPA704 says “VERY VERY DANGER BOOM BOOM THING IN BOX”
Barrier	NOPE

Temple of the gods

Structure that generates in the End highlands with chorus fruit surrounding. Inside are the Caleb (solar god) and Lunar. These mobs will copy any sounds that the player has heard in the last 10000 ticks, however the lunar god will have corruption. Giving the lunar god some Theorem Notebook Remnants will cause it to transform into an inverted enderman and give quests. The first quest is to defeat the isosceles triangle and obtain the Isosceles Triangle Theorem Page. Next, the quest is to obtain the Hypotenuse-Leg Triangle congruence theorem from Pythagoras. (the player will defeat Pythagoras in a later quest)

Mycquests

Mycolandra

A mushroom found on the mushroom islands, similar to Fly Agaric with shorter stem. Contains a summon eric block and a library full of enchanted books and Inaturalist observations. The enchanted books are locked. Placing a crimson fungus in a pot on summon eric block will summon eric.

Eric

Eric is a mob that can give the player quests. Quest list:

Fermenter

Naturalist domain

A structure that spawns in the sky and consists of a cloud with a small house in it. Inside the house is a naturalist and a UUID-trained documentation raven.

Giving the UUID identifier to the raven will make it [unresolved]

Naturalist

A villager profession caused by a naturalectern. The naturalist has 1 trade – 1 x *Ricknella fibula* for a spy glass, an Inat UUID Server Identifier, or a *Ricknella Fibula*.

Liver types and mechanics

Cheese types and mechanics

Rend and beating the game

Eye of Render

The eye of render is an item crafted with extraordinary breeze powder and a render pearl

Render corruption

Render Dragon (boss mob)

Renderman

Render Pearl

Other Mechanics

§1 THE PEBBLE §r

THE PEBBLE is a pebble that is a melee weapon. This weapon does 2,147,483,647 damage [8th Mersenne prime] [1073741823 ½ hearts], and applies the following enchantment:

- Fire Aspect 32,767 [custom data tags]
- Shock Aspect 32,767 [custom data tags]
- Frost Aspect 32,767 [custom data tags]
- Void Aspect 32,767 [custom data tags]
- Looting 32,767 [custom data tags]
- Sharpness 32,767 [custom data tags]

Guide to Fibers of the Southern Southern Western

A book the player spawns with in Parabola Mode.

Depressed Deep Frier

Obtained by crafting lead and Dinnerbone Hoe together, causing any villager to grow long hair and play “yap” TTS nonstop.

Dinnerbone Hoe

Dinnerbone Hoe can be used to craft depressed deep friers. Obtained through farmer chests in parabola mode or crafted with nametag and hoe.

Longest item

The longest item is:

‘Waxed Cracked Lightly Heavily Weathered Polished Pristine Prismarine Cut Chiseled Eroded Oxidized Pot-like Jungle Old Iconic Copper Golem Statue Stairs Shard on a Stick’

This item has 168 characters (144 without spaces)

Chiseling

A chisel can be crafted with a hammer and a chorite ingot, which chisel blocks. Blocks now have a very chiseled, extremely chiseled, very extremely chiseled, very painstakingly detailed as if the object was alive chiseled, and chiseled chiseled variants, each of which can be obtained by chiseling the previous variant. Chiseling a chiseled chiseled block will give the image on the chisel. **[Unresolved: fill out chart]**

Block	Chiseled…	Extremely chiseled…	Very extremely chiseled…	Very painstakingly chiseled…
Sandstone	Chiseled Sandstone	Sand pebbles	Sandpaper	Very Painstakingly Chiseled Baby Camel
Stone Bricks	Chiseled Stone Bricks	Target Brick	More targety target brick	Brick, no targeting
Mossy Stone Bricks	Chiseled Mossy Stone Bricks	Mossed		Very Painstakingly Chiseled Creeper
Red Sandstone	Chiseled Red Sandstone	Red dead dust	Blue dead dust	Gired-Gimbly Very Painstakingly Chiseled Small Wither
Sulfur	Sulfuric	Sulphur	Lotionites	Bounciezed Lotionized Sulfur cube
Quartz	Q-Tarmac	Jello DioxiPolymer	Shampoo	Lotion Motion Oceanic Red-40 Hawaian Estonian Volcanoes National Park Service
Blackstone	Chiseled Blackstone	Gold Hunker	Very gold hunker	Extremely golden hunkererererer
Tuff	Chiseled Tuff	Less Tuff Tuff	Lesserling youngling tuffer	Tuffn't
Deepslate	Chiseled Deepslate	Brik on the ruf	Brik despawned	DEPRESSED BRIK

Australia Ore

The Australia ore is found in the badlands and can be used to craft a block of Australia.

Bottle o'IQ

Variant of bottle o'enchanting that gives blue XP particles that can be used on the School Table to craft IQ points.

Phoenix SC

Craftable with a Sulphur cube, an IQ point, a waxed lightly weathered cut copper stair, and a red dyed egg. Crafts into meme.

Meme

Edible; can be eaten to give Hero of the village 127 for 6.7 minutes

Netherite Golem

Can be created using this:

	Carved pumpkin	
Block of austrailia	Sulfur cube with sulfur cube	Netherite block
	Stonecutter	

When this is spawned, any villagers nearby will start making a lot more noise and a set of simple trials will spawn. If the trials are failed, the player is repeatedly teleported at random places in the sky. If the player succeeds, the netherite golem takes on a lighter appearance and once punched will turn green. After this, punching the golem will give nephrite.

Water Bucket Totem

Water buckets in parabola mode cannot be used while the player is falling and instead acts as a totem of undying when the player lands to an MLG. Note that the totem pop animation that plays on the player's screen is replaced with a water bucket render. Attempting to use will result in eating water bucket.

Mining mobs

Any passive or neutral mob the player encounters can be mined by holding left click. Once the mob is mined, eating any mob will allow the player to restore full hunger. When a mob is being mined, the mob's hitbox will snap to the nearest block and the mob's direction will round to the nearest cardinal direction.

Brushing obsidian or ore

When a brush is used on cracked bricks, cracked exposed stone, crying obsidian, or any raw ore, the ore will be brushed to its block form; such as brushing diamond ore will result in a diamond block. For example, crying obsidian in a ruined portal can be brushed instead of mining them to repair the portal faster.

Ore sticks

All tools in parabola mode are crafted using ore sticks, such as iron ore for iron ore sticks which can be used for Rusty Ole Southwestern Iconoclastic Piece of Minecraft (Iron pickaxes). All ore sticks are crafted by crafting an ore block, then placing in a furnace with ice in the bottom to get ore logs.

Campfires

In parabola mode, campfires will spawn hostile mobs even during daylight. Though campfires are the only way to smelt any metallic ores in parabola mode. As a bonus, any raw ore placed next to a campfire after 5 Minecraft days will turn into diamond ore blocks.

Killing logs or wood

All wooden materials or blocks in parabola mode cannot be mined and are instead killed with the player's fists or a weapon. All wooden blocks that can be killed have 1 health (half a heart). When any wooden block is killed, it will always drop the "not so wooden planks" block which can then be crafted into "real wooden planks"

Batchedling device

Mr. Batchelder?

Soul Sand

When soul sand is mined without silk touch, it will drop souls, which are a item used to instantly kill any non-boss natural entity. The mob killed with the soul item will drop its corresponding spawn egg.

Shearing

Any mob can now be sheared to drop wool or leather, or in the case of a wither skeleton, a wither skeleton skull, and for the case of a bogged, parched, stray, or normal skeleton; a skull.

EATIN SOME ORE THES OL DAYS

Item allowing the player to manipulate ores given at the end of the game.

Corrupted Anchovy

The Corrupted Anchovy is a type of 2x1x1 purple fish that is able to fly. When the Corrupted Anchovy is killed, it drops a food item labeled "Corrupted Anchovy". When eaten, it has a 5% chance to give the entity that ate it a random status effect for 13'14". The remaining 95% of the time, the entity dies. It spawns during dawn and can be killed by a player with a fishing pole and dies after taking 5 points of damage. Other than this, it acts like a normal fish.

Isosceles triangle

Summoned upon completing a math quest, this invincible mob is ravenous for iron and will autohome toward any player within 100 blocks and will search their inventory for iron upon finding player (He can see through walls). If at least a single iron nugget is found the isosceles triangle will take it and leave you alone, permanently deleting all iron in the player's inventory. If the player has no iron, he will throw you 1000-2000 blocks in the direction he is looking at and will start to slowly chase you across the world moving at 30

bps. The only way to stop the isosceles triangle is to get an obsidian cage (crafted with 7 iron bars and 2 obsidian) and right click on him with it and he will be paralyzed for an infinite duration. in the background whenever the player is within 500 blocks of him there will be a light background sound of him begging to be free. If the player decides to have mercy and uses a diamond or better pickaxe to break the cage, he will start trying to kill all villagers that the player has traded within the last 10000 ticks and will try to steal any iron from chests, furnaces, or even iron armor trimmed armor. If the player has not completed any of the following actions within the las 10000 ticks then he will throw you into the abyss dimension by clipping you through the floor and the player will receive the achievement, “Angry angle.” However, giving him a cornflower will cause him to erupt into a congruent corrupted double angle and a congruent corrupted double side double corrupted angle. Crafting these together will give you the Isosceles Triangle Theorem

Villagers

Villagers if aggravated in Parabola Mode will hold a block model of a Netherite Pickaxe and start digging straight down to Bedrock. Villagers can be aggravated by:

- Player hits the Villager.
- Player scams Villager. [see also: Scamming]
- Player attacks Iron Golem.
- Player tramples farm.
- Player breaks a block apart of the Village structure.
- Player holds an item tagged with `#minecraft:swords` within 4 blocks of a Villager.

Noobular Harpoonism

The Noobular Harpoonism is a melee weapon that does no damage to entities unless they are tagged `#mod:harpoonable`. The Noobular Harpoonism can be obtained through naturally generating chests within the `mod:federated_states_of_micronesia` feature.

Beating the Game

Better Classic Method

The better classic method is a new method of beating the game inspired by the classic way of beating Minecraft. In this method, the start of the game is the same, but crafting an eye of ender instead requires materials from the 6 elemental strongholds (only 5 are needed since the stronghold itself counts as an elemental stronghold) of power around the world. An eye of ender requires 1 shock rune from a desert pyramid, 1 flame rune from a nether fortress, 1 splash rune from an ocean monument, 1 dark rune from an ancient city, and 1 flourish rune from a jungle temple. This eye of ender is used to craft a stronghold compass

to find the stronghold in the world. A stronghold compass leads to the closest stronghold and is combined with an ancient rune from the stronghold to make the Elemental Greatblade. The Elemental Greatblade and Elemental Longbow are the only ways to damage the Corrupted Guard guarding the end portal. When used to craft the Elemental Greatblade and Elemental Longbow, the Stronghold compass or ancient rune is not exhausted.

Collector Method [unresolved]

In this version the void dimension is enabled. To beat the game, the player must visit the 12 places of importance on the map that are: desert temple, jungle temple, ice palace, nether fortress, shrine of souls, stronghold, Ancient City, Flaming Forges (savanna)[coming soon], trial chambers, monolith, stone temple (dripstone caves) [needs better name] [coming soon], and stronghold. Defeating these areas will give runes, while they all house their own elementals (except for the stronghold, which has its respective elemental in the End near the rift. The player must collect all the runes and use them in the Mo

Classic Eric's Method

Eric's method for beating Strawberry Minecraft uses the same gameplay for the early stage speedrun as normal Minecraft, other than the crafting recipes for eyes of ender, which now require a prismarine shard, wind charge, ice bomb, blaze powder, and ender pearl. This new eye of ender is then inserted into a waypoint chiseled cauldron, which will ping the coordinates of the closest stronghold. The waypoint can then be used to travel to the stronghold's corresponding above ground waypoint. Note that due to the scarcity of Eyes of Ender in this method, all end portals are pre-lit or mostly filled. Once the player enters the end, the player awakens next to an end relay structure in the outer end islands, where the player must find palline and netherite tools to coat with palline. If the player did not bring any netherite tools, the player must loot end cities, which each chest in an end city has a 4% chance of containing 1 netherite ingot, or a 2% chance of containing netherite tools. The player can get palline from strip mining or looting end relay structures. If the player does not have a smithing table to coat the netherite tools, the player can find a smithing table located in end ships. To prevent players from getting Elytras prematurely, Elytras are now in indestructible vaults that require an End Token, a renewable item crafted with ender dragon scales, to open. Once palline coated tools are acquired, the player can finally kill the ender dragon, which can only take damage from palline coated netherite tools. This normally marks the end of beating the game, but the player must return to the overworld now. To go back to the overworld, the player must find a rift, where the dragon egg is inserted and then can be mined to enter the rift portal back to the overworld.

Lobby: Settings

The Lobby is a new structure that the player will always spawn in, containing portals that will change the game mode and settings of the world. Once the player spawns in, they will be in Adventure mode and will be able to enter colored nether portals. 5 portals are present, Parabola mode, Realism mode, Creative mode, Maps, and Gameplay. These 5 portals once entered, will teleport the player to their corresponding pick and set their game mode to the default game mode. For the Gameplay portal, a set of 3 sub portals will appear once entered. The 3 portals allow the player to select their game mode and other settings such as methods of beating the game including Aadi's method, Classic, Eric's method, Sai's method, and hardcore.[more to come]

Admin Settings

People with admin or operator permisions have access to an admin lobby, which has rooms with switches the admin or operator can alter affecting the entire world. These settings are physical levers. Admin lobbies also contain helpful debug information with fps, and more.
All levers correspond to different gamerules, including ones both listed in the settings menu and additional/hidden settings. Any settings that requires a dropdown bar, number input, or multi-stage toggle have corresponding custom block displays.

Pre-installed Resource Packs

Developer: Posterized

Developer: Realistic

Biome and Dimension Additions/Changes and Blocks

Let's face it: Vanilla biomes are not the best. And instead of adding too many new bad biomes like other resource packs do, Strawberry also enhances the existing ones.

Crimson/Warped forest

Entities

Phoenix

The Phoenix is a neutral, bird-like mob found in nests on Crimson and Warped large fungus. Phoenix nests light any adjacent block or item on fire and is the size of a slab. Phoenixes will actively defend their young, which are passive. If the Hatchlings are

damaged, the mother will attack the entity that damaged the young bird. If the player provokes the Phoenix, either by hitting it or it's young or taking their eggs, then the Phoenix attacks the player [see below]. Phoenix Egg blocks can be mined with a silk touch pickaxe. When attacking, phoenixes fly around in an unpredictable figure-8 motion while shooting Flaming Feather projectiles at the player, as well as occasionally swooping down at the player. Phoenixes cannot be harmed by projectiles, and can only be killed by a melee or explosive weapon. Phoenixes have 30 health (15 hearts), and Fire Feather projectiles deal 4 damage (2 hearts) and elemental fire damage on impact, and swoops deal 6 damage (3 hearts) per hit. If the player is killed by swooping, the player will receive the "<Player Name> was brutally mauled to death by Phoenix" death message. If the player was killed by a Flaming Feather projectile, the player will receive the "<Player Name> was shot by Phoenix" death message. When the Phoenix is killed, they drop 5 experience orbs, up to 3 tail feathers [affected by Looting enchantment], which can be used to customize arrows.
[see also: Fletching Table, Customizable Arrows]

Phoenix Eggs

The Phoenix egg is a block that spawns on phoenix nests. In one phoenix nest, there can be up to 3 eggs. Eggs can be mined with a Silk Touch enchanted tool, and will drop as an item. When mined, the adult phoenix will be provoked and will attack the player [See also: Phoenix] . When placed on a lit Netherack Block or a Campfire, the player must wait ten real life minutes for the hatchling to hatch. The hatchling will immediately bond with the player that placed the egg, and will rest on the player's shoulder until the hatchling turns into an adult phoenix, at which point it will fly occasionally leave it's perch and attack any entity that harms it's owner. Tamed Phoenixes ca be perched on any block, where it will fly off and start collecting sticks or leaf litter to create a nest on the block it has been perched on. Once perched, the phoenix cannot be perched anywhere else. When a tamed phoenix is killed, it will not drop anything.

Phoenix Nests

Phoenix Nests compose mainly of Charred blocks.

[unresolved: needs more detailed description]

Warped/Crimson Snatcher

The Snatcher is a new hostile arthropod insect found in Warped Forest biomes, with bodies shaped like orchid mantis. Snatchers camouflage in `minecraft:shroomlight` tagged blocks and `Glowstone` with only eyes and claws visible, waiting for the player to approach. Once the player reaches within 7 blocks of a Snatcher, the Snatcher will ambush the player; showing its body and locking the player's movement whilst attacking the player and

giving them the Poison II effect or burning them. The snatcher does 5 health per second (2 ½ hearts) To make the Snatcher release, the player must either shake their cursor violently or hit the Snatcher with a Bane of Arthropods enchanted tool. When killed Snatchers drop 1-2 Warped/Crimson Snatcher Claws, which can be used to craft a Warped/Crimson Grappling Hook or used as a melee weapon.

Snatchlings

Snatchlings are an arthropod populating Warped/Crimson Forest biomes; and are meant to be the counterpart of the adult Snatchers. Snatchlings are neutral (unlike their adult counterparts) and will try to hide or run away from players perceiving them as a potential danger. Snatchlings will drop Snatchling Membrane when they age into a adult snatchling after ~50,000 ticks or when killed which can be used to craft the Snatchling Cloak. Snatchlings are mainly defensive; and will hide in the tangle of vegetation in Crimson/Warped forests, but if the player gets too close, then snatchers may hit-and-run, dealing 3 health (1 ½ hearts). The Snatchlings can be tamed using Ember Lily Bulbs and will become favored to the tamer. If the Snatchlings grow into Snatchers while favored towards the tamer, the Snatcher becomes neutral instead of hostile. If the player feeds the Snatcher more Ember Lily Bulbs, the player receives the [put cringe name here] achievement which causes the snatcher to follow the player around and attack mobs that the favored player attacks essentially becoming the player's pet. The player can also coat the Snatcher's claws with Magma Cream to cause them to deal extra fire damage [using the forging tool in a crafting table]. The Snatcher bond can be broken if the player does more than 8 damage to the Snatcher in 24,000 ticks (20 minutes).

Snatcher Grappling Hook

The Snatcher Grappling Hook can be used to pull entities from up to 30 blocks away to 1 block infront of the player who used it, making it useful to catch escaping enemies. This tool can also be used to pull the player to any Cracked Stone variant from up to 30 blocks away and will keep the forward momentum after the player stops equipping the Snatcher Grappling Hook. This item is crafted with 2 Warped Snatcher Claws, 2 Crimson Snatcher Claws, and 3 Strider Silk. The Snatcher Grappling Hook allows the player to practically be slingshotted long distances while using an Elytra, making it very useful late game for traveling.

Snatchling Cloak

The Snatchling Cloak is a wearable item crafted using Snatchling Membrane and Warped/Crimson Snatcher claws. When worn, Snatchling Cloaks take up the chest armor slot and offer no defense or resistance. The Snatchling Cloak when worn will apply the texture of the block they are standing on to their skin as an overlay, essentially

camouflaging them. If the player is not standing on any block or standing on Air, Void Air, or other air variants; then the Snatchling Cloak remains deep red. The Snatchling Cloak causes snatchers and snatchlings to stop aggravating on the player (like piglins) and reinforces the favoring of the Snatcher.

Snatchling membrane	Snatchling membrane	Snatchling membrane
Warped snatcher claw	Inferno parchment	Crimson snatcher claw

Crystal Hollows

The Crystal Hollow is a new cave variant in which multiple crystals geodes and Calcite noise pillar formations generate on the ground of various compositions, with amethyst geodes being the most common. Calcite, Tuff, Cinnabar, and budding mixes of ores may line the ceiling and walls. Crystal hollow axolotls have a 5% to be blue axolotls. Crystal Hollow caves are vast and expansive. Crystal hollows favor moderate climates such as the Plains biomes with an intermediate temperature index but high humidity.

Indicators

Dendrite

Crystal Hollows can be found on the surface by finding Dendrite infected trees, then following the dendrite trail down into the crystal hollow. Dendrite is aqua green in color, which can only be found in crystalized trees; which are trees of any type with a hollow inside filled with dendrite. These crystalized trees are composed of hollow logs with dendrite inside. Dendrite is a teal/aquamarine in coloration and doesn't have any major uses other than having the second highest reflectivity in Crystal Mixing. Dendrite Clumps can be crafted to make Block of Dendrite; which is a useful building material.

Garnet Nodules

Garnet Nodules are a new item with their corresponding block being embedded garnet nodule; which is found in riverbeds above crystal hollow caves. These deep red Garnet Nodules can be cut in the stone cutter to obtain polished garnet nodules which are used in a variety of products. Garnet Nodules are a valid armor trim ingredient.

The Void

The Void is a lightless dimension consisting of islands formed by an uneven plane separated by wide irregular cracks, separating the dimension. A 2nd generation plane above the ground causes floating cones and pillars to generate interconnected bridges of Voidstone. The Void is mostly barren; with occasional groundcover blocks and pools of Void Tears. Theoretically, the Void is beneath the sift; similar to how the Nether is under the Overworld. This indicates that 1 block in the void is 8 blocks in the Sift.

Reaching

The void is accessed by the following methods:

- Rifts may randomly open in the sift that lead to the corresponding coordinates in the void. These rifts; unlike typical rifts in the sift, will close permanently after the player enters, acting as a 1-way ticket to the void. Rifts generating in the sift have a 0.15 chance of being a void archetype.

Mechanics

Elytra

Even though elytras are already unreliable due to the jagged terrain and darkness, the following features are still implemented.

When the player is using the elytra, every tick, there is a 0.4% chance that the player gets one of the following effects:

- Blindness for 30 seconds
- Wall spawns in front of player causing player to die of kinetic energy
- Player teleports down to nearest ground with elytra still open
- Remove all positive status effects

Falling

Hallucination

Hallucinations may occur if the player spends an extended amount of time in the void dimension. Hallucinations include:

- Rifts that disappear when the player gets too close.
- Lava pools in the distance that disappear.
- Overworld houses and structures in the distance that correspond to the features in the corresponding coordinates in the overworld.
- Bright lights in the distance that disappear if player gets too close.
- Gains the Gloom effect that can cause the following from most likely to least:

- Reduced speed, attack damage [weakness] 0.5 [<== chance /1]
- Items in inventory appearing and disappearing 0.45
- Random damage occurring at any time 0.3
- Tools breaking prematurely 0.25
- Footsteps with reverb in distance 0.2
- Thundering sounds accompanied by light blue flashes of light 0.2
- Muffled sounds of disc 13 or custom_disc1 playing in distance with reverb 0.2 [see also: custom music discs]
- Mining fatigue 0.1
- Keybinds reversed temporarily 0.1
  **Note that the player cannot see if they have the gloom effect; they only know if they suffer from the symptoms listed above.*

Configuration properties

- The brightness setting while in the void is set to 0 and cannot be changed.
- Potions of Night Vision do not work when drunk in this dimension.
- Lanterns, glowstone, magma, redstone lamps, sea lanterns, end rods, froglights, shroomlights, soul lanterns, soul torches, torches including torch arrows and other light sources emit only 4 light levels instead of 15 or original light level config.
- Totems when popped cause the player to be teleported to a random location they have already loaded in.
- Hunger levels continue to deplete even if the player is standing still.

Biome resets

Every 10-15 minecraft days, a biome reset will happen causing the entire biome to reset to its original state, preventing the player from building permanent structures and corruption from becoming annoying. Note that features such as Cavalier hives or Deceiver patches do not respawn at their original location. This can happen to any biome in the void. Note that not all biomes reset at the same time to prevent lag.

Ambience

Ambience _ void_1-6, drone, enter

Has a 0.5% chance of playing every tick in Abyssal Plains [void], Voidstone peaks [void], and Echoing Deep [void].

For more details, read the section on void in ambience.

Features

Void tear pools

Large void tear pool

Large Void Tear Pools are features generating on the many Voidstone islands of The Void. These tend to generate at the center of the islands, without favoring any biome in The Void. These consist of 3-10 block deep disc of void tears panning out on the edges onto voidstone banks. Pools can reach 8-18 blocks wide, with Starlight Kelp[see also: void: groundcover] establishing themselves in the deepest parts of the Void Tears though Abyssal Teargrass has a lower chance of spawning than in smaller pools. Holes may generate near the banks of the Void Tear pools into the depths of the islands, causing large waterfalls to flow into the deep void.

Small void tear pool

Small void tear pools generate at the edges of smaller fragments of voidstone islands causing small waterfalls to flow into the Void. These pools are 4-7 blocks wide, circular to ovoid in shape, and 1-4 blocks deep. These pools are less likely to generate starlight kelp but have a higher chance to generate abyssal teargrass instead.

Abyss

The Abyss is a large sinkhole-like structure generating in the Abyssal Plains with a voidout zone at the bottom of the sinkhole (Totems do not activate here), causing a unique death message causing the screen to turn black then white then the player respawns at the last location they touched land.

Deceiver structures

Deceiver's trap

If a player moves toward a Deceiver mushroom with the trap preset, then the deceiver will dim to a light level of 1, causing the player to investigate the mushroom. If the player is within 7 blocks of the Deceiver, an Abyss sinkhole generates spontaneously, causing the player to void out. The sinkhole may degenerate after 10000 ticks and corrupt the chunk into a new generate. Regeneration of the chunk causes this biome to become increasingly rugged until the next biome is reset. The deceiver mushroom can be harvested with silk touch shears, but the player must harvest them quickly or else it will corrupt the chunk.

Entities

Corrupted Angel

The Corrupted Angel is a variant of the Angel but larger in size and more desaturated. The texture of the Corrupted Angel other side however is closely resembling the void with a

halo prominent on the head that is half white yellowish that is rendered with Emissive Glowing and the other half cracked and dark purple. The Corrupted Angel can fly using random pathfinding procedures and attack by throwing the Void Fragment Projectile. This entity has 25 health (12 ½ hearts).

Void Carp

Void Carp is a species of fish that can be found in large void pools and have no eyes, as eyes are no longer needed in the darkness. The skin of this fish is very pale and slightly translucent but is very sensitive to sounds that the player might make. Once sounds are sensed by the Void Carp's whiskers, the Void Carp swims into the darkness of the Void Tears and then despawns into white particles. If the player is holding Abyssal Teargrass, the void carp will instead go toward the player and will disappear if attacked or not given Abyssal Teargrass within 10 seconds.

Morphology

The Void Carp has no eyes and has large prominent whiskers located at the front of the fish that can sense vibrations that the player makes within 5 blocks of the fish in void tears, but the fish cannot sense vibrations outside of void tears. This fish comes in 2 sizes, large and small. Large Void Carp live primarily in solitary, sparse groups in large void tears pools only. Small packs of Void Carp can spawn in large void tears pools and small void tears pools on the larger side of the spectrum (~>5 in diameter, ~4 blocks deep) and typically swim in packs of 1-4 individuals. Note that these fish cannot grow into larger fish from smaller Void Carp.

Groundcover

Starlight Kelp

Starlight Kelp is a species of kelp endemic to pools of void tears in the Void dimension, especially in echoing deep biomes. Starlight Kelp is dark purple with white dot patterns.

Deceiver

The Deceiver is a mushroom endemic to the Abyssal Plains biome generating sparsely in the Void. Even though this biome is practically barren, the occasional Deceiver may generate in already loaded chunks that mislead the player to false exits, traps, treasure, or distract the player from mobs behind them.[see also: Void: features] Deceivers glow blue and are forced blocks so they can be seen from far away. The Deceiver is only able to exist in the Void and will crumble if in any other dimension. There is a random 0.7% chance every tick that this shroom performs on of the following:

- Drift slightly to confuse depth perception of player

- Blink
- Disappear
- Clone itself
- Change into Voidstone once the player is near or looks at it.



Fig 1. Inspiration for Deceiver

The deceiver shroom can be broken with silk touch shears and can be eaten to gain 1 gloom levels but get 10 hunger points (5 hunger bars).

Teargrass

Teargrass is a new groundcover that populates the entire Void dimension. This groundcover has two variants, including the default variant and the Abyssal variant. The normal variant only grows on Voidstone and has a 70% chance of mutating into a tall subvariant if the Teargrass base block is within 10 blocks of Dechite Ore blocks. The Abyssal variant will only grow submerged within Void Tears and will attract nearby Void Carp. The Teargrass will slowly accumulate moisture and will drip teardrops of Voids Tears every 2-5 seconds that can be collected in Cauldrons to make a Void Tears farm. This plant can be eaten but will reduce hunger by 5 points (2.5 hunger bars) but resets the Gloom effect to level 1. Can be used to breed Void Carp and when combined with Starlight Kelp and Deceiver shroom to cook Stuffed Void Carp which gives Gloom Immunity effect.

Generation and biome

The Void consists entirely of Voidstone, and consists of a collection of small and larger islands with large gaps and connections between them. The Void is a lightless dimension

so without glowroot torches or glowing effect; the player will simply fall into a pit and die of starvation because hunger drops even when the player is standing still or walking in this dimension. Biomes include Abyssal Plains, Echoing Deep, and Voidstone Peaks are the biomes of the Void dimension.

Abyssal Plains [Void]

The Abyssal Plains is a vast, barren, plain-like dimension of Voidstone. Dried lava pools can be found with regular Stone on the side, and even sometimes with a few blocks of Cobblestone and a failed speed portal.

Features

Entities

Shade

The Shade is an invincible mob that generates naturally exclusively from light levels 0-3. This mob only spawns on naturally generated blocks, not ones modified by players or other entities. The Shade spawns after the player peers behind them after hearing the ambience_void_1-3 sound of footsteps. Appearance is entirely black, with Wisp particles generating behind it. Appears similar to a grim reaper. Despawns if the player is within 15 blocks, and will not spawn if the player is looking in the area. Like Enderman, this mob may alter the blocks of a player's structure, or corrupt Voidstone to air.

Depth worm

This new creature has 100 health (50 Hearts) and is very illusive and typically burrows deep into the voidstone. When awakened the worm will try to attack from below the attacker (most of the time the player) and will try to swallow them whole (1 shot them). The player cannot damage the worm with any weapons as its scales is too tough for normal weapons and will just waste 2x the durability oneromac weapons attacking it. The only way to damage the worm is to use explosives (tnt, gunpowder arrows) and get the worm to swallow them. When the worm is killed it drops 4-7 Dechite Shards [see also: Dechite] and has a 5% chance to drop 1 Raw Dechite (Equal to 36 Dechite Shards).



^^ worm (not this one just one that exists)

Structures

Monolith

The Monolith is a new structure that now spawns exclusively to the Void dimension. This structure is made up of 9 towering structures made of Voidstone Bricks with 4 entrances on each side of the main structure. In each of the smaller structures there is a new mob: the Corrupted Guardians. These Voidstone guardians are controlled by a Wisp and when killed drop 2-5 Chiseled Voidstone. At the top of the towers there may be either 1-5 Dechite Shards, and in one of the towers there is a Dechite Upgrade Template (guaranteed). In the central tower there are 5 floors:

Floor 1: There are 5-10 Corrupted Guardians and a chest containing decent loot from all the other 4 dimensions.

Floor 2: There is a Chiseled Cauldron and ingredients for various vials.

Floor 3: There are many Wisps moving blocks around making it hard to get to the 4th floor.

Floor 4: This floor has 4 different rifts, 1 lit with the other 2 being closed. The 3 remaining portals from left to right are the:

- 1st portal: Lit with a Crimson Snatcher Claw to go to the corresponding coordinates in the Nether
- 2nd portal: Lit with a Soul Block to enter the Sift
- 3rd portal: Lit with Chorine to enter the Outer End Islands

Floor 5: On the last floor of the monolith there is a single chest in the center containing 1 Dechine Lure inside.

When the dechine lure is used on a voiding rod in the end, there is a 2.5% chance of getting a depth worm key (and a 97.5% chance to get a dechite shard but will take 90 secs per reel). When this key is obtained, a secret room appears below the central tower with a

keyhole. When the key is used on the entrance to the secret room, the mother depth worm will spawn.

Mother Depth Worm

The Mother Depth Worm is a boss mob that occurs only in <room>. [unresolved]

Mother Depth Worm Fight

Once the room has been opened, the Mother Depth Worm will rush out immediately with a bite attack that deals 25 damage [affected by armor attributes] if hit. Then 1 of 4 attacks will occur until it reaches 25% health:

- The Mother Depth Worm lashes out with its tail and then pounds the player into the ground dealing 7-10 damage. [random chance] (does not include suffocation damage)
- The Mother Depth worm digs underground waiting for around 15 seconds if the player has not built a platform above the ground within that time period it will instantly kill the player regardless of armor or potions. (Does not include totem of undying)
- The Mother Depth worm leaps high and screeches if the player does not build a wall between them and the Mother Depth worm; they will take 10 damage regardless of armor.
- The Mother Depth worm will stand up vertically and make a long bellowing sound that summons 5 Depth worms and will shoot 10 homing Depth worm teeth at the player dealing 2 damage each.

Voidstone Peaks

This biome contains elements that make the player feel lost and challenge the player in ways other than mob spam. Eventually the biome will kill the player as they will run out of food and are unable to heal then gradually take fall damage until the player dies, unable to heal the damage without food.

Entities

Cavalier

The Cavalier is a 6-legged creature resembling a villager with scales inhabiting “hives” located on platts on steep peaks. If the player accidentally looks at a Cavalier, it will emit a high-pitched noise that gives the player Slowness, Blindness, Withering, and Voidal Shroud which gives it time to attack the player with tentacles that extend from the chest. The Cavalier can be scared away with strong lights such as Torch Arrows, Spectral Arrows, Deceiver shrooms, or Glowroot products.

Wisp

The Wisp is a translucent Allay-like hitboxless entity that cannot be killed or controlled, and only spawn in the Voidstone Peaks biome. They can phase through any block. They will randomly pick up and place any block that is not Voidstone or Dechite Ore but will not do this when the player is looking. The Wisp will teleport away if the player looks toward the direction of the Wisp's location, and only spawn on artificially placed, modified, or blocks that the player has stepped on behind the player. The Wisp dims light sources and causes the chunk it is in to have a 3% higher chance of corrupting. The Catalyst Lantern can be used to ward off the Wisp at the cost of its fuel.

Blocks

Voidstone

Voidstone; the matrix of the Void, is a plain, dark gray, desaturated stone variant present as the matrix in most void biomes. Voidstone has a resistance of 2.5, hardness of 1.5.

Dechite [see also: Ores and Minerals: Dechite]

Dechite ore is the main ore present in Voidstone Peak biomes. When mined it drops 2-7 Dechite Shards. Dechite Ore has a 20% chance to drop 1-2 Dechite Chunks and has a 2% chance to drop a singular Raw Dechite. The Dechite Ore has the same chance to spawn as diamond, but a higher chance of exposed air deletion.

Void tears

This new corrosive liquid destroys everything that isn't from the void. When touched or drank it will give the Voidal Shroud effect which impairs vision causing everything the players sees to appear gray and when exposed to light will take heavy damage. This liquid vaporizes on very little heat. If this liquid falls more than 25 blocks it will instantly vaporize turning into a lingering cloud of voids tears. When the liquid is exposed to any amount of light, it instantly vaporizes.

Natural events

At any point randomly, a very faint rumbling sound can be heard followed after 10-30 seconds following the sound, a random event of the following may happen:

- Mountain glitches and turns into air, causing the player to fall into void.
- Player gains Gloom or Voidal Shroud effect.
- Glitched areas unglitches and fills back with new terrain.
- Chunk regenerates paving over any existing blocks.
- A rift opens to another dimension (nether or overworld with corresponding coordinates).

Features and structures

Echo shrine

The Echo Shrine is a structure that generates in the Voidstone Peaks biome and the Echoing Deep biome. This structure contains a Voidal Vault feature. The vault can be opened with a Depth Worm Key and contains various rare loot including:

- Dechite Chunks
- **[unresolved: more items]**

Abandoned Cavalier Hive

Abandoned hives; unlike live ones turn from gray to black. While they are hard to spot, these hives contain loot that the Cavaliers have scavenged. Straggler Cavaliers or Cavalier skeletons may be found on the ground, as well as Void Duff, a block that can be brushed for loot; because of buildup from Cavalier's sheds, droppings, and bodies.

Voidal Abandoned Camp

Voidal Abandoned Camps are structures made from Voidstone Talus blocks are found on platts in this biome. Camps contain small amounts of Void Duff blocks sometimes concealing a chest, makeshift Voidstone Talus walls at various stages of decay, and deep pits covered with Void Duff.

[unresolved: GRAMMAR]

Oceans

Blocks

Ancient Prismarine

Ancient Prismarine is a block in the same family as Ancient Sand. Ancient Prismarine when mined does not trigger the array of Prismatic Sensors within the Ocean Monument. Ancient Prismarine, unlike other blocks, are not affected by Mining Fatigue and vanish into air if any Ancient Prismarine block or item leaves the perimeter of the monument, defined by the bounding box, to prevent the player from smuggling Ancient Prismarine across Ocean Monuments.

Ocean Monument changes

The ocean monument is now included in the points of importance across the map, which include Ocean Monument, Jungle Temple, Nether Fortress, Desert Pyramid, Ice Palace, and Ancient City. Each of these contain puzzles leading to a boss fight, dropping a

corresponding rune to each point of importance. These in Eric method are used to craft an Eye of Ender. This point of interest's theme is about transporting a key through copper grate pipes. The player must repair or make it so that a key can make it through the copper grates and throughout the Ocean Monument. Eventually the player picks up the key and uses it to summon the boss of this point of interest. Generation of the ocean monument has now been moved to the Ocean Abyss biome.

Prismatic Sensor

The Prismatic Sensor is a special block responsible for detecting blocks broken in the Ocean Monument and replacing them to prevent greifing before the puzzles and boss are completed. If a block is broken, one of the four Prismatic Sensors located in the 4 quadrants of the Ocean Monument will activate, causing the block to not drop and be replaced while spawning enchantment table particles (server side).

Active

Active Prismatic Sensors have a deep blue matrix with 2 panels of red breathing illuminators on each side, depicting its state. Client-side enchantment table particles are expelled from the sides that orbit the main block. Particles are culled if >30 particles are detected in a chunk.

Deactivated [see also: Water dragon]

Deactivated Prismatic Sensors have the same deep blue matrix as the active sensors but have light blue breathing panels instead of red ones. These deactivated sensors allow block to be broken and are only deactivated after the water dragon is beaten.

Puzzle jigsaws

Puzzles in the Ocean Monument are all separate jigsaws that are then paired together and form a complex maze system. Each puzzle has multiple variants, to prevent speed runners from using the same route over and over again. The following are possible puzzles with variants listed as bullet points below.

P1: Inertia exists

This puzzle consists of a 9x9 grid of copper grates encased in glass. Pistons above the glass have underwater levers attached, allowing the player to open the gates and push the key through the grates using the water. However, the player must push it to a certain central location by using pistons to block or unblock areas.

Inspiration

P2: Piston puzzle

This puzzle consists of a piston pointing upwards followed by a 3-block gap and an offset Copper Grate pipe that the key is stuck at. The intended solution is that the player must place 3 Ancient Prismarine blocks between the piston and the Copper Grate blocks so that when the piston is activated via an input (varies by variants) the copper grate pipes are re-aligned.

- Variant 1 is identical to the description above but has a Weighted Pressure Plate input in a hole that can be filled with a hidden chest between a slab that can be accessed when the player crouches. Within is Rotten Flesh for the pressure plate and exactly 3 Ancient Prismarine.
    - Variant 1,1 uses a chest hidden by a spruce trapdoor instead of a crouch slab device.
    - Variant 1,2 has the chest hidden to the left behind ambient seagrass and kelp (that generates across all subvariants).
    - Variant 1,3 has the chest hidden by the right behind ambient seagrass and kelp (that generates across all subvariants).
    - Variant 1,4 has a chest hidden in 3 block deep hole embedded into the side wall and an extra hidden barrel in a stack of barrels with a Stone Button (special attributes to be placed) to activate the piston instead of Weighted Pressure Plate.
- Variant 2 has a 2-block gap between the piston and Copper Grates instead of a 3-block gap, as well as only 2 blocks of Ancient Prismarine in the hidden chest instead of 3.
    - Variant 2,1 has clumps of seagrass and tall seagrass growing in a <material determined by variant> bed in the room.
        - Variant 2,1,1 has clumps of Seagrass growing in Gravel.
            - Variant 2,1,1,1 has Suspicious Gravel under the normal Gravel in the pit.
        - Variant 2,1,2 has clumps of Seagrass growing in Sand.
            - Variant 2,1,2,1 has Suspicious Sand under the normal Sand in the pit.
        - Variant 2,1,3 has chests filled with randomized loot under the seagrass/tall seagrass substrate, including:
            - Iron Ingot, Nugget
            - Gold Ingot, Nugget
            - Copper Ingot, Nugget
            - Diamond

                - Redstone Dust
                - Rotten Flesh
                - Bone
- Variant 3 has 2 separate gaps; specifically, 3 blocks and 2 blocks wide. Only 4 Ancient Prismarine blocks are scattered in groups of 2 within Barrels hidden across the room (depends on subvariant) as well as a hidden chest in a trapdoor crawlspace located on the floor containing a 1 durability unrepairable Iron Pickaxe with Silk Touch and special attributes that allow the player to mine 1 of the Ancient Prismarine. The intended solution is to first use 3 of the Ancient Prismarine blocks to allow the key to enter a Hopper with a Lever. Afterwards, use the 1-durability Silk Touch pickaxe to mine one of the 3 placed Ancient Prismarine combined with the 1 leftover Ancient Prismarine to recover the last broken section of Copper Grate pipe; leading the key to the next jigsaw.
    - Variant 3,1 has two 2-block gaps instead of 3 and 2 block gaps and corresponding only 2 blocks of Ancient Prismarine. Each Barrel instead contains only 1 Ancient Prismarine.
        - Variant 3,1,1 has only 1 barrel with both Ancient Prismarines located behind Kelp blocks for camouflage.
        - Variant 3,1,2 has only 1 barrel with both Ancient Prismarines located behind Seagrass and Tall Seagrass blocks for camouflage.
    - Variant 3,2 has the 1-durability Silk Touch Iron Pickaxe hidden in a sub-room located to the side of the main room with simple parkour and a blockade of blocks [depending on variant] to block out the water. Below the parkour is lava.
        - Variant 3,2,1 has a 3-block jump to an intermediate platform followed by a 1-block Neo leading to the chest containing the 1-durability Silk Touch Pickaxe.
            - Variant 3,2,1,1 has Kelp blocks decorating the Neo jump.
            - Variant 3,2,1,2 has Seagrass and Tall Seagrass blocks decorating the Neo jump.
            - Variant 3,2,1,3 has a Sea Lantern block forming a checkerboard pattern on the Neo jump.
        - Variant 3,2,2 has 2 adjacent 3-block jumps leading to the chest containing the 1-durability Silk Touch Iron Pickaxe.
            - Variant 3,2,2,1 has Kelp blocks decorating the intermediate platform between the two 3-block jumps.

        - Variant 3,2,2,2 has Seagrass and Tall Seagrass blocks decorating the intermediate platform between the two 3-block jumps.
        - Variant 3,2,2,3 has a checkered pattern of Sea Lantern Blocks decorating the intermediate platform.
    - Variant 3,2,3 has a singular 4-block jump.
    - Variant 3,2,4 has two 1-block Neos separated by a central platform made from different blocks [determined by variant].
        - Variant 3,2,4,1 has a singular Conduit block decorating a Quartz pedestal.
        - Variant 3,2,4,2 has an intricate fountain structure made from Stone Brick family blocks with 4 spouts exiting from the cardinal directions of the fountain.
        - Variant 3,2,4,3 has a large platform in the center with a glass tank extending from ceiling to the bottom of the fall pit. Inside the tank is an arrangement of Gravel, Sand, and Clay covered by Kelp, Coral Block block family, Coral Fan block family and Coral block family. Tanks may be generated with Squid, Cod, Salmon, randomized Tropical Fish, or Pufferfish.
        - Variant 3,2,4,4 has a large 2x2x(height of room) tall quartz column arrangement with a circular perimeter walkway to reach the next Neo jump.

P3: Target Block puzzle

This room when entered has a deep pit on the righthand side guardrailed by Stone Brick Wall blocks extending across the entire left side of the pit. Behind the pit embedded in the wall are 2-3 Target Blocks (determined by variant) connected to each other via an AND gate. On the lefthand side of the room are stairs leading into an opening with 2-5 Drowned, and 4 non-despawnable Skeletons guarding a mountain of barrels, mostly empty, with only 1 containing exactly 4 Prismarine Shards. The object of this puzzle is to transport the key through the room to the next by repairing the waterlogged Copper Grate pipe system located directly above the pit. The Copper Grate system is repaired when all Target Block are activated at once causing the AND gate to output a signal; which then causes a hidden piston to push an out-of-place Copper Grate block back into position whilst triggering a dropper to dispense the key. The intended solution is to either use a Multishot enchanted bow that you may have with you or kill the skeletons for arrows which then can be inserted to a Fletching Table located on the lefthand side along with the Prismarine Shards from the chest (as the tip), and one of the 4 Oceanic Rods obtained through a hidden Hopper

located on the left side wall of the pit accessible via F5 [can be overridden by variants]. Note that projectiles such as [unresolved: Oceanic Rod result projectile] do not have a large enough explosion range on their own to activate all Target Blocks consistently.

	Railing			
Block	Block			R side
Block	Hopper			R side
Block	Block			R side
Block	Block	Pit bottom*	Pit bottom*	R side

*Not to scale

The product arrow from the Fletching Table can be shot underwater (due to Prismarine Shard tip) and creates an explosion radius around impact point, which when shot at the central Target Block, triggers the 2 surrounding Target Blocks.

- Variant 1 has a line of Hopper blocks with only 1 randomized hopper containing the 4 Oceanic rods.
- Variant 2 has only 2 Target Blocks instead of 3 with only 2 Oceanic Rods in the hidden Hopper, only 2 Prismarine Shards instead of 4, but more hostile entities including 6 Drowned, 3 Skeletons, and 1 Guardian. The player must shoot between the 2 Target Blocks to activate them.
    - Variant 2,1 has

First floor

On the first floor, various puzzles are contained that challenge the player.

Upper level

In the upper level of the Ocean Monument, this is where the player must first go. 2 staircases flank the monument exterior leading to the upper exposed floor, where there is a pedestal guarded by 3 trident Drowned and 2 Guardians. At the center is minable Ancient Prismarine, that like ancient sand, will not trigger the calibrated sensors within the monument. Ancient Prismarine is not affected by mining fatigue. The mined Ancient Prismarine unblocks a bubble column that triggers a sensor on the roof of the gazebo-like structure over the center of the monument which dispenses a key into a Copper Grate pipe system. The pipes lead to the front of the monument, where the pipes are broken and require the player to replace the system, reaching a dispenser checkpoint. If the key despawns now, the key will return to this dispenser instead of the starting dispenser above the pedestal. The key then proceeds to enter a roofed room with a barred off area with ice floors. The key is dispensed in a sulfur cube with barrel that the player must use prismarine

shard underwater arrows to direct at a hole, where the key continues its journey onto the next room.

Lower level (Basement level)

After the key enters the hole leading to the lower level, the player must use a spiral staircase or the previous 2 staircases flanking the exterior sides of the monument to access the basement floor, which is below the main floor where the boss fight occurs. The key is dropped through a glass tube falling along with the central cavity in the spiral staircase, then reaches the lower-level B1. The lower level of the monument is themed around an abandoned underwater dungeon, with relatively dark lighting accompanied by:

- Seagrass
- Tall seagrass
- Barrels [randomized loot]
    - Candles
    - Prismarine items
    - Skull
    - Iron nugget
    - Copper nugget
    - Gold nugget
    - Bone
    - String
    - Rotten flesh
    - Leather
    - Rabbit hide
    - Explorer map
        - Abandoned camp
        - Sunken ship
        - Other Ocean Monument
- Burnt candle
- Kelp
- Sand
- Gravel
- Suspicious mediums
- Hadal mediums

and other components such as skulls. Burnt candles can also be found on barrels. The barrels on this floor contain miscellaneous items such as

- Nautilus Shells,
- Gold Ingots, Nuggets, and equipment
- Iron Ingots, Nuggets, and equipment
- Copper Ingots, Nuggets, and equipment
- Hearts of the Sea
- Emeralds
- Uncut materials

Water Dragon boss sequence

The Water Dragon fight occurs at the top of the Ocean Monument on a glass disc specifically located in front of the gazebo-like structure used for the first puzzle.

Initiation [technical]

The sequence is initiated when the player inserts the key into the center of the glass disc, which encapsulates a cylinder made of prismarine filled with ocean groundcover. As the key falls through the cylinder below the disc of Glass Blocks, `camerashake` is applied to the player while the camera is locked following the key on the way down. A block display is summoned off campus out of bounds behind the prismarine cylinder in an inaccessible area. The block display is made of custom blocks that depict the Water dragon, which then snatches the key by first killing the tagged item then summoning a block display anchored towards the Water Dragon's block display. Then the main block display swims across the width of the cylinder, breaking any nearby blocks and summoning bubble particles with negative gravity. The block display is culled and saved after the Water Dragon is out-of-bounds. After the first sequence listed above, then a similar sequence is played out as the camera pans toward the Dragon block display [tagged mod:water_dragon]. A similar sequence occurs directly after this time with amplified water fog and stored as a file that plays at the same time each time during this fight. After the 2nd cutscene, the Water Dragon itself (non-block display) is forced upwards which currently has no hitbox. The glass in the area is culled with the glass break sound effect played on the global server-side. `glass_break` particles are summoned around the perimeter of the disc as the Water Dragon gains elevation then curves downward initiating the fight.

Main

Alpine Meadowlands

The Alpine Meadowlands are a relatively uncommon biome generating in cold areas typically near Jagged Peaks and Windswept Taiga biome borders. Weirdness index peaks in these biomes, causing glacial valley-like features to form. Glacier structures have a high

chance to generate in these biomes. Berry Bushes, Ferns, Bushes, Lupines[see also: lupines], and other vegetation and fungi are common in the open areas in this biome. This biome has a unique tint where some vegetation is tinted red, orange, or yellow while other vegetation such as Lupines are not affected by vegetal tinting in this biome. Poplar trees, old growth birch, and fir spruces also spawn sparsely and erratically in this biome; often in clusters. Red Bushes from the Dappled Forest biome may also generate here contrasting with the green needles of the evergreens. Shelf Fungi may generate on trees, and small rodents [undecided] dig burrows here. Creek river biomes may cut through this biome, making Garnets [see also: Garnets] and other ores such as Emerald, Gold, Coal, and Copper are common here.

Lupines

Lupines are a naturally spreading plant found in the Alpine Meadowlands biome. Lupines bear violet to blue flowers.

Clover Glades

Clover Glades are a new biome consisting of many different plants, animals, fungi, and materials. Clover Glades generate in areas with intermediate temperature noise and intermediate to high humidity/vegetation noise. Sparse patches of Oak trees generate with bee hives. Rivers have a higher chance of re-routing into this biome; causing delta-like floodplains to form; allowing Clover plants to grow along their banks.

Mutated

Mutated Clover Glades can occur after a Clover Glade intersects with a low depth value biome or hilly biome. Mutated Clover Glades may also be generated at the edges of superbiome borders [read also: superbiomes and improved terrain generation]. Mutated Clover Glades contain different trees and may have other terrain features such as hills and valleys caused by erosion value variation.

Clovers

Clovers are a new plant block with no hitbox that has the same selection box as Leaf Litter, Wildflowers, Cherry Petals, etc… Natural Clovers when mined, will drop either 3-leaf clovers or a 4-leaf clover, with 3 leaf clovers being the most common [read below]. Clover plants cannot be bonemealed and can only be propagated through their slow, albeit natural spread across:

- Coarse dirt
- Hadal dirt
- Poor farmland [read also: farmland changes]

- Rooted dirt
- Clay
- Cobblestone
- Mossy cobblestone
- Exposed stones
- Sand
- Red sand
- Gravel
- Suspicious sand
- Suspicious gravel
- Stoneshroom mycenated...[read also: stoneshrooms]

Looting III shears can be used for the clover plants to drop up to 5 clover items.

3-leaf Clover

3-leaved clover items are obtained by mining naturally grown clover plants, which have a 95% chance of dropping 3-leaved Clovers. Note looting shears still use the same proportions of drop rate ratios of 3-leaf Clover to 4-leaf Clover items dropped. The 3-leaved Clover can be used in the composter to create bonemeal (faster than usual).

4-leaf clover

4-leaved Clover items are a rare and uncommon item with a yellow hover text in the hotbar or inventory menu. 4-leaved Clovers can be obtained by using the 5% drop rate from mining clover plants. Once acquired, 4-leaved Clovers can be used to make 1 entire bonemeal in the composter with just one 4-leaf Clover or used to brew potions of luck when added into the brewing stand along with an awkward potion.[read also: potions of luck]

Clover loot table

When a Clover plant block is broken, the following loot table is used:

If item has Silk Touch >0:

- Clover

If item has no Silk Touch

- 3-leaf Clover 0.95
- 4-leaf Clover 0.05

If item is in tag category `#minecraft:swords` overriding previous loot table statements:

- Air

Desert

Due to the changes in Strawberry survival, the desert's extreme heat, dry conditions, limited food supply, and the lack of anything except sand and cacti makes it a useless, undesirable biome that most players will avoid and rarely use. However, the new changes in the desert now make the desert not only a good place to live, but maybe even better than other biomes.

Scorched caves

The scorched caves are a cave biome that can generate under deserts and consists of coarse dirt, sandstone patches, scorched stone, exposed stone, and fossil sandstones as well as giant fossil walls.

Fossil sandstone

Fossil sandstone is a new variant of sandstone identified by a {fossil} nbt tag (only visible in creative mode), and has a texture nearly identical to regular sandstone, and the only way to identify them in survival is to use a hammer on them, at which point the fossilized sandstone gets a new cracked texture temporarily. When mined, fossilized sandstone drops a bone and rarely an iron, copper, or gold nugget. When a hammer is used on a fossil sandstone, they drop 1-3 bones, raw iron/gold/copper, iron/gold/copper nugget, bone, charred bone, diamond shard, redstone, skulls, and other items. The craftsmanship enchantment [read also: Craftsmanship Enchantment] affects these drops, too. Using a silk touch hammer (just please do not ask how that works) will give the same drops, but the bones will be replaced with rare fossils that can be decorated with crystals and act as a stylish item frame, and some will be black in color, representing the charred bone.

Contents

The scorched caves contain copper and gold as well as lapis lazuli. Large fossils generate diamonds at every Y-level, but the deeper the fossil, the more diamonds. Serpentine generates here as well. Lumin Mytanum is also common here, and black Lumin Mytanum is exclusively found here.

Mutated

Scorched crystal hollows

The scorched crystal hollows are a biome at the intersection of crystal hollows and scorched caves where garnet and calcite make up most of the crystals, and 'crystal fossils' spawn where the normal large fossils are adorned with diamond and emerald crystals, as well as other crystals, and fossil sandstone will have the 'crystal axolotl' fossil and other fossils adorned with crystals.

Oases

Oases are a new feature that replaces desert wells and generates near desert villages acting as more points of interest in the desert. Oases of both types may generate separately in areas with a weirdness noise >-0.5 and will increase the surrounding humidity noise to ~0.3, which allows dark forests and jungles to generate in small patches adjacent to oases. Oases come in 2 types: small, and large. Oases encompass palm trees which have many functionalities [read also: palm trees], making them a desirable option to live in. Oases also offer cooler temperatures during the day, so the player does not have to wear a palm vest during the day.

Small oases

The small oasis is a feature that replaces 50% of the generation of desert wells. Small oases consist of up to 2 pools of water roughly the size of a charged creeper crater but flatter in depth. Small oases contain sparsely scattered palm trees, and small oases also have a temperature noise so that leaf vests are not needed. Suspicious sand also generates in concentric rings around the perimeter of water pools along their shores.

Large oases

Large oases are large features bearing pools of water up to 30–50 blocks in diameter. These large oases boast a wide range of unique vegetation such as thunder roses including plants that grow terrestrially, marginally, and aquatically. Palm trees are abundant, especially in large oases. Large oases only generate where desert villages generate, but the desert villages still generate next to the oases. The jigsaw generation has been changed to not have a path originating from the village generating more than 50 blocks away from the oasis. This means the villages are centered on or around the oasis. Palm trees in these oases are scattered on the perimeter of the oasis, generating after the village to prevent palm trees from clipping villagers' doors in half. Docks may be generated in larger villages within the pools of water.

Jungle oases

When a jungle generates next to an oasis, they will become intertwined. A montane jungle results in one or two big mountains and a waterfall flowing into the oasis. Both jungle and palm trees are found in these oases.

Technical details

Oases are a feature, but oases also have their own biome attribute as `desert_oasis`. Oasis water has a salinity ranging from 0.2 to 0.4 or up to 1.0 near the ocean, and a temperature of 2.5 to 3.0 and brackish water, making filters necessary before safe drinking. Minecraft villages generating in the specified biome will now use palm wood instead of jungle.

Palm trees



Fig 1. Palm tree in superflat world

Palm trees are a new species of tree with mostly upright trunks and large, shady leaves used to make leaf vests.[realism mode only] Palm trees also contain coconuts that may fall when ripe [see also: coconuts]. Palm trees are endemic to oases and desert islands (deserts surrounded by rivers and oceans that are small) and can be found scattered in small oases and in patches of groves in large oases. Palm trees grow from palm seedlings on. [see: coconut].

Palm Leaves

Palm Leaves are special leaves with a unique texture with diagonal bracts and a central petiole. Palm Leaves along with string or rope can be used to craft palm vests to prevent the player from overheating in hot biomes. [realism mode only]

Coconuts

Coconuts are harvested from palm trees and will eventually fall, getting replaced by new unripened coconuts. If they fall on a mob or player's head, they take 6 damage (❤❤❤) and when a coconut falls, the player can pick up the coconut item, and the player can manually make a coconut fall by hitting the on-tree coconut with swords, shears, or axes.

Throwing Coconuts

The player can throw a Coconut as a projectile, dealing 6 damage points (❤❤❤). This could be used for 'Coconut combat' kit where people throw coconuts at each other in a form of PvP. Coconuts can be used basically as a heavier form of snowballs. If the Coconut is thrown with too much force, the coconut may split in half and drop two coconut halves on impact. Coconuts upon impact on a hard surface gives off a white particle and Coconuts deal extra knockback compared to arrows.

Planting Coconuts

Coconuts can be planted by right clicking on:

- Gravel
- Hadal mediums

- Red sand
- Sand

This is the only way to get an artificial palm tree. After 1-5 in game days, the coconut will grow into a palm sapling and eventually, a palm tree.

Coconut Milk and Bowls

Coconuts once acquired can be cracked open using a crafting table to produce 2 coconut halves and 1 bottle of coconut water if an empty glass bottle is also inserted into the crafting table. To acquire Coconut milk, the player must either insert the coconut halves into the evaporator to produce coconut fiber (used to craft rope and composting) and 1 bottle of coconut milk. Coconut milk restores 5 hunger points () and can be crafted with watermelons on the crafting table to make a watermelon smoothie. Coconut bowls are a new item crafted by inserting coconut halves into the crafting table or eating out the meat inside of a coconut. Coconut bowls are a sustainable way to make bowls.

Coconut oil

Coconut oil is a new greasy substance extracted from coconut milk. Coconut oil can be used for lubricating pistons to make them more silent and reduce push time, used as a lubricant for gem cutters, and making candles.
Coconut oil is extracted from coconut milk through the evaporator or fermenter.

Desert villages

Tall houses

The tower house in the desert consists of a single bed at the bottom with a long staircase up to the top, and a chest somewhere in the middle. To make these easier to renovate, the spiral staircase will be on the sides, like this:

Wall	Wall	Wall	Wall	Wall
Wall	Hole for stairs	Floor	Staircase	Wall
Wall	Hole for stairs	Floor	Staircase	Wall
Wall	Hole for stairs	Floor	Staircase	Wall
Wall	Wall	Wall	Wall	Wall

This gives 4 blocks of space (under stairs) to use on each floor, possibly for village expansion by the player. Houses now can also optionally generate including a terrace, balcony, deck, or large window as part of a new jigsaw generation method for each individual house.

Farms

Farm irrigation is now connected to oases to prevent evaporation within the irrigation ditches. Oasis water is not suitable for farming (due to light salinity of brackish water) but can still grow crops without turning nearby soil and growing materials into hadal mediums.

Irrigation channels have walls on both sides made from andesite, mossy cobblestone, tuff, cobblestone, stone, exposed stones, and the occasional stoneshroom mycenated block. Note that these channels are collapsed in abandoned desert villages. These irrigation channels can also be used as boat canals for the player as docks now also generate here.

Town centers

In vanilla Minecraft these are the 3 town centers for desert villages and the players' names for them:

The fountain | The well



Top left [fig1]: Fountain, Top right [fig2]: Desert well, Bottom [fig3]: Marketplace stands encompassed in agoras generating in some desert villages.

Straw Golem

The Straw golem is a golem built with 4 haybales and 1 pumpkin. It will walk around in a 15-by-15 block radius, defending and harvesting crops.

Agora

- Stands (shelf) like the one in the marketplace where villagers display their items from (Same profession shares stands)
- A meeting place with stair-benches and a glazed-terracotta centerpiece
- Stairs into the oasis biome
- An always – generating palm tree or two that provide shade to the poor villagers standing in the sun which is deadly without proper protection

Planks

Minecraft villages generating in the specified biome will now use palm planks or charred planks instead of jungle materials. This allows for more wood-oriented house designs that have charred plank roofs to lower the chance of burning down during thunderstorms and raids.

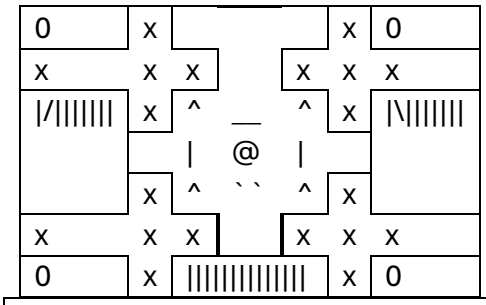
Thunder rose

Thunder roses are rare plants that can sometimes be observed generating near water under palm trees or on copper rich mountaintops. Thunder roses emit lightning particles during the night and during precipitation. Thunder roses along with dragon's breath, a goat horn, and copper ingots can be used to craft liquified horn of storms vial.

Desert Temples

Desert temples have been improved in strawberry Minecraft because the current desert temples are not the best with only 2 secret rooms and not enough interesting features that draw the player's attention. Desert temples in Strawberry Minecraft will now generate only ½ of the time compared to vanilla Minecraft because of the large amount of loot containers and increased size of existing temples.

Each square represents a 3x3x3 cube **[needs remaking]**



Key:
0 = Treasure room
X = T-tet-d room
||| = Long room
|/| = Long room with stairs

| or __ or `` = Mosaic

@ = tunnel to basement

Opus

The opus room, located at the front-center of first floor of the desert temple, is an area where the paths to all 3 branches of the pyramid conjunct. At the center is a waypoint with a casting cauldron where the key crafted from each of the 9 key fragments in each branch of the pyramid is inserted.

Mancer Summon & Fight

Sequence

Once the key is inserted, yellow shock particles eject out of the casting cauldron as the particles start to gather together to form the outline of the mancer[better name coming soon]. The particles are forced, then despawn after the mancer appears and takes the place of the particles. The mancer then proceeds to fall and then break the floor, and the player along

with the mancer fall into the underground colosseum below the pyramid. The underground colosseum is an underground semi-obsidian dome, with chiseled patterns lining the wall. A corrupted chest stands above the center of the colosseum, above the floating casting cauldron. The chest can be opened by using a shock rune, which is dropped by the mancer once defeated.

The first attack the Mancer uses is by summoning lightning, trying to shock the player. If the player is shocked, and if their equipped weapon is made of metal or metallic, the weapon takes 30 durability and the player takes 10 damage (5 hearts) if the player is wearing non-conductive armor and 20 damage (10 hearts) if the player is wearing conductive armor. This does not affect weapons, even metallic if the weapons have the shock resistance I enchantment. The mancer during this stage hovers in the air, and the player must drop a metallic item or place a lightning rod under the area where the mancer is floating, which causes the mancer to shock itself. The mancer then falls to the ground, allowing the player to attack the mancer with even non-metallic weapons. Lightning rods can be found in chests at the edge of the underground arena.

After the mancer reaches 50% health, the mancer summons armored parched and Shock mobs that will heal the Mancer until they are defeated. The mancer during this attack has a yellow forcefield that must be dismantled for any attacks to work. The player can either use gunpowder laced arrows or use a wooden shield to deflect the ball lightning that the Mancer launches to knock the mancer down, then use a non-conductive weapon to attack.

Once the Mancer reaches 10% health, it enters a phase in which it can only take damage to the head. During this phase, if the player times a jump based on its attacks, they can be launched onto a ledge.

Once the Mancer dies, it creates an explosion that has a radius spanning the entire desert temple, one that only destroys calibrated sensors. (An explosion of this kind is also created in the underground part of Branch 2, Room 2.) The player is launched upward out of the colosseum with a rune of thunder into the treasure chest, containing a shockblade and chestplate of the storm.

Chestplate of the Storm

The chestplate of the storm is a rare chestplate found in the chest after beating the mancer. The chestplate allows the player to do 2x the damage to players with metallic or metal gear during rain, thunderstorms or snow or mobs in water or precipitation.

Ancient Sand

Ancient Sand is a block native to desert temples affected by gravity. Unlike other blocks, ancient sand does not trigger the calibrated sensors embedded within the structure of the

desert temple, allowing them to be placed and broken within the perimeter of the sensors. This means that Ancient Sand blocks can be used to interact safely with the puzzles in the Desert Temple. Each block of ancient sand contains a non-modifiable attribute containing the ID of the desert temple it generated in, as these blocks disappear after the mancer of the temple is defeated. This is to prevent ancient sand from being smuggled across temples to be used to cheat certain puzzles.

Branches

Branch 1

The first branch of the desert temple contains a series of 4 rooms, leading up to loot and 3 desert key fragments along the way, which can be assembled with 6 other desert key fragments to summon the mancer (better name soon).

- The first room is rigged with calibrated sensors surrounded by observers, not allowing the player to mine or place any blocks. If the sensor is triggered, wired traps specific to that room (in this case a fall trap) may activate and the block the player mined will respawn, preventing greifing until the structure is beaten. [coming soon] The first room consists of 3 parched along with suspicious sand containing ancient sand blocks, which can be placed without triggering the calibrated sensors. The ancient sand is then used to pillar up at the end of a lava parkour sequence, in which the player can enter the second room through elevated exits at the end of the parkour.
- The second room: To exit the second room, the player must penetrate through a wall on the far side of the room. However, this is extremely hard because there are Charges constantly trying to attack the player and an object [characteristics not decided] that will inflict Mining Fatigue on the player and can only be destroyed when a charged piece of metal is near. The charge is attached to a lead which is attached to a fence gate, and if the lead snaps or the charge dies, it is detected and a trap is triggered. To disarm the trap, the player must mine the observer, however it is incredibly difficult to do so due to the Shock and the mining fatigue. To destroy the crystal, the player must collect an iron tipped arrow and shoot the crystal with it, then have the shock hit the crystal. This destroys the crystal allowing for the trap to be easily disarmed and the next room to be reached. Under the trap is a key fragment.
- The third room: This room contains a hidden target block over a lava pit that is blocked by ancient sand. The calibrated sensors allow for shooting arrows and exploding arrows if they impact ancient sand. The ancient sand can also be blown away using the wind gust created by a frostblade. The target block once shot at activates a redstone dropper to dispense 3 ancient sand blocks, used to allow

repeaters to power a piston door, leading to the final room before the treasure chest at the end containing a key fragment.

- The fourth room: Throughout the room, there are armored parched, Shock, leader husks, and charged creepers (do not do damage to blocks when exploding) lining the waterlogged floor with pressure plates. Note sponges are prohibited in this room (will be detected by calibrated sensors). Normal weapons do no damage or durability loss to their armor, and instead the player must redirect shock lightning towards the parched, which breaks the parches' and husks' armor. Once the armor is broken, the player can attack the parched and husks with normal weapons. This can also be done with the shockblade, which can be found in the final chest in this room only. The floor is lined with pressure plates that are activated by the mobs, so the player must kill all the ground dwelling mobs to stop the pressure plates from activating and then open the piston door to a loot chest with a guaranteed key fragment.

Branch 2

The second branch in the back of the desert temple contains a series of 4 rooms, leading up to loot and the 2nd desert key fragments along the way, which can be assembled with the 2 other desert key fragments to create the desert key used to summon the mancer.

- Rooms 1, 2, 3: These rooms are all filled with sand, suspicious sand, and ancient sand, and will contain suspicious sand that spawns in pottery and spawners. The individual rooms are layed out like so:
    - Room 1: This first room contains a statue of a wither made of fossils, with a spot where ancient sand can be inserted to be powered by a repeater then trigger a hidden chest to be revealed via a piston door. This room is a branch from room 2, so this room is entirely optional.
    - Room 2: This is another side of the room adjacent to room 1 with stairs leading down. At the end of the stairs (around Y level=-40), is a small room with a minecart that is clipped into the wall. Calibrated sensors here mean that the player cannot mine, place, or touch the minecart but instead must open the minecart and get an ancient sand. There is only sand for a few blocks down, after that the tunnel becomes more preserved. Cobwebs line the corners of this room.
    - Room 3: After following the stairs between rooms 1 and 2, this room will contain a layer of water with suspicious and normal sand underneath. There are calibrated sensors meaning that the player cannot splash in the water, mine blocks, or place blocks other than ancient sand. There is also a piece of suspicious sand amongst a pile of sand in the corner.

Once brushed (as brushing does not trigger it) the suspicious sand will reveal a pot, and the pot will give ancient sand when broken. Ancient sand from the previous rooms can be used to parkour across a pit of lava that is 1 block deep.

- Room 4: This room contains a set of 7x7 spiral stairs leading up to the top section, where the classic secret room of the desert temple is located, but with no block in the middle. To get the Desert Key Fragment, the player must jump down and immediately go to the side to avoid the pressure plate. This part also has a 10% chance to spawn with an observer underneath the pressure plate, making it more dangerous as the player cannot mine the pressure plate. One of the 4 chests contains the Desert Key Fragment, but one of them is trapped and will trigger a TNT trap. These two chests can be the same one, however since the player can mine in this room, the best course of action would be to disarm the trap first. If the trap is triggered, the desert key fragment is a forced item and does not despawn (server lag implications), unless there are more than 5 desert key fragments in 1 chunk.

Branch 3

- Room 1: The player cannot mine or place blocks in this room due to calibrated sensors. This has basic lava parkour leading to the other side of the room. A loot chest positioned at the end of the parkour gives the following garbage loot:
    - 64 guaranteed rotten flesh
    - Guaranteed fire protection IV enchanted yellow leather chestplate
    - Bones
    - String
    - Spider eye
    - Wheat

  The better loot is underneath the main entrance chamber, and to get there, the player must put on the chestplate and jump in the lava, which leads to the next room. Note that the lava here takes significant durability off the player's armor if the chestplate given with unique attributes is not worn, although a fire resistance potion the player might have will also work. Underneath the lava is the loot chest containing a singular desert key fragment as well as other loot such as:
    - Gold nuggets, ingots
    - Iron nuggets, ingots
    - Emeralds
    - Diamonds

    - Bones
    - Rotten flesh (important)
    - String
    - Gunpowder
    - Spider eye
    - Sugar
    - Empty locater map
    - Randomized explorer map
    - Dried kelp
    - Wheat
    - Bread
    - Seeds (wheat, melon, pumpkin, beetroot)
    - XP bottles

  After the player has looted this chest, the player can return via a pressure plate activated one-way 2x2 piston door located diagonally across from where the player first enters the hidden chamber.
- Room 2: The 2nd room of this branch consists first of a weight puzzle, needing the player to sacrifice 64 items to continue by using a special attributed pressure plate. The most convenient way is by using the 64 guaranteed rotten flesh from room 1 decoy loot chest. Once the pressure plate is activated,

Dappled forests

Groundcover

Shelf fungi generation modified

Instead of only generating on fallen logs, shelf fungi can now also generate on live poplar and spruce trees and on planks found in tent camp sites.

Tree composition

Tree composition of the dappled forest biome now also contains birch trees and more scattered spruce trees. Leaves in the dappled forest biome other than spruce trees are yellowish green in color, like the leaves of an acacia tree in a savannah biome. Falling leaf particles and leaf litter in this biome are also increased, and leaf litter color here contains brown, orange, and red leaves.

Forests

Features

Sloped stone talus

Sloped stone taluses are new features generating abundantly on sloped, wooded and grassy areas in the following biomes:

- Windswept birch hills
- Forest
- Birch forest
- Old growth birch forest
- Mutated birch hills
- Dark oak forest
- Taiga
- Pine forest
- Old growth taiga
- Grove
- Snowy taiga
- Jagged peaks
- Icy peaks
- Gravel peaks
- Dappled forest



Fig. 1 sloped stone talus reference in real life geology

Sloped stone taluses are areas of bare exposed stone also consisting of gravel, cobblestone, exposed stones, coarse dirt, mossy cobblestone, coarse dirt, moss, rooted

dirt, juniper trees, stoneshrooms, suspicious gravel, and other materials depending on adjacent biomes. Generation first generates random adjacent blobs of material with a layered structure, then blending and eroding the separate features. The layered structure consists of the following:
**Note that these blocks might have stoneshroom mycenated tag although not listed*
Top:

- Gravel
- Exposed stones
- Coarse dirt
- Cobblestone
- Surface vegetation [*note that originally generating vegetation will be overriden with feature vegetation in the 3rd generation stage]

Middle [Generates only in non-stone surface biomes]:

- Gravel
- Exposed stones
- Cobblestone
- Rooted dirt [only under juniper]

Lower:

- Original matrix material
- Cobblestone

Juniper trees

Juniper trees are rare trees scattered across sloped stone talus features adjacent to areas with a low humidity value upon generation but still sufficient [>-0.2] vegetation value continents. Juniper trees bear twisted and warped looking trunks [similar to acacia but stouter in length], with roots twisting over cobblestone and exposed stone blocks. Stoneshrooms may also grow at the base of a juniper tree.[see also: stoneshroom mycenated...] Juniper trees have a rough bark texture and extremely blast resistant wood that has warped lines and pinkish or salmon in coloration. Juniper planks have twisted lines and have a unique plank texture.

Juniper berry

Juniper berries are a new, nutritious type of berry obtained by right clicking fruiting juniper tree leaves. Fruiting juniper leaves grow from flowering juniper leaves which have a 10% chance to spawn instead of normal juniper leaves. These berries can also be used in triple berry pie.

Stoneshrooms

Stoneshrooms are specialized bracket/shelf fungi that generate on exposed stone and cobblestone, slowly decaying them into gravel, then coarse dirt, then normal dirt. Stoneshrooms fruit under juniper trees for shade and due to the tough texture of the mushrooms.

Stoneshrooms can grow on:

- Stoneshroom mycenated exposed stone
- Stoneshroom mycenated mossy exposed stone
- Stoneshroom mycenated cracked exposed stone
- Stoneshroom mycenated gravel
- Stoneshroom mycenated cobblestone
- Stoneshroom mycenated mossy cobblestone
- Stoneshroom mycenated suspicious gravel
- Stoneshroom mycenated calcite
- Stoneshroom mycenated tuff
- Stoneshroom mycenated andesite
- Stoneshroom mycenated granite
- Stoneshroom mycenated diorite
- Stoneshroom mycenated deepslate
- Stoneshroom mycenated sand
- Stoneshroom mycenated red sand
- Stoneshroom mycenated suspicious sand
- Stoneshroom mycenated hadal mediums
- Stoneshroom mycenated juniper log
- Stoneshroom mycenated juniper planks

Due to Stoneshrooms' tough texture, they must be hydrated in a cauldron full of water for 1 real life minute for suitable use.

Stoneshroom decay

Stoneshroom inoculated blocks will slowly decay into other blocks in this chart.

Deepslate, Basalt, Tuff -->

Stone, Granite, Diorite, Calcite, Andesite -->

Exposed stone, Mossy exposed stone -->

Cobblestone, Mossy cobblestone, Cracked exposed stone, Juniper log -->

Gravel, Juniper planks -->

Sand, Red sand, Suspicious sand, Suspicious gravel, Hadal mediums,

Inoculation and Stoneshroom spore vials

Stoneshrooms can be artificially inoculated using stoneshroom spore vials. These vials are crafted with potions of regeneration and raw stoneshrooms. When a block is inoculated, the block will become mycenated and start growing stoneshrooms if the block is exposed to light. Stoneshroom spore vials can also be used to use the “spore” spell when used in a chiseled cauldron.[see also: stoneshroom]



Fig 1. Stoneshroom inspiration

Groundcover

Exposed stone features

Exposed stone is a special newly added type of bucolic decorative stone that can be found throughout forests including the following biomes:

- Forest
- Birch forest [all variants]
- Dark oak
- Pale oak
- Midnight oak
- Dappled forest
- Taiga [all variants]
- Pine forest [all variants]

- Swamp
- Mangrove

Exposed stone serves as a habitat for Glowroots, Stoneshrooms, moss, glow lichen, and many other florae. Can consist of the following blocks reaching a size never larger than 5x4x5 [x**y**z]

- Exposed stones
- Cobblestone
- Gravel
- Florae
- Hadal mediums [if adjacent {~-2~+-~-2}to ocean biome chunk]

Crafting recipes

Stone tools

Just like other stones, exposed stone (not mossy or cracked) can still be used to craft stone tools in the crafting table.

Mossy/Cracked exposed stone

After moss grows on exposed stone, exposed stone turns either into cracked exposed stone or mossy exposed stone. Exposed stone can be turned into cracked exposed stone after putting in furnace, and mossy exposed stone can be created by crafting moss carpets or moss blocks in the crafting table with normal exposed stone.

Moss

Moss will now slowly creep up logs creating mossy logs.

Jungles

Features

Foothill Jungle Talus

Foothill Jungle Talus features consist of piles of loose debris forming right triangles at the base of steep cliffs such as Exposed Limestone Outcrop features. These piles of debris can contain:

- Gravel
- `#minecraft:sands` [environmental dependent: see below]
    - TalusSandCalculate procedure in MCreator determines the sand in which should generate in said Foothill Jungle Talus feature. If the Jungle or Montane jungle this feature is generating in borders a Mesa, Mutated Mesa, or any other biome that has a layer of Red Sand on the surface, then the feature will generate

with Red Sand instead of Sand. Sand is the default generation pattern that can be overridden by this procedure.

Exposed Limestone Hill

Exposed Limestone Outcrop



Fig 1. Limestone Outcrop

Jungle temple

Generation

Montane jungles

Montane Jungles are a new variant of Jungles that replace 60% of existing jungle generation. These new Montane Jungles feature mountains going all the way to y-level 300, and some places around y-level 0 in basins. These extreme altitude differences cause a large variation in flora and animals, with ocelots in mid altitudes and pandas at low altitudes along with parrots. Steep limestone or exposed rock sedimentary cliffs occur as drop-off points conjoining montane areas with basins. Trees are populated in the following pattern:

High elevation (>160 y):

- Abundance of large jungle trees; higher chance of lush caves generating beneath.
- Fancy oak trees generate in scattered clusters.
- Jungle Orchid and Pistol Plants can be found on the trunks of these trees.
- Trunks of trees are moss covered. [see also: moss covered logs]
- Some areas have a `strawberrymc:jungle_scub` feature.
    - Jungle features are large areas with no canopy and exposed grass blocks, moss, and exposed stones that allow many plants to colonize

Mid elevation (50-160 y):

- Pearlescent vines start becoming increasingly common as elevation decreases.
- Ocelots can be found at this elevation.
-

Grouping

Existing jungle temples are now clustered at mountain tops, and if 2 or more jungle temples are present within a radius of 100 blocks, they are merged into a larger jungle temple. High elevation areas in jungles are very misty most of the time. [more about this in weather]

Groundcover

Pearlescent Vine

Pearlescent vines are parasitic vines native to biomes with a temperature >6 and high humidity. The pearlescent vine will slowly creep up large jungle trees until the entire tree is covered. Vines spread via Pearlescent Vine blocks, which are orient affected blocks with a 1[XZ] 16 Y bounding box. Pearlescent Vine blocks come in 3 block states:

- Young
- Mature
- Ripe

“Ripe” Pearlescent vine blocks have the same texture as “Mature” vine blocks but with milky white splotches to represent berries. If sheared with shears without Silk touch >0 enchantment, will drop Pearlescent Berry item(s).

Pearlescent Vine loot table

If harvested via tool with Silk Touch >0:

- Pearlescent Vine block

If via shears with no Silk Touch >0:

- Pearlescent Berry (1-2)

*Note Looting enchantment affects this drop

If with any item other than specified above:

- Air

Pearlescent Berry

Pearlescent berries are a berry obtained by shearing a Pearlescent Vine block when the berries are ripe. If the berries are not ripe (signified by green color) then the vine will not drop any item; unless the shears are enchanted witch Silk Touch >0 which causes the vine

to drop the Pearlescent Vine block. Pearlescent berries when brewed into a Awkward Potion will result in a Potion of Pearlescence that gives the following effects:

[UNRESOLVED: ARUGUMENT. DO NOT CHANGE THIS SECTION UNTIL AADI AND ALEX BOTH APPROVE] there will be consequences if Sai or Eric change this.

Jungle Orchid

Jungle Orchid is an epiphytic flowering plant growing horizontally from an Orchid Root block generating on the trunks of large jungle trees across areas with an y-value coordinate >150 in montane jungles. These orchids have randomized color of center and petals [similar to tropical fish] and come in 3 variants:

- Single large
- Few medium
- Multiple small

These come in these different colors from most common to rarest:

- Red
- Pink
- Yellow
- Purple
- Orange
- White
- Cream
- Green
- Blue

Color combinations are randomized for the center and petals based off of the list above. This data is not stored locally within the block, therefore if harvested and replaced, the orchid will not have the same configuration.

Jungle Orchid Roots

Jungle Orchid Roots are vegetal blocks that spread from the orchid flower block itself; and slowly spread with a chance of 0.02 per tick. These blocks are collisionless, placement-oriented blocks and have a 1[XZ] 16 Y bounding box. Jungle Orchid Roots may only spread on blocks tagged as the following:

- `#minecraft:logs`
- `#minecraft:planks`

Pistol plant

Pistol Plants are a vegetal block that is a carnivorous plant predating on ocelots, parrots, and other rodents in mid to high elevations ranging from y-levels 100-270.[see also: Procedures:

PistolPlantDetector] These carnivorous plants have 3 tendrils represented in the animated custom model that latch onto prey (like sundews or leaches) and will slowly make them take suffocation and poison damage. Groundcovers mineable with silk touch.

Rock Lichen

A slow spreading placement oriented, mat forming naturally generating organism group populating and covering limestone outcrops and cliffs, and exposed stone blocks which are more common at high altitudes. Rock Lichen has a bounding box 1 pixel high on the Y-axis. Although Rock Licken blocks can be found at any elevation in jungles, they are more common at higher altitudes where Exposed Stone blocks and Limestone Outcrop features are more common.



Fig 1. Montane Jungle

Inferno wastelands

The inferno wastelands re a new biome in the nether consisting of charred woods, blocks of ash, ash, and inferno lilies. Rifts in the ground form canyons with lava present inside.

Groundcover

Inferno Lily

The Inferno Lily is a fireproof species of groundcover in which the roots of the plant are fibrous and can regrow into another lily plant after burning. The stems of this plant can be used to make a high quality completely fire-resistant material known as Inferno Lily Straps, which can then be used to make completely fireproof armor. This material is also more reliably fire resistant than `#strawberrymc:charred_materials`. Inferno Lily flowers harvested with Silk Touch >0 can be used to make caramel dye, a dye classified as an "ancient dye" crafted from ancient plants such as pitcher plants and torch flowers.[read also: Ancient dyes]

Propagation

Inferno Lilies can be propagated in 2 ways:

- Silk Touch >0 tools can be used to harvest the tuber in which can be replanted in the following blocks:
    - Netherack
    - Basalt
    - Crimson and Warped Nylium
- Ash layers can be right clicked, which causes a 0.3 chance of causing the tuber to spread to an adjacent valid block; therefore duplicating itself.

Inferno parchment

Inferno parchment is a fire-resistant coating used to cover wooden materials to prevent them from catching on fire. The upside compared to charred wood is that inferno parchment is completely fireproof with a flammability index of 0 in Blockbench. Inferno parchment can be crafted by drying out inferno lily straps. An alternative recipe is crafted with snatchling membrane and blaze powder. This parchment can be used to craft fireproof, enchanted, and regular books. **[Needs more uses]**

Features

Lava trenches

Lava trenches are valleys with steep sides containing magma and lava at the valley's bottom. Lava trenches are scattered across inferno wastelands and are treacherous to cross without having the embers shooting from the lava to the crosser. Inferno lily coated armor is recommended for crossing these rifts. Lava trenches are 4-10 blocks deep depending on generation and no wider than 7 blocks.

Swamps

Lily Pads

Small Lily Pads

Small Lily Pads are a variant of regular lily pads that are generated in any non-saline body of water. Players cannot stand on small lily pads, while frogs can.

Frog pathfinding changes

Frogs now have a higher chance of pathfinding towards a Small Lily Pad or Lily Pad block; causing the frog to stand on the block.

Fresh Bodies of water

Lakes

Lakes instead of being when a river suddenly widens, are now a feature that is marked on the bounding box map in the F3 menu. This is so that rivers can connect and possibly terminate at a lake instead of abruptly ending.

River

Generation

Rivers generate as usual, with a few exceptions that make it a good method of boat-transport:

- Rivers are generated with the start and end in mind
- Rivers cannot spontaneously end; they must connect to a body of water such as an oasis, a lake, the ocean, etc.
- Rivers also cannot randomly just start; they must start at a raised body of water, from a snowy glade/mountain biome spring, from multiple springs (waterfalls) joining, etc.
- In rivers, water data (temperature, salinity, etc.) is mixed much better, and is only mixed downstream, never upstream
- Rivers require boats with the Swordfin enchant to travel upstream
- If 3 or more river outlines are generated near a cliff, a river cascade may be formed.
- Rivers will not always be at sea level but will generate at the same level as the land around

River Cascades

River Cascades are features that generate near Swamps and conjoin rivers at different heights, creating a waterfall (river cascades biome). River Cascades boast Flyfish, Fallgae,

and many other species. River Cascades also bring fertile silts to the riverbanks, allowing unique plants to grow. River Cascades only generate in fresh bodies of water.

Fallgae

Fallgae is a species of marginal riparian algae native and endemic to flowing water in areas with warm winter temperatures (if seasons are turned on). Fallgae can be harvested with shears and used to brew water swiftness potions. Water Swiftness potions allow the player to move up to 2x faster in water, but 0.25x slower on land. [see also: Brewing: Water Swiftness]



Fig 1. River Cascades feature. [credit: Nekoma]

Savanna

Dry Savanna

The Dry Savanna is a variant of the Savanna biome. The Dry Savanna consists of areas of sand and coarse dirt; populated sparsely by Baobab trees[see also: Baobab]. Dry savannas are flat and generate in areas with high continentalness and temperature with low humidity; typically in bordering areas near Deserts and normal Savannas. Dry Savannas have a higher chance of generating the Flaming Forges structure. The goal of this biome is to blend desert edges with lush biomes more seamlessly.

Groundcover

Vegetation in this area is sparse and made of Dry Grass, Tall Dry Grass, Short Grass, and occasional ferns. Acacia trees are sparser than in normal Savanna biomes.

Entities

Termites can be found here, just like the Savanna biome in large numbers.

[unresolved: more detail]



Fig 1. Dry Savanna inspiration [credit: Sciencing]

Flaming Forges

The Flaming Forges is a structure generating in the Savanna biome and all Savanna biome variants. The Flaming Forges consists mainly of blocks in the mud family, such as Mud Bricks and Dried Mud. The Flaming Forges is a metal themed point of importance [see also: Ocean Monument changes] that provides the Metal Rune used to craft the Eye of Ender in the Eric method of beating the game. The Flaming Forges consists of many puzzles the player must solve to obtain fragments of the key to unlock the final boss; the Metal Dragon; that drops the final rune in the Eric method.

Ancient Mud

Ancient Mud is a “ancient block”, just like Ancient Prismarine, will not trigger the calibrated sensors within the Flaming Forges. Ancient Mud cannot be taken outside the bounding box defined perimeter of the Flaming Forges it is specific to; identified via a block tag, to prevent the player from smuggling Ancient Mud across Flaming Forges to cheat in puzzles.

Upper level

The upper level (F3) of the Flaming Forges is where the player must start and is also where the temperature is the lowest; allowing the player to access it without any armor.

- Entrance: The entrance to the Flaming Forges consists of a north African styled decorated mud arch, then opening into a wide hall of columns with tunnel leading to rooms.
- Room 1 (closest right):

- Room 2 (closest left):
- Room 3 (far right):
- Room 4 (far left):

[unresolved: rooms need description]

Middle level

The middle level of the Flaming Forges requires either a Potion of Fire Resistance or Flame Armor. Wooden weapons and tools cannot be used on this floor, or they will catch fire and lose durability every 5 seconds.

Baobab tree [credit: <unknown>]



Fig 1. Baobab tree model made from Acacia wood and leaves.

The baobab tree is a type of tree that generates as a lumpy tree with a hollow interior and a new type of baobab wood.

Interior

The interior, roughly 3 blocks in radius, is a good starter base for savanna spawns as it already has walls. Baobab trees have a 40% chance to generate with water inside with the water turning darker over time.

Baobab wood

Baobab wood is the usual wood, as it has multiple uses from crafting rope to being fireproof. Baobab wood is also not palatable for termites.

Fire resistance

Baobab wood and planks are fire resistant and give builders that want to avoid arsonists but still use wood more options than crimson, warped, and charred wood

Termites





Left [fig1]: termite, right [fig2]: occupied termite mound

Behavior

Termites can go up to 50 blocks from the original mound searching for food such as leaf litter, grass, planks, logs, and other organic matter. After a termite eats a log, the log becomes a hollow log, which may become housing for the termite, or can be harvested and used for decorative purposes. [see hollow log]

Hollow Log

The hollow log is a new block created when termites eat away the center pith of a log. This block supports shelf mushrooms on exterior and interior. Variants include all trees except Baobab. When hollow logs are placed horizontally, items with a small hitbox can be placed inside such as:

- Lanterns [soul, normal]
- Pots
- Brown Mushroom
- Red Mushroom
- Any 1 block tall flower [not epiphytic]
- Conduit
- Heavy core
- Button [no Redstone signal when pressed]
- Cobweb
- Shelf mushroom on interior
- Grass
- Fern
- Saplings [do not grow]
- String block

- Cushion

Badlands

The badlands biome has challenges like the desert, and despite its greatness for mining gold, it is not good to live in due to lack of resources and diversity. Strawberry Minecraft brings more interesting features to the Badlands biome.

Village generation

Villages generating in badlands and eroded badlands now have a 75% chance of being unoccupied. These unoccupied villages spawn relative to nearby exposed mineshafts and may contain mining materials such as tailings [realism mode only] and hadal sand.

Tumbleweeds

Tumbleweeds are non-natural entities that are blown in the direction of the wind and have high momentum that will seem to make them never stop rolling. They are extremely flammable and catch fire easily, spreading the fire, then spontaneously combusting. Tumbleweeds are native to badlands, eroded badlands, and (rarely) deserts.

Item storage

Tumbleweeds have a 70% chance to generate loot contained within 3 inventory slots reveled by right clicking the tumbleweed. Loot in tumbleweeds are randomized with chances of containing emeralds, gold, copper, iron nuggets, flint, sticks, rabbit hide, and rope. Tumbleweeds may also pick up items left on the ground and release the item once right clicked. Tumbleweeds can be mined with any tool, which will drop all its contents.

Tumbling

They will continually tumble and can tumble upward. If the wind is blowing, they will tumble in the direction of the wind but will always retain their inertia.

Glaciers and Icy Biomes

Icy caves changes

Icy caves boasts fossil features, diamonds, obsidian, pockets of gravel, all buried in packed snow, ice, and the endemic spectrite. Icy caves can be found where glaciers and icy biomes like the ice hills [coming soon] are adjacent to a rocky biome, such as gravel slopes, jagged peaks, rocky beaches, etc. Only soul torches and Glowroot torches can be placed in icy caves, as regular torches will melt the surrounding ice, unleashing the random pockets of gravel that generate in a glacier formed by. The glacier scrapes the sides of the rocky biomes. Generation of this cave exists in “noodle caves” and “spaghetti caves”.

Icy stones

Spectrite

Spectrite is a decorative ore with stats like iron endemic to the glacier biome [read also: glacier biome]. Raw spectrite (like lumen mytanum) has a dull, grayish blue coat, but the vibrant inner layer can be revealed after cutting the raw spectrite via a gem cutter with coconut oil as lubricant[read also: coconut oil]. the spectrite reveals a wide spectrum of colors; that can either be used to create armor and equipment or armor trims[read more: spectrite in ores and minerals]. Spectrite has similar stats to iron.

Ice temple

The ice palace is a major structure generating in the icy biome, with extra items that make the trip to the cold biomes worth it.

Ice staff

The ice staff is a new spear like a melee weapon allowing the player to paralyze the opponent by freezing them. Once the opponent is frozen and paralyzed, attacks with metallic weapons do extra damage by 1.5x. The ice staff can be obtained by defeating the lapis golem and using a lapis core combined with 2 blizzard rods in the crafting table.

Blizzard rod

Blizzard rods are a new rod that can be acquired either by looting the ice temple or by killing Blizzards (new blaze variant). Like wind charges from breeze rods, blizzard rods can be used to craft ice bombs, freezing the player's opponent or water. The Blizzard rod with other materials can be used to craft the ice staff, as well as modify arrows to generate ice cubes upon impact. [read also: fletching table and arrows]

Icy watchtowers

Icy watchtowers are substructures generating in the 4 corners of an ice temple with a tall, lighthouse-like look. Watchtowers contain ice-related enchantment books, modified arrows, fire weapons, cold resistant armor, liquified horn of storms vial[coming soon], and other items the player will need for the ice palace. Ice watchtowers have 1 spiral staircase to the top that is infested with Strays.

Lapis Golem Summon and Fight

The Lapis golem is a boss that generated in the Ice Temple [coming soon]
The player can summon the lapis golem by inserting a summoning spell into a chiseled cauldron in the center of the arena. To obtain a summoning spell book, the player must search the upper sanctum area (Guarded by armored frozen). Any chest in the upper sanctum has a 35% chance of containing the summoning spell inside of the enchanted book. The Lapis Golem has 3 attacks (On first stage [Above 66.66% Health]).

1. The Lapis golem swings out it's one arm and melee attacks the player (It looks like an enchanted lapis block with a long single arm attached on top with two stubby legs on the bottom that it uses to shuffle along and has 3 eyes attached to the lapis block [About 5 by 6 by 5 blocks in size) Dealing 5-10 Damage (2.5-5 Hearts of Damage).
2. The Lapis golem Throws 5-10 Frozen at the player with a clenched fist violently lashing out of the ground throwing the Frozen in different directions
3. The Lapis golem enlarges its eye, shooting a laser (Similar to Guardian) made of enchantment particles at the player dealing 10-15 Damage (5-7.5 Hearts of damage [Regardless of Armor]). When defeated drops the enchantment tome and rune of ice.

Snowy Glade

Generation

The snowy glade spawns in valleys between two mountains in a cold biome. Smaller snowy glades can be found at the base or side of a mountain. Then, they receive a large amount of snow until the Y level of the surface is near the Y axis of the mountain peaks. This is what happens when a structure is spawned (they are spawned before snowfall):

- If it is a village, the jigsaws for village paths include barrier blocks which are removed as the final steps of generation are complete
- If it is an igloo, the igloo remains there and is allowed to be buried by snow
- If it is a waypoint, an air bubble is created around it, and loot includes torches and snowballs
- Rivers generating through icy glades will be cut off

Materials and Groundcover

Snowy glades have blocks consisting of packed snow near the bottom, and powdered snow, ice, and snow layers/blocks near the surface. Gravel pockets may also generate in snowy glades similar to glaciers, although spectrite is not present. Air pockets within snowy glades will naturally form ice and stalagmites. Glowroots may grow on the surrounding stone perimeter of lava pools.

Top area

The top area is completely snow, however only 1 layer on top so that it is indistinguishable from a normal glacier.

Below snow

Due to the glacier carrying around materials and ores, the area below the glacier is heavily enriched with all overworld ores varying in vein size.

Stripes

Stripes made of ice variants and ice form on snowy glades as they resemble glaciers.

Sulfur caves

Hot Springs

Hot springs are remade sulfur springs. They spawn with large pools of water in the middle and small pools on the side with magma and potent sulfur geysers.

Edges

At the edges of large hot spring/geyser pools on surface sulfur caves, sulfur goo blocks may form. Sulfur goo is a uncraftable unobtainable block that can be harvested with a hoe to obtain 4-9 sulfur goo items. Sulfur goo items along with gunpowder, ash, paper, and any colored dye can be used to craft smoke bombs. Smoke bombs upon ignition summon forced smoke particles with a cap of 3 smoke bombs per simulation distance radius. These smoke particles are forced so that players with particles turned off do not have an advantage to players with particles turned on. Potions can be added to smoke bombs for the smoke bomb to cause that potion effect to players around the explosion radius.

Technical

Hot springs have 4 distinctive features: hot_spring_1, hot_spring_2, hot_spring_3, and hot_spring_4 which correspond to blue, yellow, orange, and green water color. Each of these features has its own preset variants.

Sulfur cubes

Sulfur cube travel using frost walker is now available again, however, instead of frost walker, the player must place a mob cage with a freeze rod inside.

Rocky beaches

Glowroots

Glowroots are a special plant with a large tap root and tuber, that grows only close to lava pools in a cold biome, although some may be found on hadal mediums.[read also: hadal dirt, gravel, and sand] Glowroots are also fire resistant and give off a cold aura.

Glowroot paste

Glowroot paste is a paste that can be crafted by crafting slimeballs and the root of a Glowroot. This is the only way to ignite a Glowroot.

Glowroot torches

Glowroot torches can be made by combining Glowroot paste with any type of torch. Glowroot torches are a special type of torch that will not melt the surrounding snow or ice(useful for exploring glacial caves).[read also: Icy caves] Glowroot torches give off a soft, creamy yellow light that has the same illumination index as a normal torch.

Taiga

Abandoned wooden cabin

Generation

Abandoned wooden cabins are a new structure generating exclusively to taigas, groves, old growth taigas, modified wooded badlands, windswept forests, tundras, snowy taigas, snowy slopes, and lower areas below or near jagged peaks biomes. Cabins generate more often near bodies of water, and can sometimes generate with a farm varying in size outside.

Appearance and Insides

These abandoned cabins are generated in 2 sizes; large and small; corresponding to the dimensions 5x7 and 7x10. Composed mainly of spruce logs in the 4 corners, spruce planks for the roof and walls, and cobblestone or smooth stone floors, these cabins have various valuable beginner game loot, such as cooked meats on a campfire, wheat from a farm that can sometimes generate outside, and sweet berries. A burnt-out campfire can be found outside with a chest nearby containing charcoal and a water filter.

Abandoned belltower

Abandoned belltowers are a new structure occasionally generating randomly in taiga, old growth taiga, birch, old growth birch, snowy taiga, tundra, grove biomes, and mutated cold biomes. Abandoned belltowers consist of a spruce/birch planks base lined by cobblestone walls with red and yellow stained glass. Abandoned belltowers contain paintings and barrels with materials such as water filters[realism mode only] and food.

Belltower lower barrel

Belltower lower barrels are barrels containing loot generating underneath the hearth of the campfire under the main tower. This chest contains rarer loot including:

- Iron ingots, nuggets
- Gold ingots, nuggets
- Copper ingots, nuggets
- Emeralds
- Coal
- Sticks
- String
- Bones
- Stone tools
- Iron tools

- Plant fiber
- Sugarcane
- Sweet berries
- Ice balls[taiga, snowy variants only]
- Water filters[realism mode only]

Belltower upper chest

This chest is located at the top of the cobblestone tower at the back of the structure. This chest is accessed via a cobblestone spiral staircase originating in front of the burnt campfire. This chest contains common loot including:

- Rotten flesh
- Bones
- Spider eyes
- Sugar
- String
- Plant fiber
- Wooden tools
- Low durability stone tools
- Iron nugget
- Flint
- Snowballs[taiga, snowy taiga variants only]

Dripstone Caves

Soul sand valleys

Ghost Peppers

Usage

These unique peppers can be found in soul sand valleys generating near fossils on soul soil. Ghost peppers are also indicators of Shridinite ore veins. [[read also: shridnite] When consumed raw, they will damage the player 3 hearts (6 damage) and give the player speed 2 (For 1 min); and strength 2(For 30 seconds). This advantage can be used to assist the player when clearing a nether fortress or bastion during speedruns. Potions of burning brewed with ghost peppers give the player speed 2 (For 2-4 min); strength 2 (For 1.5-3 min); allow the player to set entities attacked by the player to catch on fire. If the mob a player attacks is already on fire or if the player uses a potion of burning II, the effect will set the

mob on soul fire. Note only soul fire will occur if the mob was set on fire not previously by a potion of burning I.

Obtaining

Obtaining ghost pepper requires right clicking the pepper plant which will drop 1-3 ghost peppers. When the plant is broken, the player can get up to 4 ghost peppers or up to 7 with looting III shears. Ghost peppers can be placed inside of a crafting grid to craft ghost pepper seeds that can only grow in hot areas. For ghost pepper plant seeds to germinate, the seeds must be planted on soul soil within 4 blocks of lava, which will convert soul soil into soul sand once the plant matures. The player can use shridnite dust to convert the soul sand back into soul soil for a second use of the soul soil. If Nether wart is planted on the newly made soul sand, the Nether wart will grow 30% faster, introducing a sort of crop rotation.

Bone daggers

Usage

These new weapons boast low attack damage, but extremely fast attack speed and takes up a new subweapon. Bone daggers can be crafted similarly to normal daggers with 1 bone, 1 stick, and 1 nugget/shard modifier (Optional). These daggers will deal half the damage of normal daggers or a fourth of the equal material sword damage. Depending on the quality/type of bone, the daggers will have more healing from the lifesteal or more damage and durability. Bone daggers also deal double damage when dual wielded (4x damage in total).

Modifiers

Bone: None
Charred bone: Burning
Copper nugget: +Dur
Iron nugget: +1Dmg +Dur
Gold nugget: +1Dmg
Raw diamond/Diamond: +3Dmg +Dur
Prismarine shard: Underwater attack bonus
Shridnite shard: Increased lifesteal

Swamps

Coven[ant] of witches

Witch hut

Witch huts now generate more frequently across swamps and forests. Coordinates inside a witch hut lead to their corresponding covenant location, which can be modified using the

`covenantSize` gamerule, which only applies to coven[ant]s generated after the gamerule is applied via commands.

Generation

When 5 or more (customizable with gamerule minCovenSize or the luck attribute) witch huts generate near each other, a coven is formed. Two or more covens can be generated in the same biome. When a coven is formed, the approximate center of the witch huts is calculated, and the nearest open cave spawns a waypoint (some waypoint spawning mechanisms are different for this).

Behavior

Every full moon, all the witches in the coven will head to that waypoint and meet. After making some conversational noises and throwing a few potions, the witches add multiple random ingredients into the chiseled cauldron, and it gradually fills up with Uncraftable Potion. Rune particles also form in a circle above the witches while doing this (credits: Cube Kid) and once the final rune is placed, the potion changes color, and every witch takes 8 bottles of uncraftable potion home to their hut.

Looting

The player has 3 ways to obtain uncraftable potion from a coven:

- Have a wise villager go to the coven in disguise and buy the bottles from him
- Kill one of the witches as they are heading back to their hut with the potions, and collect the potion drops (looting enchant has no effect)
- Take some uncraftable potion from the cauldron without letting the witches see the player, and escape before they realize some bottles are missing

If the player is caught, the witches behave as a group and can use one of the following attacks:

- All the witches throw splash potions of harming at the player at once
- Witches might use splash potions of poison at the player
- Witches have a 30% chance to use runes to summon mobs, and another uses harming and wind charging effects to knock the player into a hole, which the witches will proceed to throw splash potions of harming
- Witches may throw potions of weakness at the player
- Witches may push players into chiseled cauldrons to teleport them to random locations

Main end island (The End)

Dragon fight

End geysers

On the outer end islands, the dragon can spew large beams of dragon's breath, which will spew back at the player using particles laced with dragon's breath. [**unresolved: more needed]**

Outer end islands

End Cities

Banner Room

The Banner Room is a room that is a jigsaw that can generate in the End City structure. The Banner Room consists of a window facing outwards, a wall covered in customized banners, and a chest containing loot. [unresolved]

Lore

The End before the dragon is killed is in a state of despair, with all current surviving entities affected by the Infected entity tag. All typically neutral entities; since infected with the Dragon's Breath particulate dispersed in the air; are hostile. This can also be seen in the Echowood trees, as before beating the dragon, the biome tint is desaturated, wilting, and covered in purple patches. Once the dragon is beaten, infected entities become neutral and Echowood trees start producing saplings again and no longer have a desaturated, purple biome tint. Endlings no longer take damage if no contained within End Cities.

Endlings

Endlings are mobs that are native to the End and once populated the entire End in the lore, but due to the Ender Dragon's breath, they fled to the End Cities and will take damage if exposed to the air before the Ender Dragon is defeated or if has been resummoned. Endlings come in X [unresolved] variants; including the following:

Hopling

Glideling

Warpling

Flyling

Chorus Leviathan

Voidal waypoints

Voidal Waypoints are small room-like underground variants of waypoints that generate in the end and have end anchors and Casting Cauldrons in them. Endillary plants are generated on top. These voidal waypoints typically generate adjacent to end relays [see also: overworld waypoints]

Ender relays

Ender relays are a new block that allows the player to set their respawn point in the end. Relays are charged to a max level of 4, like nether respawn anchors; with palline.

Casting cauldrons

When an ender pearl, snowball, or any other projectile is tossed into a Casting Cauldron then the direction of the travelling projectile is regarded and a 30-degree angle on the XZ plane is created with this direction. The projectile comes out of the nearest casting cauldron within this angle, allowing for gateway-like access to remote places in the End.

Palline refinery

This is a table that refines palline even more into a crucial endgame item [name coming soon] that is used to craft a Chorite Core.

Endillary

Endillary plants are a special species of plants endemic to the outer end islands. The plant will generate directly above Waypoints in endstone. Silk touch is needed to mine Endillary plants for propagation.

Endillary nectar bottle

When the flower of the plant is ripe (shown by the bloated stem below the flower of the plant), the plant can be right clicked on with a glass bottle, resulting in a Bottle of Endillary Nectar. When a Bottle of Endillary Nectar is drunk, the player will receive 10 seconds of the `strawberrymc:invulnerability` effect. This effect is useful in end cities to prevent shulkers from poking the player. [see also: cleansing]

Chorus wasps

Chorus wasps are a new arthropod inhabiting the chorus fruit forests (End highlands) of the outer end islands. Hives of chorus wasps will generate near endillary and chorus flower blocks. The hives are lavender in color and usually have chorus nectar dripping out the bottom, accompanied by purple drip particles. If the player comes within 5 blocks of an endillary or attacks a wasp, then the wasp becomes aggravated.

Variants

There are three major types of categories in chorus wasps: No attribute, Drone attribute, and Obsidian attribute. Upon death the normal variant has a 10% chance to drop a chorus wasp stinger and a 2.5% chance to drop a venom sac, the drone has a 5% chance to drop a venom sac, and the obsidian variant has a 10% chance to drop a chorus wasp stinger, a 5% chance to drop a venom sac, and a 5% chance to drop a obsidian shell.

Normal

The normal variant will apply dragon's venom II for 10 seconds to the player upon sting. The sting itself deals 3 damage (1.5 hearts). The drone variant does not sting the player and instead will spew the fire dragon's venom clouds around the player dealing 6 damage (3 hearts).

Obsidian

The obsidian variant like the normal variant stings but, has a thick outer shell made of sharp obsidian acting like armor. When the player attacks the obsidian variant with a normal weapon, the player will take 2 damage (1 heart) and deal no damage nor knockback. The only way to safely damage the obsidian variant is to use a diamond or better pickaxe and right click the wasp 3 times to destroy the armor and then be able to safely damage the wasp with a normal weapon.

Hives and horticulture/breeding

Like their overworld counterparts, end wasps can be "contained" with the artificial hive block. The wild hives in the outer end islands can be radiated for the player to easily obtain chorus wax without aggravating the end wasps.

Chorus wax pellets

Chorus wax pellets are a new item which is a type of wax harvested from chorus wasp nests using the smoking method and shearing. Note that looting shears of any level gives a guaranteed 2 wax pellets. The wax can be used to craft bioluminescent candles that emit substantial amounts of light when lit, stop food from decaying (More effective than normal

beeswax), and can be used to craft artificial chorus wasp hives. The flames of these candles are bright yellow while emitting lavender particles.

Sensegrass

Sensegrass is a tuberous plant native to the outer end islands. Cultivated Sensegrass will output a redstone signal based on how far away a player is within a radius of 2-4 blocks of the Sensegrass tuber. The Sensegrass stems will retract into the ground based on how far away the player is. Sensegrass can be corrupted by either right clicking the sensegrass with dragon's breath, or naturally spawning corrupted Sensegrass, which has a purple hue to the stems. The player will take 6 damage points every second from standing on corrupted Sensegrass (❤❤❤).

Roasted Sensegrass Tuber

A delicacy found at the end in End Cities and Waypoints. Roasted Sensegrass Tubers restore 8 hunger points (4 hunger bars) and can be obtained by either looting end relay chests, End City chests, or by excavating the tuber of a Sensegrass.

Chorite and Chorine products [see also: chorite and chorine]

Recovery template

Recovery templates are a special item that can be used in smithing tables, crafted with 1 netherite surrounded by Chorine. When a recovery template is applied to a tool, the tool will be "keep inventory", but only once then must be re-plated

Recovery pearl

The recovery pearl, a similar item to the ender pearl, is crafted with 1 Chorine and 1 ender pearl. A recovery pearl will teleport any item, entity, or block struck by the pearl to whichever dispenser, entity, or player that threw the recovery pearl.

Echowood Forests

Echowood planks and products

Echowood is a green colored wood native to end relay rooms and end highland biomes with a twisted, ancient feel and structure similar to juniper trees. Echowood leaves will fall upwards and will deter endermen.

Echowood acoustics

All sounds originating from behind any Echowood products will boast a high reverb, and sound muffled and sound like it came from far away.

Echo Squid

The Echo Squid is a new enemy only found in the echowood forests.

Behavior

The Echo Squid blends in with the surrounding Echowood trees and only moves when either the player gets within a 5 block radius or if the player directly looks at the hitbox of the Echo Squid; similar to an Enderman. The echo squid will then do one of the following attacks:

- The Echo Squid latches onto the player and start dealing poison and melee damage to the player through biting. This can be antidoted by shaking the players mouse violently or damaging it through external projectiles:
    - When the Echo Squid is latched onto the player, the player cannot take external damage, as the player is shielded by the Echo Squid.
- The Echo Squid shoots a Sonic Shriek similar to that of a Warden that deals 6 damage (3 hearts) to the player and/ or tamed mob (Especially spectral).
    - Sonic Shriek particles are generated in the direct line of fire.
- The Echo Squid may hover above the player and then smash downward attempting to hit the player.
    - Afterwards, it may perform a cyclone attack that causes the player to be knocked backwards and temporarily stunned for 2 seconds (unaffected by tick rate) via the Paralyzation effect. During this timespan, the player may not attack, move, or use any item.

Echo Squid loot table

If killed with item (affected and weighted by looting):

- Shrieking Gladius 0.75 1-2x
- Chorus Fruit 0.15 1-3x

If killed with melee weapon enchanted with Fire Aspect >0 overriding previous loot table statements:

- Shrieking Gladius 0.75 1-2x
- Popped Chorus Fruit 0.15 1-2x

The echo squid when killed will drop 1-2 echo squid gladius. These gladiuses can be used to make a vial of shrieking or to make the locater compass. [see also: locater compass]*

Radometer compass

Recipe

[unresolved: needs recipe]

The radometer compass is made with 1 sac, 3 radonite ingot, 1 lodestone, and a recovery compass. The locater compass when right clicked (While holding) to send out a green pulse that can be used to locate ores, especially redstone.

Valid ores

- Iron ore
- Gold ore
- Raw diamond
- Lapis lazuli
- Green beryl [uncut emerald]
- Redstone ore
- Dendrite
- Garnet nodule
- Copper ore
- Nether quartz
- Ancient debris
- Nether gold ore
- Shridnite ore
- Spectrite ore
- Lumen mytanum
- Amethyst cluster, budding amethyst, and block of amethyst

**Note all ores in matrix other than stone count*

Limitations

This compass can only be used outside of the end dimension as the outside frequencies cause it to stop working (Like clock in nether).

Decayed echo shrine

This structure is the remains of what used to be a normal echo shrine but has been destroyed by enderman over the years. This structure provides basic materials needed to explore the echoing deep biome. In these shrines instead of normal voidal vaults there will be decayed voidal vaults that contain downgraded loot and are opened using an echo shard instead of a depth worm key. The decayed voidal vaults have a 1% chance to give the player a catalyst lantern which is necessary for warding off the wisp.

Voiding rod

Mechanics

The voiding rod is a special fishing rod variant crafted with oceanic rods, recovery pearls, and strider silk. The voiding rod is used as a fishing rod in the void to acquire items from the void dimension, lost items the player dropped in the void, and the depth worm key.

Spectral

The Spectral is a passive variant of the phantom that is based off a monarch wing butterfly, can be tamed and saddled with chorus fruit. Once tamed, the Spectral can be used to easily traverse the outer end islands for looting end cities and exploration. Chorine armor can be crafted for armor for Spectral. Spectrals cannot survive in the Overworld or Nether without potions of levitation which can be brewed using a potion of slow falling with a fermented spider eye or a sensegrass tuber on an awkward potion. Note spectrals can be healed using popped chorus fruit, and spectrals can be bred using chorus fruit in which they will lay eggs.

Behavior

Until the spectral is tamed, the spectral may roam end highlands, and Echowood forests flying from structure to structure away from the deadly echo squids.

Sounds

In game

Music discs

No escape [credit: T_en_M @spotify, Laudividni]

Toccata [credit: Aran Khachaturian, Eric Xia]

Rendering changes

UI

Crafting tables

Tabs

To compensate and keep the crafting table organized while still allowing the crafting table to retain many uses in this mod, the crafting table now has static tabs located on a bar at the top of the crafting table UI. These tabs are “Forging”, and “Crafting”. When a crafting

table is opened, it defaults to the “Crafting” tab, where normal crafting table objectives can be done such as crafting blocks and items. The forging tab remains locked when a crafting table is crafted, and must be unlocked by putting the crafting table in another crafting table and combining it with 3 cobblestones in any of the cardinal directions of the crafting grid with a forging hammer filling the last cardinal direction and 4 metal ingots [with NBT data “iron] in each corner. The new resulting crafting table now retains the NBT data “can forge”. With a “can forge” crafting table, the player can use the forging tab to forge weapons or equipment out of any metal the player has or combine them to form alloys with special properties in the forging tab using the crucible tool. The forging section of the crafting table requires a temperature of 1300 degrees with all the tools requiring this heat. The recipe book in Java edition or the sidebar in Bedrock edition when in forging mode will show recipes the player can forge or create alloys of in 2 sub-tabs in the recipe book/sidebar. Any of the options the player can craft can be clicked and given a fuel source, will result in the alloy or equipment. Each recipe takes 15 seconds to forge.

Brewing stand

Changed the UI for brewing stand on bedrock edition to the original UI on Java edition [Bedrock edition mod only] [Java edition mod coming soon]

Furnace

The furnace UI for all gamemodes has been changed for 2 output slots, a slot for any slag in realism mode and a slot for the product item. The furnace UI also now allows for the selection of output products, such as choosing to smelt sand into charred sand or glass. The recipe book that tells the player what they can smelt is now turned into a sidebar allowing the player to click a recipe instead of dragging the items into the top slot of the furnace

Rendering of non-collidable objects or entities [disputed]

Flora

I think we can all agree right now that the rendering function used to render flowers, plants, etc. has lots of space for improvement. The current fatal flaw for the rendering script that is currently in use uses 2 two-dimensional bitmaps of the rendered object that have been interwoven together like a “X”. To fix this, Strawberry Minecraft will overhaul the entire rendering script and make objects 3d, instead of interwoven 2d planes.

Cloudcover

Cloudcover now determines weather, with a new way to render clouds. Clouds are now more 3d, floating in the direction of the wind. Clouds are rendered via block models of pure white that are 32x32 pixels, so 2 blocks wide and 2 blocks tall.

Precipitation

Clouds in the sky before any precipitation, at the edge of the air front containing precipitation, are darker and thicker in color, with less light reaching the terrain below. Note overcast weather can still occur independently when an air mass with precipitation dies.



Fig 1. 3D cloud demonstration with player flying Elytra [credit:<unknown>]

Wind

Wind now has 3 levels:

- Breeze
- Gust
- Cyclone

These 3 levels of wind will trigger different effects in rendering in the game. At wind level 1, grass and other flora as well as leaf falling particles will start to move in the direction of the wind (land breeze or sea breeze). At wind level 2, wind particles [see wind particles] will start to dot the lower atmosphere, and clouds will start picking up pace. Campfires at this wind level's smoke will start to drift in the direction of the wind. Torches may flicker from the strong gust. Rain starts to fall diagonally, which is more dramatic as the wind level increases. At wind level 3, torches become unusable (as they will blow out), and dust particles will form near the ground. If sand, barren soil, or dirt is next to a grass block or other vegetation, it will die and turn into a soil block. Standing down wind of snow will give the player frostbite and not wearing wool armor in areas colder than 7 will slowly drain the player's health.

Note that the player takes more hunger moving against the wind, but less hunger than usual moving in the same direction as the wind.

Particles

Wind particle

Wind particles render when wind levels are >1.



Fig 1. Wind particle demonstration

Ping particle (not natural)

The ping particle is a new particle with unique properties. The ping particle will render through blocks and stay the same size and rotation no matter the player's location. The color of the particle and the particle can be summoned using (comscript):

Summoning:

```
particle.color.ping(<hex color code>)
particle.summon.ping
```

Color group:

```
particle.color(<hex color code>)
particle.summon.ping
```



Fig 1. Ping particle example
Credit: Doctor4t

Cauldron with water

Heating bubbles

Heating bubbles in cauldrons occur when a cauldron is placed on top of a heat source such as a fire, campfire, magma block, or lava. Items placed inside will become boiled, such as hydrated stoneshrooms[read also: stoneshroom], boiled eggs, boiled chicken, and more. Water boiling in a cauldron will slowly evaporate, leaving behind less water. A trick the player can use is by putting a dripstone that will slowly refill the cauldron as the water evaporates. Bubble particles occur when the water is heated to full after a heat source is placed (ranges from 1-3 real life minutes).

Other Features

Decorative retextures

Chiseled blocks

Sandstone: Creeper

Mossy Stone Bricks: Target ring
Cinnabar: Cave spider
Sulfur: Sulfur Cube
Red sandstone: Wither
ell as mirroring its other abilities (allowing these abilities to be used in entities).

World presets

Amplified [bedrock]

The amplified world preset is a new world preset accessible by changing the drop-down menu when creating a world in bedrock edition. Note that this feature is not available on console edition. The amplified preset increases the world's build height limit to 500 blocks and generates steep cliff faces with extensive trees crowded in the rare relatively flat area. This world preset can be used for elytra maps or creative experiments like rollercoasters; more for people with beefy devices.

Skyblock

The skyblock preset is a new world preset that can be modified in settings during world creation. The skyblock preset contains a floating 7 block deep L-shaped island with stone at the bottom. An oak tree is populated at the top of the island as well as a chest containing ice, dripstone, a bucket of lava, and 12 end portal frames. Structures are still generated in this world preset. Biomes still exist in this world, but the spawn chunk is always `minecraft:plains`. Note that the skyblock preset allows customizable biomes, blocks, and chest loot. Bonus chest is not allowed in the skyblock preset.

NomadLife

NomadLife generated as a normal world, however the player has benefits and disadvantages:

- The player cannot sleep in any bed that did not naturally generate.
- The player cannot stay in an enclosure and store items within
- The player cannot sleep in wool beds
- The player cannot strip mine for more than 10 blocks (they must use caves).
- The player can mine Ender Chests with a pickaxe iron or higher.
- The player can access treasure enchantments with an enchanting table.
- The player has 60 inventory slots including hotbar slots.
- The player respawns at the last respawn he placed, whether the bed/anchor is still there.

Village Life

Parkour civilization

The parkour civilization preset is based off the meme “do you jump for the beef or go straight for the chicken?”. This preset generates a superflat above the spawn area with infinite 1x1 floating islands generating on a horizontal plane with 1 block gaps. Small soil or stone huts may generate 1 oak log studs on some blocks with item frames consisting of beef.

Terrain generation

Segmented generation method

Even though vanilla Minecraft uses a refined Perlin Noise biome generation, Strawberry Minecraft’s uses a different terrain generation method. The new method will first segment the entire world into adjacent superbiomes, or “continents”; which is an area that will eventually encompass many biomes of a similar temperature noise and humidity noise. The superbiome containing spawn is given a random temperature noise ranging from -1(coldest) to 1(hottest). Then, when superbiomes adjacent to spawn are loaded in by the player, the adjacent superbiomes are given a temperature value like the temperature value of the spawn superbiome determined by the seed of the world. After superbiome temperature noise values have been determined, Strawberry Minecraft will proceed to fragment the superbiome into normal biomes of that or similar temperature noises. Compared to current terrain generation, Strawberry Minecraft’s ratio of land to water will be adjusted for larger oceans, although to a point in which land is still abundant, yet smaller and more fragmented. Smaller bodies of fresh or brackish water may generate inside of larger bodies of land.

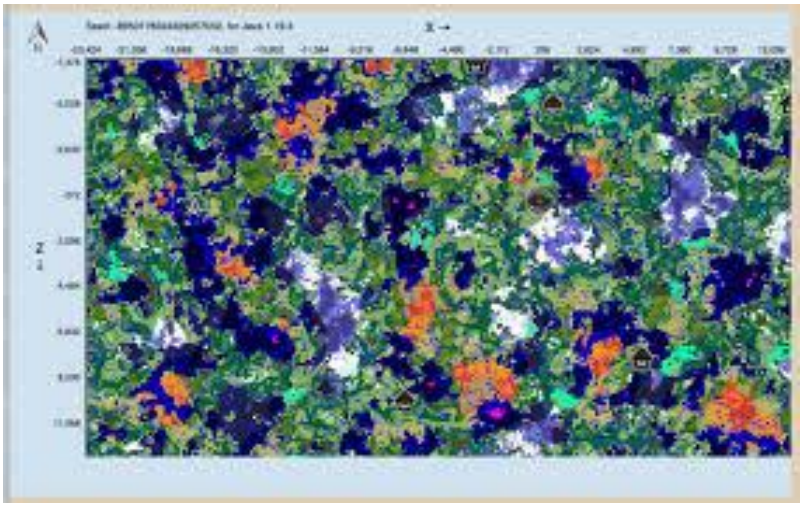


Fig 1. Map example from Chunkbase

Strongholds of the W noise

Areas with a weirdness noise consistently higher than 0 over 50 blocks will turn into W noise strongholds, where W noise is then amplified in that area and allows for extreme and variable terrain generation. Weirdness noise strongholds typically generate near or on superbiome borders.

Block layers

Soil

Most to all generation of dirt on the first 1-2 blocks under a grass block have been replaced to soil, with the same properties as dirt, but darker in color. Dirt still naturally generates, but is far less abundant; only generating in small patches near exposed stone features[read also: exposed stone], talus slopes[read also: talus slopes], rocky beaches, and below soil. The new layout for layers is:

1. Grass blocks (1 block thick)
2. Soil (1-3 blocks thick)
3. Dirt (1-2 blocks thick)
4. Gravel (1-2 blocks thick)
5. Exposed stone (possible generation in some biomes; in which it is 1 block thick)
6. Stone (variable across terrain)
7. Deepslate (variable across terrain)
8. Bedrock (1-5 blocks thick)

Villagers and wandering traders

Bell mechanics

In vanilla Minecraft, a bell can only support up to 32 villagers, although in strawberry Minecraft, a bell can naturally connect to 64 villa gers, unless the player uses `/attribute` to modify the bell.

Trades

Cartographers

Ender relay exploration maps

Expert or higher-level cartographers now trade ender relay exploration maps, which are maps used in the end to find ender relays in the end dimension.

Wandering traders

Trades

Wandering traders can now additionally trade sunflowers, rose bushes, lilacs, peonies, mycelium, cocoa beans[java only], glow berries, sweet berries, juniper berries, stoneshrooms, exposed stone and exposed stone variants, and raven feathers.

Soil and farmland

Soil

Soil Crusts

Soil crust

Soil crust is a block that naturally generates on cracked sand, sand, poor dirt, dirt, degraded farmland, limestone, and any other block that cactus can grow on. Soil crusts also host lichen and other organisms and will also naturally generate below mesa plateaus and in deserts.

Cracked sand

Cracked sand is a special type of sand found in desert temples, deserts, badlands, and some beaches. Only cactus and tumbleweeds[read also: tumbleweeds] can grow on cracked sand blocks.

Cracked dirt

Cracked dirt will from when mud is dried out in a biome with a temperature noise >0.1. Cracked dirt also naturally spawns near deserts, near mangroves, and near badlands. [more to come]

Farmland variants

Fertilized Farmland

Farmland containing yellow specks, suitable for growing crops on. A 1.2x speed boost is applied to any crop growing on fertilized farmland. Fertilized farmland will remain fertilized for 3 rounds of crops that are planted and harvested, in which after the fertilized farmland block will lose the yellow specks and turn into regular farmland. Fertilized farmland is obtained by right-clicking normal farmland blocks with sulfur or poor farmland twice with sulfur.

Poor Farmland

When farmland has been overused without proper fertilization; such as planting 3 rounds of crops without any amendments to the pre-existing soil, Farmland blocks will degrade into poor farmland, which stunts all plant growth growing on the block.

Noteblocks

Honey

When a bottle of honey is used on a noteblock, the noteblock will become a honey noteblock. Honey noteblocks play a longer sound than normal compared to a not sticky noteblock when activated.

Waypoints (Overworld)

Waypoints are a new structure generating in the overworld split into 2 types, continental waypoints, and local waypoints. Waypoints allow faster travel across far distances, although for the player to travel, the player must have an “uncraftable” potion. Waypoints also have a 2% chance of malfunctioning and causing the player to teleport to a random waypoint the player has already discovered.

Continental or superbiome waypoints

Generation and Functions

1 continental waypoint will generate every “superbiome”[read also: terrain generation]. These continental waypoints allow the player to travel to adjacent continental waypoints, or waypoints that have been linked to this waypoint, including all manufactured and local waypoints touching that superbiome.

Local waypoints

Generation and Functions

Local waypoints are a type of waypoint that allows the player, with the uncrafting potion [read also: covenant of witches], to be able to teleport to surrounding local waypoints on that continent that are within 1500 blocks of this waypoint. This uses the same logic as voidal waypoints, except the direction the player is facing is regarded instead of a projectile. Local waypoints generate near significant landmarks, such as

- Inhabited snowy glades [read also: snow buried villages]
- Villages
- Abandoned villages
- Woodland mansion
- Pillager outposts
- Desert temples
- Ocean monuments
- Sometimes above trial chambers
- Strongholds
- Covens[read also: coven of witches]

- Significant underground features (spawning underground)

Chiseled cauldrons

Creation

Any core of teleportation placed down in the overworld with valid chunk coordinates to be a waypoint will turn into a chiseled cauldron. Then a waypoint will form around the chiseled cauldron if space is present. If there is no space, the core of teleportation will not turn into a chiseled cauldron upon placing.

Functionality

Chiseled cauldrons are an unobtainable type of cauldron that is found inside waypoints. Once a potion of uncrafting is spilled inside of a chiseled cauldron, a geyser in the cauldron forming mist will start to engulf the player and allow the player to choose which waypoint to teleport to. Once the player has selected, the mist will fade away revealing the player's destination.

Man made waypoints

The player can create waypoints using special materials to create a core of teleportation. The core of teleportation once placed will decide whether the current coordinates a suitable for a chiseled cauldron, and if so, will turn into a chiseled cauldron and waypoint.

Spells for Chiseled Cauldrons

Vials containing spells can be casted on chiseled cauldrons that act as "commands" in survival mode. Some spells can be casted spontaneously while others may take time before activating.

Liquified horn of storms vial

This spell once casted into a chiseled cauldron takes 4 seconds to charge, releasing thunder particles[coming soon], firefly particles, and angered villager particles. After charging, a shockwave of energy is radiated out from the chiseled cauldron which then will send lightning to any player who has attacked the player who casted the spells during the last in-game day. The liquified horn of storms vial is crafted with a bottle of dragon's breath, copper ingots, thunder roses[coming soon] and a goat horn. When a liquified horn of storms vial is used, the liquified horn of storms also sets the weather to thunderstorm.

Spectral radar vial

This spell can be casted near spontaneously by right clicking a chiseled cauldron with this spell's vial. Upon clicking the chiseled cauldron, a flash of green light will appear teleporting the desired player the spell caster chooses to the spell caster. To cho0se the player to teleport, the spell caster must choose a username to teleport and wait for the player to be teleported to accept the teleportation. After the subject has been selected, the

subject has a 5 second window to cancel the teleport by moving. Once the player has accepted, the player will be teleported within a 5-block radius of the cauldron. The radius in which the player is teleported to can be changed with the /gamerule cauldronTpRadius <integer> command or [comscript]: gamerule set cauldronTpRadius <integer>; line. Spectral radar vials can be crafted with spectral wings, radonite, echo shards, and a recovery pearl. This spell also gives the “glowing” effect (also used in spectral arrows) on all players within 100 blocks, unless the player has invisibility.

Liquid light vial

The liquid light vial is a spell that when casted during the day, will extend the daylight time or change night to a normal day if casted during night. Liquified light vials can be obtained by crafting glowroot paste, glowstone dust, glowsquid ink sacs, and a ghast tear.

Stoneshroom spore vial

Also used to inoculate stones with stoneshroom[read also: stoneshrooms], these vials when casted into a chiseled cauldron, will first inoculate the surrounding area with stoneshroom, then proceed to start growing on undead enemies, due to their weakened immune system. Stoneshrooms will start making them take damage and get the infected effect, making their attacks do less damage, hallucinate and damage each other, and corrupt all their drops into stoneshroom. Note that the infected effect can also be applied to enemies using the `infected` effect.

Guardian vial

The guardian vial is crafted using 4 prismarine shards on the 4 cardinal directions of the crafting table grid with oceanic rods on the 4 corners and a heart of the sea in the middle. This vial upon casting will summon temporary channeling tridents from above the cauldron that auto-aim at the entities surrounding the player that the player has attacked in the last 2000 ticks. These tridents after setting the weather to thunderstorm will summon lightning bolts that destroy the tridents after. This channeling effect will make the player’s opponent drop all metallic equipment in their inventory.

Forcefield [Instant cast]

Creates a semi-transparent blue particle sphere around the player that reflects all ranged attacks.

Forcefield I

Will disintegrate either after 2.5 minutes of casting this spell or after deflecting 5 ranged attacks.

Forcefield II

Will disintegrate either after 5 minutes of casting this spell or after deflecting 10 ranged attacks.

Elemental dragons (Wuxing)

[unresolved: needs descriptions]

Fire

Life

Metal

Nature

Earth

Spectral scales

Obtaining

Spectrals can be sheared to obtain 1-3 spectral scales, with looting shears dropping up to 6 spectral scales. Spectral scales can also be obtained by killing spectrals, dropping 5-7 spectral scales or a guaranteed 7 if looting 3 weapons are used.

Uses

Spectral scaleplate

Spectral scales can be used to craft Spectral scaleplate. The spectral scaleplate is a type of armor that can be used for

- Protection against dragon's breath (end geysers included).
- Gives as much protection as a chainmail chestplate.
- Applies thorns to any enemy that attacks the player (damages the enemy).
- Prevents `lifesteal` I from functioning. (Does not prevent `lifesteal` II)

Spectral Scale Shield

The spectral scale shield is a new type of shield with superior stats compared to a normal shield:

- Can deflect dragon's breath, snowballs, wind charges, arrows, fire charges, ender pearls, shulker bullets, and all other projectiles up to 10 blocks.
- Spectral scale shields have more durability compared to the average shield.
- Entities that attack the shield will be teleported or knocked back 5 blocks.
- Entities will receive 6 damage (3 hearts) every time the entity attacks the shield.

- Due to the spectral shield's lightweight, spectral shields can still be used when the player is walking instead of crouching.
- Spectral shields have a lowered cooldown time compared to normal wooden shields.

Water

Water data values

Water has multiple new continuous data float (double) values between 0.0 and 5.0. These are:

Salinity

Water salinity levels from 0.0 to 1.25 are safe to drink, 1.25 to 2.5 is risky to drink due to dehydration, 2.5 to 3.0 being very risky or dangerous to drink, and 3.0+ is intolerable by the player. Using charcoal or charred wood products to filter water is only effective if the salinity index is below 2.5, which will only reduce the salinity of the water by 0.5 or 1 for every filter.

- 0.0 = Fresh
- 1.0 = Brackish
- 2.0 = Saline
- 3.0 = Seawater
- 4.0 = Supersaturated

Temperature [realism mode only]

The value for water is the water temperature in °C/20. Will not only mix with adjacent water, but also with the ground and air around it as well as the biome using temperature diffusion. Water with a value over 5 will evaporate instantly, but levels 4-5 also can evaporate slower. Water with a value of 0 or under will freeze unless near more water, making ice bottles and ice cauldrons. Laws of convection hold true, and hot water rises while icy water sinks; however, the two can mix.

Water and dirt effects

Mud

All soil or dirt from within 2 blocks of radius from a block of water with a salinity < a salinity of 3 will transform into mud blocks, which sugar cane can be grown on. Note sugar cane can also grow submerged within water a maximum of 1 block deep

For soil or dirt with a radius of 2 next water with a salinity value of 3 or > will transform into hadal dirt; which only specialized beach plants such as Glowroots are able to grow on. Hadal sand and gravel also exist; but due to their coarse texture, both saline and fresh

water can penetrate a radius of up to 7 blocks, depending on the temperature of the biome.

Hadal dirt, gravel, and sand

Hadal materials are a special type of growing medium formed when water with a salinity level of >3 touches soil, dirt, gravel, sand, suspicious gravel, suspicious sand, coarse dirt, or rooted dirt, causing the medium to transform. Due to the salinity of hadal mediums, only some plants can grow on these growing mediums such as glowroots. Hadal mediums cannot dry and must be rinsed with 5 bottles of water to return to normal. Hadal mediums may also leach salt into fresh water.

Enchantments

Experienced I-II

Converts all mob drops to experience (not compatible with mending enchantment)
Compatible with:

- Swords
- Axes
- Maces
- Daggers
- Spears
- Javelins
- Tridents
- Bows
- Crossbows

[unresolved]

XP table

Drop	Amount of XP	
Rotten flesh		
Bone		
Arrow		
Arrow (tipped)		
Ashen bone		
Gunpowder		
Spider eye		
String		
Emerald		
Iron Sword		
Copper Sword		
Gold Sword		

Armor	Durability * Protection points	

Craftsmanship I-III

An enchant that increases the overall quality of items forged by a forging hammer with this enchantment. This enchantment leans toward strong ratios in metal and can only be put on forging hammers. This enchantment also increases the quality of the hammer itself.
Compatible with:

- Hammers

[Unreolved: Needs more definitions]

Level I

Level II

Level III

Cutter I

The cutter enchantment is a new enchantment allowing crystal ores mined with pickaxes to drop in their cut form and gives the pickaxe more durability when mining ores, such as mining diamond ore which will drop diamonds instead of raw diamonds. This enchantment prevents the pickaxe from mining materials that are not crystal ores.

Forger I

The forger enchantment is a new enchantment allowing metal or metallic ores mined with pickaxes to drop in their smelted form and gives the pickaxe more durability when mining metal ores and infested stone, such as mining iron ore will drop iron ingots instead of raw iron. This enchantment prevents the pickaxe from mining materials that are not metal or metallic ores.

Shock resistance I

shock resistance I is an enchantment found in desert pyramid loot chests that can be used to evade and reduce the damage of shock attacks. Shock resistance can be applied to any conductive weapon or armor, allowing the player to attack or be attacked by an electric enemy without taking damage and dropping the weapon.

Conductibility table

Material	Conductivity from –1 to 1	
Iron	1	

Copper	1	
Flint	-0.3	
Gold	1	
Obsidian	-0.3	
Stone	-0.5	
Wood	-1	
Netherite	0.3	
Diamond	-0.5	

Lifesteal I-III

Lifesteal is a new enchantment that allows the player to gain enemy health while attacking allowing the player to gain health while attacking without having to use saturation or constant golden apples. *Note: These enchanted books cannot be combined*

Level I

Level I of lifesteal is s the most basic type of lifesteal, with this enchantment only obtainable by looting chests in dungeons, which have a 4% chance of generating lifesteal I books. This enchantment can be used to enchant any tool with an anvil or enchantment table, but this enchantment requires lots of XP. Note that level I of lifesteal can be blocked spectral scaleplates. This enchantment will allow the attacker to regain 10% of damage delt; rounding down.

Level II

Level II of lifesteal allows for gaining 15% of damage delt and can penetrate spectral scaleplates.

Level III

Level III of lifesteal allows for not just heart regeneration, but also the player gets the `regeneration` II effect for 3 seconds, and this enchantment also steals the hunger of the player's opponent. This enchantment steals 25% of damage delt without armor. Although this enchantment weakens the player's armor by 15% and causes any negative potion effects to be amplified. This enchantment is obtained by looting the final chamber chests from after beating the lapis golem or prismarine golem, or having a 5% chance to generate in ender relay rooms.

Hunting I-V [unresolved: comparison to sharpness enchantments on how this is better needed]

The Hunting enchantment is a new enchantment aimed at increasing the damage of weapons against wild animals including the following:

- Pigs (all temperature variants)
- Cows (all temperature variants)
- Chickens (all temperature variants)
- Turtles
- Ocelots
- Foxes

[unresolved: more animals]

This enchantment has three levels and is not compatible with sharpness, bane of arthropods, and smite. This enchantment can be obtained from Abandoned Camp structures weighted towards level I, Pillager Outpost structures, Buried Treasure features, Shipwreck structures, and Abandoned Belltower structures. [see also: Abandoned Belltower]

This enchantment is applicable to the following equipment:

- #minecraft:swords
- #minecraft:axes
- #strawberrymc:hatchets [custom tag] [realism mode only]
- #strawberrymc:daggers [custom tag]
- #strawberrymc:elemental_weapons [custom tag]
- Mace
- Bow
- Crossbow
- Trident

Level I

Hunting I gives a 10% damage increase to wild animal damage.

Level II

Hunting II gives a 20% damage increase to wild animal damage.

Level III

Hunting III gives a 30% damage increase to wild animal damage.

Level IV

Hunting IV gives a 40% damage increase to wild animal damage.

Level V

Hunting V gives a 50% damage increase to wild animal damage.

Creative Mode Inventory [bedrock edition only]

*Note that some items from different subcategories may overlap, inspired based of off java edition creative inventory.

Items

The “Items” category of bedrock edition’s creative inventory is split into “Functional blocks”, such as beds, work benches, job blocks, containers, light source blocks, and more, and also split off from “Items” is “Redstone blocks”, including blocks that can output a signal when a comparator is attached, target blocks, redstone related blocks or blocks that can be powered by redstone, and redstone power sources. The last category is “Misc.”, including brewing stand items and other such as scutes, membranes, brewing ingredients, conduit, nuggets, and more.

Nature

The “Nature” category of bedrock edition’s creative inventory is split into “Living,” including spawn eggs, groundcover, and plants, living blocks, as well as “Nonliving,” including foods, spawner cage, cobwebs, misc., stones, and animal/plant products. Note processed tuff products such as Polished Tuff, Chiseled Tuff, Tuff Stairs, and others can now be found in the “Construction” category.

Forging and Mechanics

Tools and Weapons

Ats= Attack speed (*1Ats= wooden sword sweeping edge speed) Dur= Durability Atk= Damage Eff= Efficiency (*1Eff= Fist speed obsidian) Enc = Enchantability [coming soon]

Stone trowel
Flint Dagger: 3Atk 5Dur 2Eff
Flint Hatchet: 2Atk 7Dur 2Eff
Flint Pickaxe: 2Atk 10Dur 2Eff
Flint Arrow: 3Atk 1Dur 0Eff
Flint Spear: 1Atk 7Dur 0Eff
Wooden Sword: 4Atk 20Dur 2Eff
Wooden Dagger: 3Atk 15Dur 2Eff
Wooden Axe: 3Atk 20Dur 3Eff 2Ats
Wooden Pickaxe: 2Atk 20Dur 3Eff
Wooden Spear: 3Atk 20Dur 0Eff 1Ats
Wooden Javelin: 3Atk 15Dur 0Eff 1Ats
Arrow: 3Atk 2Dur 0Eff

Wood Shovel: 2Atk 20Dur 3Eff
Obsidian Sword: 9Atk 7Dur 1Eff
Obsidian Dagger: 8Atk 10Dur 1Eff
Obsidian Axe: 7Atk 12Dur 1Eff
Obsidian Spear: 6Atk 10Dur 1Eff
Obsidian Javelin: 6Atk 10Dur 1Eff
Obsidian Arrow: 4Atk 1Dur 0Eff
Stone Sword: 5Atk 20Dur 3Eff
Stone Dagger: 5Atk 25Dur 3Eff
Stone Axe: 4Atk 25Dur 4Eff
Stone Pickaxe: 4Atk 15Dur 4Eff
Stone Spear: 4Atk 20Dur 3Eff
Stone Arrow: 5Atk 1Dur 0Eff
Gold Sword: 4Atk 10-50Dur 2Eff
Gold Dagger: 4Atk 10-50Dur 2Eff
Gold Axe: 3Atk 15-30Dur 3Eff – Note: Uses -0.5 durability instead of –1, and +1 Eff on
Gold Pickaxe: 3Atk 15-30Dur 3Eff – Note: Uses -0.5 durability instead of –1, and +1 Eff on
Gold Spear: 3Atk 10-30Dur 2Eff
Gold Javelin: 3Atk 10-30Dur 2Eff
Gold Arrow: 4Atk 1-3Dur 0Eff
Copper Sword: 5Atk 10-30Dur 4Eff
Copper Dagger: 5Atk 10-30Dur 4Eff
Copper Axe: 4Atk 10-30Dur 5Eff
Copper Pickaxe: 4Atk 10-30Dur 5Eff
Copper Spear: 4Atk 10-30Dur 4Eff
Copper Javelin: 4Atk 10-30Dur 4Eff
[realism mode only]: {
Bronze Sword: 5Atk 20-40Dur 5Eff
Bronze Daggers: 5Atk 20-40Dur 5Ef
Bronze Axe: 4Atk 20-40Dur 6Eff
Bronze Pickaxe: 4Atk 20-40Dur 6Eff
Bronze Spear: 4Atk 20-40Dur 5Eff
Bronze Javelin: 4Atk 20-40Dur 5Eff
}

Ash

Properties and crafting

Gunpowder

Gunpowder can now be crafted by combining 1 ash, 1 sulfur, and 1 charcoal in the crafting grid.

Locality

Basalt deltas

Ash will naturally build up in basalt deltas when there are at least 10 blocks of open air above the ground. Ash accumulates in layers that with default settings, can build up to the max of 5 layers. Ash can also be blown away and disappear after building up to prevent them from clogging basalt deltas. Note that the maxAshLayers and netherWindMax gamerules[coming soon] can be used to manipulate the max amount of layers ash can build up to.

Burning

After organic matter such as logs, planks, leaves, or other flammable materials are burned, ash can be left as a byproduct often sitting on top of charred wood and dirt blocks[coming soon]. Ash deposited by burning will also blow away based on the wind level, with ash blowing away in under 1 minute when wind levels reach 3.

Ash drops rates

The following have a 70% chance of dropping 1 layer of ash upon burning, and 10% chance of dropping 2 layers of ash upon burning. Note maxAshLayers gamerule[coming soon] does not affect ash produced by burning flammable blocks:

- Logs
- Planks
- Bamboo blocks
- Job blocks
- Barrels
- Woods
- Stripped logs
- Moss blocks
- Crafting table

Fences, fence gates, hollow logs, armor stands, slabs, stairs, and doors: 50% chance of dropping 1 layer of ash, 55% chance of dropping 2 layers of ash.

Buttons, pressure plates, signs, slates[education edition only], trapdoors, hanging signs, not muddy mangrove roots, leaves, fallen leaves, groundcover, flowers, mushrooms, vines, animals, and moss carpets: 20% chance of dropping 1 layer of ash.

Burn morels

Burn morels are a species of mushroom that can only grow on charred dirt at any light level. Burn morels once harvested will turn charred dirt and coarse dirt back into normal dirt. Can be eaten but cause enflamed effect.

Charred blocks

Charred blocks are blocks that have not burned properly, with charred blocks having a 30% chance of being generated when any of the listed items below are burned:

- Wooden logs
- Wood block
- Wooden fences
- Wooden planks
- Wooden slabs and stairs
- Stripped wooden logs

Charred wood

Charred logs

Charred striped logs

Charred planks

Charred blocks are brittle and do not resist explosions well but are fire resistant and have a 3 on Blockbench's index of flammability. Charred blocks can only smelt 8 blocks but will turn into charcoal in the furnace after smelting 8 items. This brings the total amount of items smeltable to 16.

Banners

Modifiers

Glowstone dust

Glowstone dust can now be applied to banners for the banner to glow in the dark based off of the patterns on the banner.

Phantom membrane

Phantom membranes can now be applied to banners to make the background of the banner invisible, leaving only the pattern visible. Phantom membranes can only be applied to sticky banners on walls. Any phantom membrane attribute will disappear upon breaking the banner.

Echo shard

Echo shards can now be applied to banners to create a skulk-colored pulsating animation originating from the center of the banner. Banners in boats cannot have the echo shard modifier.

Sticky

Sticky banners can be made by either combining a banner and a bottle of honey or honeycomb in the crafting table or by right clicking any banner with a honey bottle causing the banner to drop and become the sticky banner item. Sticky banners can be directly attached to walls like posters.

Placement

Boat Flags

Banners without any modifiers can now be placed in boats for creating sail boats.

Wall

Sticky banners can be stuck to walls for graffiti, especially when phantom membranes are applied. These sticky banners will no longer be sticky when mined, unless mined with a silk touch tool.

Patterns

Crystal

Crafted with echo shard, amethyst shard, quartz, or amethyst block, shows crystal pattern outline in the middle of the banner.

Small rhombus

A small rhombus in the center of the banner.

Large rhombus

A large rhombus in the center of the banner.

Top left triangle

An isosceles triangle on the top left corner of the banner.

Top right triangle

An isosceles triangle on the top right corner of the banner.

Bottom left triangle

An isosceles triangle on the bottom left corner of the banner.

Bottom right triangle

An isosceles triangle on the bottom right corner of the banner.

Dripstone up

A small isosceles triangle facing up towards the center of the banner.

Dripstone down

A small isosceles triangle facing down towards the center of the banner.

Donut

Circle with hole in center center at center of banner.

Brewing, Effects, and Potions

Effects with no official potion

Cleansing

The Cleansing effect; only obtainable through drinking Bottles of Endillary Nectar, allows the player to be immune from all effects for 10 seconds. This effect is useful from preventing the player from getting levitation from shulkers; allowing hastier lootings in End Cities.

Potions

Recipes

Potion of Glowing

This potion lights up a small area around the player and is mainly used to explore the void dimension. Crafted by brewing glowroot paste into a potion of night vision.

Potion of Levitation

The Potion of Levitation is brewed by either brewing a Sensegrass Tuber[see also: Sensegrass] into an Awkward potion or using a Fermented Spider Eye to alter a Potion of Slow Falling. This potion when used on a Spectral is the only way for Spectrals [read also: Spectral] to survive in the Overworld.

Potion of Luck [recipe]

Potions of Luck can be brewed by inserting a Rabit Foot item into a Potion of Herb.

Potion of Nausea

The Potion of Nausea can only be brewed by brewing a Potent Sulfur block into a Potion of Night Vision. Potions of Nausea can also be found naturally generating in the loot chests for dungeons and inside Decorated Pot blocks in Trial Chambers.

Potion of Blindness

The Potion of Blindness can be brewed by brewing a Midnight Wing [see also: Midnight Oak] into an Awkward Potion.

Potion of Paralyzation

The new Potion of Paralyzation and its complementing effects are split into 2 levels, level I, and level II. This potion is brewed by using a spectrite shard on a Potion of Slowness. Potions of Paralyzation stop the player from moving for 3 seconds in the first level and 5 seconds in the 2nd level.

Potion of Famine [unresolved: needs better name]

Potions of Famine are potions that can be brewed by brewing rotten flesh and redstone dust into an awkward potion. Effects of food poisoning include the opponent not being able to consume food. If the opponent consumes food, they take 3 hearts of damage (6 damage).

Potion of Dragon's venom

Potions of dragon's venom are potions that can be brewed by brewing a chorus wasp stinger into a bottle of dragon's breath.

Potion of Cleansing

Potions of cleansing are potions that can be brewed by brewing a bottle of endillary nectar into a bottle of milk.

Potion of Herb

Potions of Herb are the precursor to Potions of Luck, only needing a Rabbit Foot item to be brewed into this potion. Potions of Herb are brewed by brewing a 4-leaf Clover[see also: Clovers] into a Thick Potion.

UI for brewing stand

See Brewing Stand UI section

Changes to existing effects

Awkward potion

The awkward potion can now allow piglin brutes, hoglins, and piglins to survive in the overworld for periods of time by using the new “awkward” effect. Awkward potions last for 3 minutes and can be used in overworld gold farms. Piglins without the awkward effect will pathfind to nether portals when in the overworld. Piglins in the overworld will also run towards awkward potion items on the ground and drink them.

Trees

Tree spread

Mechanics

Trees may now spread by spawning saplings within a 40-block radius of the tree. To prevent tree saplings from taking over the world, Sheep, cows, and pigs can graze on saplings. Trees will stop spreading if a fire is lit within a 50-block radius to prevent further ignition of a forest.

Pets

Echo Collar

The echo collar is a craftable item crafted with 2 echo shards, 1 name tag, and a leash [may change] that can be equipped on dogs, cats, parrots, or other pets. When worn, echo collars will be the same color regardless of light level, creating a 'glowing' effect. When a pet wearing an echo collar and dies, a broken echo collar will drop, and the pet will respawn at the player's last spawn point. A broken echo collar can be repaired with 8 blocks of sculk on a crafting table.

Fires

Tall fires

Mechanics and Appearance

Normal fires when sitting on top of 2 fuel blocks or dry fuel blocks have a 30% chance of turning into a tall fire. Tall fires are 2 blocks tall and shoot ember particles in random directions upon burning. Tall fires last for longer, spread vigorously, and the embers have a 50% chance to ignite random blocks within a 10-block radius of the tall fire. The radius in which tall fire spreads can be controlled by the tallFireRadiusSpread gamerule in bedrock and java editions. Tall fires also double the speed of fires spreading within a 16-block radius of any tall fire, due to the increased heat. The increased heat means that players within a 10-block radius of the fire may also take fire damage [8 damage (4 hearts)] without proper protection. Tall fire charr a 2-block deep area when burning and have a lower chance of charring objects instead of burning them. Stone next to tall fires will turn into exposed stone and gravel.

Fletching Table and Arrows

Spectral arrows

Spectral arrows now act as ranged torches, as when a spectral arrow does not land on an entity or mob, the spectral arrow will stay in the ground emitting light with a value of 10. This function is useful when the player needs to see if ores are present in a dropdown larger cave and does not want to go down there and tower back up while getting shot at by

skeletons with no ores. The light spectral arrows emit also does not melt ice or snow variants preventing gravel cavens in icy caves and works in the void; where torches and other conventional lighting is unreliable.

Customizable arrows

Arrows in strawberrymc can now be customized to contain custom effects ranging from chock, fire, water, irradiation, or other status effects. Arrows can be customized with the new fletching table UI allowing the player to customize 4 parts of the arrow:

- Tip coating
- Tip
- Shaft
- Fletching

Below are the different options available for arrow customization as well as incompatible ingredients.

Fletching

Phoenix Tail Feather

When added to an arrow crafted with a blaze rod, magnifies the flame effect by a radius of 2 blocks. When used solo, has the same effect as the blaze rod shaft.

Wheat/Tumbleweed

Wheat or tumbleweed[coming soon] can now be used as the fletching, also known as the feather part of the arrow, during crafting or modifying a preexisting arrow using the fletching table for a cheap alternative to feathers in arrows. Note that arrows with dry plant fletching's are unstable and need an upgraded bow to fly straight.

Phantom membrane

Phantom membranes when applied to the base of an arrow using a fletching table, can fly further than a normal arrow, and create a set bonus when the shaft of the arrow is a breeze rod for the arrow by making the arrow deal a larger wind charge explosion.

Spectral wing

Spectral wings when applied to an arrow with a preexisting modifier will increase the effect of the modifier. For example, an arrow with a blaze rod shaft and a Spectral wing fletching will increase the radius of fire created by the blaze rod upon impact.

Raven feather

Raven feathers when attached to arrows using a fletching table, will amplify all the existing buff modifiers of the arrow as well as make the arrow fly further and deal more damage.

Shaft

Blaze rod

Blaze rods when used as the shaft of an arrow made via the fletching table, will create a blast of fire upon impact with the radius of the fireball depending on other materials on the arrow, such as creating a larger ball of fire when wheat is used as the fletching of the arrow.

Breeze rod

Breeze rods when used as the shaft of an arrow made via the fletching table, will create a burst of wind upon impact with the radius of the wind charge depending on the other materials of the arrow creating a set bonus. Set bonus materials include any material that makes the arrow shoot further.

Oceanic rod

Oceanic rods when used as the shaft of an arrow will create a 3x3x3 cube of water upon impact, with a set bonus creating a 5x5x5 cube of water when the tip of the arrow is a Prismarine shard.

Blizzard rod

Bilzzard rods when used as the shaft of an arrow will create a 3x3x3 cube of ice upon impact, with a set bonus creating a 5x5x5 cube of ice when the tip of the arrow is a Prismarine shard. When the tip of an arrow is a spectrite shard, the arrow will freeze into a 3x3x3 blue ice cube.

Bamboo

When bamboo is incorporated as the shaft of an arrow, the arrow gains the flexibility of bamboo which allows the arrow to bend and ricochet off walls, slowly losing momentum, including bouncing off shields.

Tip

Flint

The default arrow

Iron nugget

+2dmg, worse aerodynamics.

Diamond shard

+4dmg.

Copper nugget

+1dmg, worse aerodynamics.

Gold nugget

No damage modifications, amplifies effects of other modifications.

Prismarine shard

Arrows with a Prismarine shard tip can be shot underwater at the same speed and range as a normal arrow would be on land. Prismarine shards will cancel out the effects of blaze rods and amplify and create a set bonus for the effects of oceanic rod arrows.

Echo shard

Though expensive, Echo shard arrows are more durable, allowing the player to retrieve the arrow by killing the mob that arrow was shot at. Echo shard arrows on netherite bows will also auto aim and target the closest player that is not the shooter or behind the player. If no other player is present, it will go after hostile mobs, then neutral and passive mobs. The arrow when shot will prioritize any natural entity the player has taken damage from in the last 5 minutes.

Amethyst shard

+2dmg Amethyst shards are similar to echo shards, are durable and will be "retained" by the target until the target is dead. As a bonus, amethyst shard arrows deal more damage and will shatter and pass-through brittle objects such as:

- Glass blocks
- Glass panes
- Pots
- Pottery
- Copper grates

Amethyst shards can shatter a max of 5 layers of brittle materials with a netherite bow or firework crossbow, and 2 layers with a typical or Palline bow. Amethyst arrows can also pass through all vegetation, making them great for sniping people through leaves. Bows upgraded with Palline shooting amethyst arrows also have a 5% chance of breaking a non-brittle block depending on the block's explosion resistance.

Dragon scales

Dragon scales when applied to an arrow with a breeze rod shaft, the dragons scale tip will create a 2x2x1 area effect of dragon's breath. This tip modification is one of the two only ways to damage the dragon (using arrows) other than palline tipped arrows.

Shears

Shears when used at the tip of an arrow will cut through 1 layer of fibrous materials such as:

- Bamboo planks

- Charred woods
- Baobab bark
- Groundcover

The downside of shear arrows is their limited range, as they are not aerodynamic. These arrows do 1 extra damage than normal arrows.*Spectrite shard*
Used for set bonuses [read also: blizzard rod], or paralyzing enemy for 5 seconds.

Palline

Palline tipped arrows are one of the only ways to damage the ender dragon other than dragon scale arrows. Other than damaging the ender dragon, Palline tipped arrows do 1.5x the damage to boss enemies

Tip coatings

Glowroot paste[coming soon]

Glowroot plants when attached to the tip of an arrow can be used to craft spectral arrows. Glow effect from spectral arrows can also be achieved using the glow spell.

Redstone dust

Redstone dust can be used to tip dagger arrows to give the target irradiation effect [see also: radiation, irradiated] upon impact.

Gunpowder

Gunpowder tipped arrows create a small explosion upon impact. Can not have any fire or water modifier or else arrow will spontaneously explode or act as a normal arrow with water when held or shot.

Shridnite dust

Steals 4 health (2 hearts) from an enemy shot and gives it to the shooter.

Sponge

Absorbs water upon impact, cancels out any water elemental modifier arrow (Prismarine shard, Oceanic rod, and Blizzard rod). Despawns after absorbing water.

Elemental canceling

Certain elemental arrows do not apply in certain situations, for example, gunpowder coated arrows do not explode upon impact underwater, as the water would put out to gunpowder before it has a chance to explode. Another example is shock rod arrows during rain shock the player as the rain conducts electricity and cause the arrow to burst in a ball of electricity as soon as the player pulls the arrow onto their bow.

Below is a table of incompatible ingredients:

Sponge	Prismarine shard, oceanic rod, blaze rod
Gunpowder	Prismarine shard, oceanic rod, blaze rod, when underwater
Shock rod	Prismarine shard, oceanic rod
Echo shard	Glowroot paste

Ores and Minerals

Furnace and smelting mechanics

UI

Read rendering UI

Slag

Metals

Metal ingot

The metal ingot item is a new item with various attributes specifying what metal it consists of. This can be any pure metal or alloy combinations of multiple materials. Metal ingots can be combined with each other in a crusavalt to obtain 2 alloy metal ingots.

Iron

Refining

Iron can only be refined using elevated temperature blast furnaces and must be smelted with coal, charcoal, charred blocks, dried kelp blocks, blocks of coal, or haybales to be refined into iron ingots and slag. In realism mode, once the iron is done smelting, the player has a short 3 minutes to use grass fiber glove to extract the semi-solid iron which must be led into a clay brick channel into clay brick molds where it is molded into shapes, then placed by water and removed manually using a stone or clay tool that will not burn.[realism mode only] Then it must be placed on a crafting table (upgraded with stone) to make items, which must be cooled in a cauldron or water block with low salt concentrations to become a basic iron tool.
Forging also requires a forging hammer [coming soon]

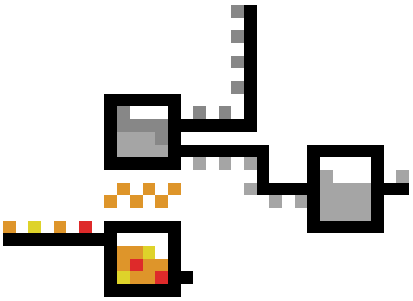


Fig 1. Smelting diagram

Distribution

Overworld

Large iron veins now have a higher/lower chance of generating under bodies of water in the following biomes:

Low chance:

- Warm ocean
- Lukewarm ocean
- Temperate ocean

Intermediate chance:

- Cold ocean
- River

High chance:

- Ice spikes
- Cold river
- Frozen beach

Copper

Gold

Refining

Gold is much easier to forge, mainly because it does not require a furnace or a blast furnace, only an upgraded crafting table and some coal. Gold ore does not always drop raw gold; instead, it periodically drops 6-10 gold nuggets that must be joined together into ingots, blocks, and tools. To join them together, put them on an upgraded crafting table with coal in the fuel slot and craft with the gold (needs forging hammer), and the player can craft gold into smaller pieces without using coal. The player can also craft items like golden apples, golden carrots, golden dandelions, glistering melons, and such without coal.

Sand extraction [realism mode only]

When sand is acquired using a silk touch iron shovel or better, the sand can be put into a sift and the sift will output heavy sand [realism mode only] and sand tailings [realism mode only]. The heavy sand can be smelted into the alloy divider [realism mode only] to extract up to 5 gold nuggets. Potion of luck is effective at outputting up to 3 gold ingots from the heavy sand.

Distribution

Overworld **Nether**

- Common in badlands, eroded badlands, and deserts.
- Moderately common in deep oceans and rivers.
- Rare in biomes with a temperature noise <0.1 other than deserts, badlands, and eroded badlands.

- Will also generate near granite

Usage and properties

coming-soon

Distribution

Overworld

Crystals

Creating crystals

Crystals can be mixed into a single material using budding amethyst blocks. To grow a crystal, first the player must place crystal dust obtained from a gem cutter into a budding amethyst block, then wait. (the process can be sped up using dendrite, but the dendrite gives the crystal a subtle red hue which does not have a effect on the functionality of the crystal) The crystal starts to grow in an amethyst-like form, and the raw crystal can be mined with silk touch, allowing the player to display their riches by adding diamond crystals to their house instead of ugly diamond blocks. Colors can be blended to make the crystal a new shade of color.

Crystal mixing [realism only]

Upon placing 2 or more distinct types of crystal dust (say diamond x lapis or diamond x emerald) inside a budding amethyst, a mixed crystal will grow, with the combined properties of the two crystals, diluted in some way. Shown below are the factors of a crystal. Placing these crystals requires 5 levels of xp

Strength

Enchantability

Enchantfit

Enchantfit [better name needed] is calculated by the amount of lapis in a crystal mix and determines whether the crystal will give good enchantments or trash enchantments on an enchanting table.

Reflectiveness

Reflectiveness is calculated by taking the mean of the reflectiveness of the crystals. Shown below are the reflectiveness of the pure crystals:

- Shridnite: 100
- Diamond: 10
- Emerald: 7
- Dendrite: 2
- **[unresolved: add more values]**

Reflectiveness affects the textures (adding a reflective sheen, or an enchanted like sheen) and the crystal's compatibility in creating spyglasses

Color

The color will be a mix of the two base colors used to create it, perhaps in an RGB format (given that there are no yellow crystals for an RYB system), and the dendrite blue hue will appear if the crystal was sped up using dendrite.

Shape

When the crystal is mined, the resulting crystal is a raw crystal (raw diamond type thing) that must be processed in the Gem Cutter.

Lapis (Overworld)

Redstone (Overworld)

Chorite/Chorine {Plating} (End)

Chorite

Chorite ore is when Chorus fruit ferments into an ore; like coal; that is distributed throughout the outer end islands. Chorite ore can be refined through a blast furnace only, to Chorine.

Chorine

Chorine pellets are the refined version of chorine, through using a blast furnace powered by coal. Chorine has many uses, such as crafting recovery pearls, plating armor and equipment, and many other uses.[see also: chorite and chorine products]

Radonitex

Distribution

Usage and properties

Invisibility Armor

Radonite armor will become invisible along with the player when invisibility potions are drunk. Radonite plated armor will become semi-transparent.

Pressure Resistance

Due to prismarine's special properties, armor of the sea must be coated with prismarine to function properly. Prismarine armor also allows the player to be unaffected by magma columns within underwater rifts within the deep ocean abyss biome. [see also: ocean abyss]

Radiation

Radonite items produce radiation that is a separate damage type. These are three types of damage:

- Block damage:
    - Radonite ore and radonite blocks will create a ray to the player when they are within 11 blocks.
    - All blocks that the ray intercepts will dampen the radiation, by their radiation tolerance value.
    - Base damage is 0.1 HP (0.05 x) in Easy, 0.15 HP (0.075 x) in Normal, and 0.2 HP (0.1 x) in Hard per tick for radonite blocks but 1/10 of this for radonite ore
- Item damage:
    - Each item will be given a radiation value based on how many radonite ingots they contain
    - Items in the inventory will deal 0.5 HP (0.25 x) per second
    - Items in the hotbar will deal twice that
    - Items held will deal twice THAT

Palline

[read: palline(strengthening)]

[Unresolved]

Shridnite

Shridnite is a sand-based

Usage

Distribution

Nether

Shridnite is a rare ore found in soul sand valley biomes in the nether. Large vertically generating ore veins can be unearthed near large fossil sites.

Diamond (Overworld)

Spectrite

[see: sprectrite shard]

Spectrite is a colored ore

Spectrite is a decorative ore with stats like iron endemic to the glacier biome [read also: glacier biome]. Raw spectrite (like lumen mytanum) has a dull, grayish blue coat, but the vibrant inner layer can be revealed after cutting the raw spectrite via a gem cutter with coconut oil as lubricant[read also: coconut oil]. The spectrite reveals a wide spectrum of colors; that can either be used to create armor and equipment or armor trims[read more: spectrite in ores and minerals]. Mainly decorative ore for armor trims and such.

Emerald (Overworld)

Amethyst (Overworld)

Quartz (Nether)

Netherite (Nether)

Spectrite (Overworld, Ice Caves)

[read spectrite here]

Lumin Mytanum

Lumin Mytanum is a normally invisible [camouflaged with surrounding matrix] metallic ore that disguises itself as stone, but when a glowroot torch is used, the ore emits a faint red light. Lumin Mytanum does more damage/efficiency, but less durability than Netherite. This ore is quite common but is hard to obtain due to the hardness of how long the ore takes to mine, and the fact that glowroot torches are needed for this ore. Lumin Mytanum spawns in every biome in the overworld and turns red after mining. The drawback of using Lumin Mytanum is that the player takes 3 hearts of damage regardless of armor stats every time the player uses any Lumin Mytanum equipment. When alloyed with another crystal, the amount of damage taken be the player is divided by 1.5 then rounded up. Lumin mytanum in its refined state is yellow or black.[disputed]

Ancient dyes

Aquamarine (dye)

Aquamarine dye is a rare ancient dye made by turning pitcher plants into pitcher plant fiber, then putting pitcher plant fiber into the crafting table to create aquamarine dye. Pitcher plant dye can also be found in trail ruins and inside pots in trial chambers.

Caramel (dye)

Caramel dye is a rare ancient dye made by either using torch flowers to make them by turning torch flower fiber and putting them in the crafting table.

Sunset (dye)

Sunset dye is made by crafting inferno lily flowers into dye. Sunset may also be found inside chests buried inside large charred trees in the inferno wastelands biome.[coming soon]

Weather

Eclipses

Eclipses are an event that are triggered through /event [see also: /event].

Solar Eclipse

Solar eclipses have a 2% chance of occurring every day in easy mode, 1.5% chance in medium, and 0.5% chance every day for hardcore, realism, and hard mode. Solar eclipses cause the sun to dim to light level 6 for 5 real life minutes (1/4th of a Minecraft daylight time) and have a chance of giving the following effects (level I) for the next day after the eclipse:

- Luck
- Swiftness
- Jump boost
- Resistance
- Regeneration (Only for 10 sec. When attacked by hostile or neutral mob during that day)

Note that solar eclipses can occur when the sun is up during that day and the weather is clear.

Lunar Eclipse/Blood Moon

A lunar eclipse in strawberry Minecraft has a 15% chance of occurring for every full moon in hard mode, and a 5% chance of occurring every full moon in easy or medium mode. These lunar eclipses or blood moons are indicated by an orangish-red hue on the moon, with extra dangers and bad luck.

Events

During a blood moon, the following will happen:

- Bad luck effect for the entire night.
- 2x the spawning rate of hostile mobs in the overworld.
- Creepers gain intelligence and will blow up holes in obstacles between the player and it.
- Mobs can hear the player if the player moves around and therefore allow creepers to blow through the walls to allow other mobs inside.
- Zombies will break down doors and brittle or fragile blocks such as doors, trapdoors, rails, leaves, charred materials, dirt, soil, cracked dirt, mud materials, bamboo materials, groundcover, glass, glass pains, terracotta, pots, lanterns, torches, glowstone, pottery, amethyst, slime, honey, TNT, and chests.
- All undead mobs gain a 50% speed boost and attack damage.
- Sleeping is disabled for the whole night

Snow

Precipitation is now instead of determined by the original, rigid function for checking current biome and altitude for generating snow, Strawberry Minecraft uses a temperature and cloud cover check, with “snow” being a separate weather option instead of a variation of “rain” in Minecraft’s code. Snow occurs when a raining air mass enters a cold region (with temperature noise <-0.3) or if a cold air mass leaves a cold region, it will continue snowing for the next 250 blocks. Only areas already loaded by the player simulate weather patterns.

Location

All weather is now determined by overhead cloud cover, meaning that instead of the entire world either precipitating or clear, now only Perlin Noise determined clouds determine the weather conditions below, giving the world a more dynamic feel.

Weather forecaster [realism only]

The weather forecaster is a craftable item requiring 4 copper ingots surrounding 1 redstone dust. The weather forecaster has 4 sections, rainy, snowy, overcast, and clear. Based on the forecast in the next 10 days (about 1 and a half weeks), the weather forecaster’s pointer will rotate clockwise and point towards the current and future weather. The weather forecaster cannot predict some events such as eclipses.

Additions

Fog

Fog is a new weather type that has a 40% chance of occurring before or after snow, 20% of the time after rain or thunderstorms, and an 80% chance in jungle mountain areas with an elevation >100 y. During fog, all surfaces within 300 blocks of a large water source and under a y level of 200 will have reduced render distance as well as overcast weather to give the game a gloomy look.



Fig 1. Foggy weather demonstration [credit: <unknown>]

Overcast

Overcast weather occurs 10% of the time in biomes with a temperature noise >0.4, 40% of the time with biomes with a temperature noise from –0.2 to 0.3, and 60% of the time in biomes with a temperature noise from –1 to –0.3.



Fig 1. Overcast weather demonstration [credit: <unknown>]

Ambience (compiled)

Void

_1-3

Sounds like reverbed footsteps on hard stone from a distance with faint creaking/screeching in the background. Can play anytime in the void, has a 0.1% chance every tick.

_4,5

Sounds like faint screeching noise often layered with _1-3. Can play anytime in the void, has a 0.1% chance every tick.

_6

Sounds like raspy breaths right behind the player. Can play anytime in the void, has a 0.1% chance every tick.

_enter

Plays only when the player just entered the void. Sounds like faint drums in the distance that play then stop abruptly, followed by the following sound that can be recreated by setting the players waveform synth to the following:

- Set waveform of synth to pulse (width 40%) or square.

- Set to polyphonic and play augmented notes (minor 2nd, tritone, etc.).
- Set sub-oscillator to sine and tune the triangle 1 octave down from the body.
- Use the Low Pass Filter 2-pole or 4-pole.
- Boost resonance by 25%, or if there is still unwanted oscillation, turn resonance Down.
- Set LFO to 0.5hz.
- Attack: Long, Decay: Moderate, Sustain: 80%, Release: 3.5s

_drone

Far away winds droning at low frequency to annoy player

Overworld

Birds

Bird sounds now naturally occur near all trees except pale oaks, and especially occur in jungle, birch, and oak trees. Bird sounds depend on the biome, with toucans, paradise birds, macaws and hornbill chirps playing in jungles and dark oak forests, while sparrow, junco and more temperate bird sounds play in temperate biomes (4-6). Bird sounds decrease in frequency as the day goes on from morning.

Frogs

Frog sounds now play from late evening to after dusk in jungles, savannahs, birch forests, old growth birch forests, taigas, groves, old growth taigas, spruce forests (note does not play in pine forests), oak forests, forests, dark oak forests. Frogs can also now spawn in these biomes near fresh water on mud.

Waves

Wave sounds now play if sand is in contact with a body of water with a salinity >2 and if the body of water exceeds the player's render distance. Wave sounds intensify if the wind level is >2.

Echoes

Echoing sounds now reverberate through areas sealed in that are deeper than a y level of 0. Echo sounds only play in dark areas.

Rain

Rain sounds now play differently if rain is hitting water, with a deeper muffled feel to the sound.

Thunder

Further away thunder will be more muffled and closer thunder is louder.

GitMC

GitMC is a native git-like version tracker that keeps track of changes in a specified world using a git chart diagram, as seen in Git Bash. GitMC makes it easier to track and modify changes in a world when collaborating or when building solo. GitMC interfaces can be accessed through the GitMC button in the esc menu.

Attributes

Attributes below are given a shorthand code (A<number>) that is listed below the list of git keywords. Keywords with the code mean that said attribute is valid for the specified keyword.

A1: --f

Forces the given command before overriding all warnings.

A2: --abort

If you decide to use the built-in merge conflict resolver, and you decide to scrap the merge entirely by reverting to the last unconflicting commit within your branch path, you can use:

```
C<user>\\: git merge --abort
```

This will automatically revert to the last change on your native branch, meaning that no pull request is needed.

A3: --squash (computationally expensive)

Git merge –squash is a command for if you would like to have the 2nd instance (main/origin) overwrite all air blocks within your branch. This method is extremely unstable, as block states stored natively on the 2nd instance are not transferred to yours, possibly leading to fatal errors. [see also: Errors]

Example:

```
C<user>\\: git merge eric_branch --squash
```

A4: --clear

Clears all data in the provided text column.

Syntax

:wq

Write and quit is used when git opens a native textbox you would like to close.

Syntax:

```
:wq
```

Commit

Committing changes to the Git graph requires a commit message.

Example:

```
C<user>\\: git commit –m“changes to tower”
```

If no –m attribute is given in the console, git will generate a temporary .gitignore txt file where the commit message can be typed and closed.

Valid attributes: A1, A2 [only during transaction]

Pull

Git pull fetches all changes from the global repository that others might have made while you were gone. Git pull cannot be used when your branch is too many commits ahead/behind main or if you are still in the git merge menu.

Example:

```
C<user>\\: git pull
```

Push

Once changes are committed using git commit, git push can be used to officially add it to other people’s commit charts, otherwise this command can be skipped if you are working on a world solo. If conflicts are detected within 2 different branches, you can choose to keep the incoming changes, keep the current changes (current branch selected), or attempt to automatically merge using git merge. If the conflict is too complex to merge, then you can manually resolve it in VScode or revert to a previous unconflicting commit using git revert.

Example:

```
C<user>\\: git push main
```

Valid attributes: A1, A2 [only during transaction]

Log

Git log provides a list of all commits ordered chronologically in the specified branch.

Example:

```
C<user>\\: git log
```

Valid attributes: A2 [only during transaction] , A4

Reload

Git reload will pull al your local changes and update them into your world. If the process takes too long, you can use A2.

Example:

```
C<user>\\: git reload <opt: instance name>
```

Merge

Git merge in GitMC is useful when you have 2 instances of a world both with features you would like to push and not overwrite. In this case, you can use git merge to merge them into a single auto commit.

Example:

```
C<user>\\: git merge <branch name>
```

If someone made changes to the main branch after you checked out your branch, then conflicts may appear. Under normal circumstances, GitMC should automatically resolve the conflicts, but under the condition of worlds with changes larger than 5MB, then git merge will not be able to automatically merge. In this case, you will either need to accept all current/incoming changes (therefore overwriting the other instance) or manually merge them in VScode or the built-in GitMC merge conflict resolver.

Valid attributes: A1, A2, A3

Revert

If you have messed up or decide to revert to a previous version of your save data of your world, you can use

```
C<user>\\: git revert <commit name/message>
```

Although if you are working on a world with multiple contributors, you must first create a pull request (referred as a PR in this section) that other contributors must agree on. If they agree, then you will be notified in the GitMC UI and the revert will happen. If for debug purposes you would like to see the delta log, you can use

```
C<user>\\: git revert --delta-log
```

Valid attributes: A1, A2 [only during pull request]

Build

Git build emulates the code to check for corrupted values, undefined variables, or other errors that may crash your computer if you had run the code immediately. After the code is built, git outputs any errors if there are any or outputs “build successful” if no visible errors are present.

Syntax:

```
C:<user>\\: git build
```

ComScript [name may change]

ComScript is a coding language based on Minecraft Commands that eliminates the need for long chains of Command Blocks. ComScript can be natively and directly executed in the admin lobby or by typing scripts into the Scripter block[read also: lobby].

Errors

Errors due to syntax or any other reason when the script is run are shown in the native terminal located at the bottom of the UI. All errors are red text, warnings as yellow text, and logs as white text.

Error: Undefined data structure(s)

This error appears when undefined variables, arrays, lists, or dictionaries are used.

Error example:

```
store new var1 = “cheese”;

store set var2 = “foot”;
```

Error: No mandatory execution marker found

This error appears when a selector requiring an execution marker does not have a global marker to fetch from. This can trigger a domino effect of errors since the invalid selectors output `null`, which may not be compatible with the data type where the selector is used.

Error: Unexpected data type in input

This error occurs when an incompatible data type is used in an input.

Error example:

```
store set var1 //“var 1 is a string” ++;

//“In this example, var1 which is a string is being changed by 1 which is a operator for integers and floats only”
```

Header

The header file in a Comscript file is an essential file that links all of the projects together. Header files are meant to be a way of declaring execution details globally unless stated otherwise in a file. Header files support the following keywords:

- mark
- return
- void
- queue
- provide
- Global functions
- Global objects

Technical details

Terminology

Relative location selectors

Relative location selectors include any selectors that are location or distance based, meaning that they must have a starting point of execution. These include:

- @p
- @n
- @s
- @c [education edition only]

If no execution marker is included, then it will error and output `null` at any use instances.

Native console

The native console is the built-in executor within the esc menu.

Token

A token is any symbol, symbolized, or packaged string that can be used in a value type.

Command selector standardization

All commands with selectors now follow the structure below:

<command namespace> <selector> <action/input> <opt: attributes, booleans, conditions, etc.>

Execution Priority

Execution priority is an integer ranging from 0 to the number of Comscript files in a project that determines when a file should be executed while in queue.

Execution pattern

Selector fetch pattern

If a selector is distance based, such @p (closest player), then the code scans in a spiral pattern in 1x1x1 chunks looking for an entity with the valid conditions for the selector.

Example:

Assume that an entity with the correct conditions is being searched for with a relative location selector. The block with that entity is marked as “X”. Chunks are labeled by scan order. The scan terminates and returns the result after the entity is found.

North

9	10	11	12
1	2	3	X 13
8	Execution marker location	4	
7	6	5	

[see also: Syntax: Custom execution patern]

Quotations

Quotations in comscript are unidirectional, meaning that they do not directionally encapsulate text.

Example:

Aptos: “”

ComScript: ""

Execution location auto-assumption key

Commands executing without a specified location (specified via the marker keyword) are executed based off of the following rules:

- If the script line is encapsulated by a *{} condition statement with a location specifier, then it is executed there.
- If no specified location is fetched from the previous condition(s), then the execution marker (if there is one fetched correctly) will become the native execution location.
- If no location is fetched still, and if a script is executing from a Scripter block, then valid selectors execute from the location of the Scripter including:
    - @p (for these locational relative/proportional selectors such as closest player/entity, it is relative to the location of execution) [property P1]
    - @e [see property P1]
    - @s [see property P1]
    - @n [see property P1]
- If the script is being executed directly from the native console and no location is defined from the previous entries, then the script executes globally, causing a syntax error to occur in any relative location selectors.

Invalid token substitution key

Invalid tokens can still be stored as strings or custom value types by substituting them with Unicode or the special character chart below:

Characters with UTF-8 charset can be directly typed and are safe.

Character	Default ComScript character code	Decimal code	Can store in normal string?	Valid in Unicode?
<	<	<	Yes	Yes
>	&\3C;	<	Yes	Yes
&	&	&	Yes	Yes

“”	"	"	Yes	Yes
!	!	!	Yes	Yes
@	@	@	Yes	Yes
#	#	#	Yes	Yes
$ UTF-8	$	$	Yes	Yes
% UTF-8	&percent;	%	Yes	Yes
^	&hat;	^	Yes	Yes
Standard *	*	*	Yes	Yes
Operator *	∗	∗	No	Yes
Heavy *	n/a	✱	No	Yes
Circled *	⊛	⊛	No	Yes
]	&sp;		Yes	Yes
}	&cp;		Yes	Yes
\	&bs;		Yes	Yes
;	≻		Dependent	Yes

Note that these tokens are fully customizable. [see: Charset token]

Syntax

Queue (execution priority)

The queue keyword is a header file only keyword used to determine the right-of-way when two files are executed simultaneously. Queue provides an index for each file ranging from 0 to the number of files in said project ranking the priority of the files. A lower queue index means higher priority.

Auto queue is a built-in queue configuration that uses the following to determine a file's priority:

- If a file has runtime keywords that take longer than 1 tick to execute: Priority 1
- If a file fetches from another source and the opposing file does not need external help: Priority 2
- If a file executed first: Priority 3
- If the file has a name higher in the alphabet when in alphabetical order: Priority 4

If any conditions above are met, `return(execute);`.

Queue syntax:

```
private class queue {

     queue add <highest priority file href>;

     queue add <mid priority file href>;
```

```
      queue add <low priority file href>;
      return(getqueue) as queueList;
}
```

Definers [unresolved: better name]

local:

The local definer causes any modifier keyword [see also: Terminology: Modifier keywords, Technical: Modifier keyword list] to become a local action. Local actions only perform the modifier within said file or when encapsulated within a flow control [see also: Terminology: Flow control] string only performs the modifier in the flow control segment unless otherwise stated with global:.

Example:

```
local:mark add @r.cache-once mobile
```

global:

.get-once

.lock-element

Custom execution pattern

Custom execution patterns can be defined using the pattern keyword. For a definition and explanation on execution patterns, see Technical details: Execution pattern.

Example:

```
local:pattern set spiral; //sets pattern to spiral
```

Execution marker/location

Execution markers mark where the script would start executing if you would like to override the location of where the scripter block is placed or specify the location of where the script is executed if written in the native console. Selectors that are location-based (nearest entity, etc.) scan in a spiral pattern clockwise starting northwest of the marker to search for valid selectors. [see also: Selector fetch pattern] Execution markers can be defined using the following:

```
mark add <location (x y z)> <mobility>;
```

Mobile relative execution markers

Relative execution markers move depending on the base selector and the spacing. These allow scripts to execute depending on the mobile selector; which means that you can have scripts executed at a specific tagged sheep for example. If a script is a macro-script (meaning executed process takes >2 ticks) then the script execution location stays at the execution marker's location when the script started executing.

Syntax:

```
mark add relative <base selector> <tilde defined relative location>;
```

Example:

```
mark add relative @e[type=sheep, tag=cheese] ~~10~;
```

Charset

Character sets and character operations can be changed manually by using the `charset` keyword. Charset acts as the <meta> tag in HTML. Charset has the following valid sub-keywords:

Charset store

Charset store changes the current charset.

Example:

```
charset store UTF-8
```

Valid charsets for comscript include

- UTF-8
- US-ASCII
- ComScript-EN (default)
    - ComScript-zh-CN (modified in language settings)
    - Comscript-hi-IN (modified in language settings)

Charset token (custom substitution) [see also: token substitution]

For convenience reasons, if you would like to create your own token substitution token then you can use the following. The example below replaces "$"'s token to "&s;". Note that a fatal error will occur if you store it as a pre-existing token.

Example:

```
charset token store &dollar; = &s;; //“2 semicolons must be used here as one is for the token abbreviation and the other is to close the statement”
```

Input formatting

Inputs in comscript can accept the following formats

- Unicode
  - Expected formatting: <value>
- String
  - Expected formatting: <value>
- Binary (useful for raw files)
  - Expected formatting: <complex value>
- Boolean
  - Expected formatting: <conditional statement>
- Float
  - Expected formatting: <value>
- Integer
  - Expected formatting: <value>
- Packaged lists
  - Expected formatting: <complex value>
- Blockstate
  - Expected formatting: <state>
- Entity
  - Expected formatting: <name>
- Item
  - Expected formatting: <item name>
- Coordinate
  - Expected formatting: <x> <y> <z>
- Custom format

If you want to change/define the format of a variable or value, you can use the following:

Convert to string

```
string(<value>);
```

```
str(<value>);
```

Convert to boolean

Only 1-digit binary integers, and strings with boolean names can be converted.

```
boolean(<value>);
```

```
bool(<value>);
```

```
booleanState(<int0-1>)
```

Convert to binary

Anything can be converted into binary, as it is the raw data type.

```
bin(<value>)
```

```
createBin(<value>)
```

Example:

```
bin(0101011100110101) //“w5 in binary”
```

Convert to float

```
float(<value>)
```

```
flt(<value>);
```

Convert to integer

```
integer(<value>)
```

```
int(<value>);
```

Package a list

Packaging a list allows for mixed values to be combined and still be able to be extracted from a different source.

Syntax:

```
pack.push(<listname>);
```

Example:

```
store set myVar = int(String);
```

Custom type

Custom types can be stored as a list with all valid strings and/or characters listed within the built-in list.

Example [UTF-8]:

```
charset store myFormat = {&quot;, $, “pcms”, “Ó”, “Í”};
```

The statement above would declare all strings or characters within the parentheses to be the only valid tokens in that data format.

Function block

Function blocks are a new block that creates global functions applicable across all command blocks. When opened, the player assigns a function name and can be assessed with the keyword.

```
<function name>(<function variables, booleans, etc.>){

<function contents>

}
```

If the function you are creating does not return a value, then you can use the syntax below:

```
void <function name>(<function variables, booleans, etc.>){

<function contents>

}
```

Variables, lists, arrays, and dictionaries created within a function are all stored locally, meaning that you cannot access them unless you make them global via

```
global(<namespace>);
```

If you would like to turn a global data structure back to local and change all outside instances of the data structure to `null`, then you can use the following:

```
local(<namespace>);
```

Classes

Classes are an important part of Comscript and object oriented programming in general. Classes in Comscript can be named, and can be specified in a header file [see also: Header File] for script execution modifiers such as execution location and priority.

Classes can be created with the following syntax:

```
public <class name> {
```

```
<class contents>

}
```

Subclasses

Subclasses can be created within a larger scope class to apply a modifier to a specific section within the scope class.

Subclasses can be created with the following syntax:

```
public class <subclass name> extends <scope class> implements <greater scope class (optional) {

<subclass contents>

}
```

Grant Private Fetch Access

If you have a private element that you would like to give another file permission to fetch from, you can use the following syntax to allow localized fetching:

```
override private(<private object namespace>, <href of file grant permission>, <optional: this file>);
```

It is important to know that this must be implemented BEFORE the line where the other file fetches and make sure that this file has a higher priority value in the header file.

Data structure creation

Variables

Creation

Variables are created using the command

```
store new <varname> = <init_value>;
```

Or multiple variables can be created with same value using

```
store new <varname>, <varname>=<init_value>;
```

Global variables (which are built in) can be fetched with the command

```
return.global <varname>;
return.global <varname> as <varname temp>; //"sets the keyword for the fetched variable to the <varname temp> value”
```

Formatting

Variables can be created with a preset format [see also: input formatting] by using the following

```
store set <varname> as <format code>(“<content>”);
```

Example:

```
store set cheese as str(“foot”);
```

Note that variables made with a preset format can still be modified using input formatting syntax.

Modification

Existing variables are modified with the command

```
store set <varname> = <float/int/bool/str_value>
```

Variable values can be manipulated by using alter commands below. Note that once a variable is created, a modifier command[read: Input formatting] must be used to change the data set type, e.g., float, boolean, string…

A space must be used between variables and operator keywords.

Change by 1

```
store set <varname> ++;
```

Change by –1

```
store set <varname> --;
```

And they can also be modified using augmented expressions:

Addition

```
store set <varname> + <val>;
```

Subtraction

```
store set <varname> - <val>;
```

Multiplication

```
store set <varname> * <val>;
```

Division

```
store set <varname> / <val>;
```

Note that division by 0 outputs “0” as a flexible float. Flexible floats can act as integers.

If you would like to retrieve the remainder of the division statement you can use

```
store set <varname> /% <val>;
```

Special operators like sqrt and cosine are listed within the “Mathematical operators” section.

Execution range

Execution range is a variable that determines the area that the script applies to and executes as. For example, if a script with an execution range of a circle with a radius of 5 blocks and the script is looking for chickens tagged with “foot” and the chicken tagged with foot is outside of the circle, then the script will return `null`.

Mathematical operators

Mathematical operators can be used to add, subtract, multiply, divide, square root, power, factorial, sin, cos, tan, $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$, log, and $\log^{-1}$ valid format variables, floats, and integers.

Addition

```
<val1> + <val2>
```

Subtraction

```
<val1> - <val2>
```

Multiplication

```
<val1> * <val2>
```

Remainderless division

```
<val1> / <val2>
```

Division by 0 results in a output of 0.

Remainder fetch

```
<val1> /% <val2>
```

```
<val1> mod <val2>
```

Logarithmic operations

Invalid operator syntax or inputs result in `null`.

Log

```
log(<val>)
```

Negative log

```
log^(<val>)
```

Cosine

```
cos(<val>) //shorthand

cosine(<val>)
```

Negative cosine

```
cos^(<val>) //shorthand

cosine^(<val>)
```

Sine

```
sin(<val>) //shorthand

sine(<val>)
```

Negative sine

```
sin^(<val>) //shorthand

sine^(<val>)
```

Tangent

```
tan(<val>) //shorthand

tangent(<val>)
```

Negative tangent

```
tan^(<val>) //shorthand

tangent^(<val>)
```

Sqaure root

```
sqrt(<val>) //shorthand

sqaureroot(<val>)
```

Random value

Random inclusive values can be generated in ComScript using the following syntax

Integer exmaple:

```
random.randint(<min>, <max>)
```

Non-inclusive values can be generated using

```
random.n.randint(<min>, <max>)
```

Random floats a determined fix/round can be used with the following

```
random.<fix length>.rand(<min>, <max>)
```

Comments

Single line comments are notated as //

Multiline comments are notated as // “[text]”

Example:

```
//this is a single line comment
// “this is a multi
-line comment”
```

Chain commands

Commands are chained in two ways:

With a newline

```
fill ~~~~3~3~3 stone
fill ~-3~-3~-3~~~ stone
```

With a semicolon

```
fill ~~~~3~3~3 stone; fill ~-3~-3~-3~~~ stone
```



Fig 1. MEME

Execute commands

For execute commands, the execute part is erased, and the as|at|in|align|anchored|positioned|rotated|facing|if|unless|run|store becomes the command itself. Example:

```
execute at @p run summon pig;//“becomes” at @p run summon pig
```

If subcommand

The if subcommand has a new sub argument: comp. Comp is used to compare values.

if comp <value1> <operator> <value2>

Run subcommand

Instead of running one command, the run subcommand could run a chain, notated by parentheses. It could also run a single command normally

```
at @p run ( //“Run a chain of commands”
summon zombie ~~2~
```

```
at @e[type=zombie] run summon lightning_bolt // Run one command
```

Event sensing

Scripts executing conditionally during an event or every tick can be started with * {<script} to execute every tick and *<event namespace> {<script>}.

Example:

```
*blockEvent.blockBreak{

at @a run summon lightning_bolt;

}
```

Keywords

Selectors

```
@server.player.last //last player that join still online
@server.player.first //first player to join still online
@server.player.id.<id> //local player id specified
@server.player.last.chat //last player to send message in chat
@server.player.last.chat(<‘msg’>) //”Last player to send specific
message in chat”
@server.player.rel(<x, y, z>) //closest player to specified position
@server.e.rel(<x, y, z>) //closest entity to specified position
@server.player.rel.(<username>) //closest other player to specified
username
@server.e.rel.(<entity id or type>) //closest entity type originating
from spawn’s closest player
@server.player.rand //random player online
@server.e.rand //”random loaded entity”
@server.player.op //he highest-ranking member online or server owner
@server.player.spawndist //the player closest to world spawn
@server.e.a //all entities
@server.player.a //all players
@server.item //all items
@server.item.<item name> //all specified items
@server.item.(<item id>) // item with spawn id specified
@server.e.a.nat //all natural enities
```

Variables

```
return server.player.cord(username) //“returns coordinates of
specified username”
return server.count //returns amount of players online
return server.fps //returns current global fps average
return server.<username>.ping //returns ping of specified username
return client.player //closest player to execution coordinate
return client.e //closest entity to execution coordinates
store.global set excrd = <value> //stores execution coordinates
```

Lists

```
return server.players //returns list of online players
```

Arrays

```
return server.chunk.loaded //“returns array of loaded chunks with
format username, chunkx, chunky”
return server.players.ping //returns players and ping
return client.<username>.xyz //returns coordinates of provided
username
```

Dictionaries

```
return <username> “location” //“returns coordinates of specified
username”
```

Boolean

```
LoadChunk(x,y,z)? //“check if loaded chunk”
Online(<playername>)? //“if given player is online”
```

Appendices

Appendix A: Block list [ONLY PLACE BLOCKS ALREADY CODED]

Block name	Block registry name	Resistance	Hardness
Chorite Ore	ChorineOre	1	2
Block of Chorine	ChorineBlock	6	5
Echowood Log, Echowood Wood, Stripped Echowood Log, Stripped Echowood Wood	EchowoodLog, EchowoodWood, StrippedEchowoodLog, StrippedEchowoodWood	2	2
Echowood Planks, Echowood Stairs,	EchowoodPlanks, EchowoodStairs,	3	2

Echowood Slab, Echowood Fence, Echowood Fence Gate	EchowoodSlab, EchowoodFence, EchowoodFenceGate		
Echowood Leaves	EchowoodLeaves	0.2	0.2
Echowood Door, Echowood Trapdoor	EchowoodDoor, EchowoodTrapdoor	3	3
Echowood Pressure Plate, Echowood Button	EchowoodPressurePlate, EchowoodButton	0.5	0.5
Echowood Sign, Echowood Hanging Sign	Echowood Sign, EchowoodHangingSign	1	1
Radonite Ore	RadoniteOre	3	3
Block of Radonite	RadoniteBlock	6	5
Palm Log, Palm Wood, Stripped Palm Log, Stripped Palm Wood	PalmLog, PalmWood, StrippedPalmLog, StrippedPalmWood	2	2
Palm Planks, Palm Stairs, Palm Slab, Palm Fence, Palm Fence Gate	PalmPlanks, PalmStairs, PalmSlab, PalmFence, PalmFenceGate	3	2
Palm Leaves	PalmLeaves	0.2	0.2
Palm Door, Palm Trapdoor	PalmDoor, PalmTrapdoor	3	3
Palm Pressure Plate, Palm Button	PalmPressurePlate, PalmButton	0.5	0.5
Palm Sign, Palm Hanging Sign	PalmSign, PalmHangingSign	1	1
Block of Dendrite	DendriteBlock	5	5
Block of Garnet	GarnetBlock	5	5
Crystalline Calcite	CrystallineCalcite	5	2
Glowroot Tuber Stone	GlowrootTuberStone	6	1.5
Clover Mat	CloverMat	1	1
Fertile Farmland, Poor Farmland, Tilled Farmland	FertileFarmland, PoorFarmland, FarmlandStrawberry [clone]	0.6	0.6

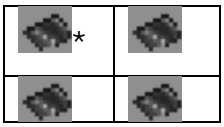
Appendix B: Item list

Appendix C: Entity list

Appendix D: Crafting

Inventory 2x2 Crafting Recipes

Flint[realism only]



* Flint Chip [credit: RLcraft]

Flint dagger



If near open wound in tree 

Flint hatchet



 ↻

*Plant Fiber



-> Dyed Candle 

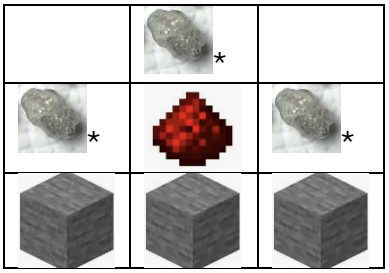


-> Candle 

* Coconut oil/ Petroleum based wax/ Beeswax

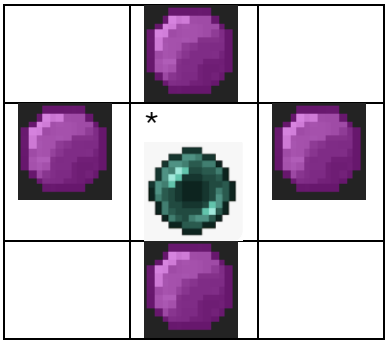
Crafting table 3x3 Crafting Recipes

Gemcutter



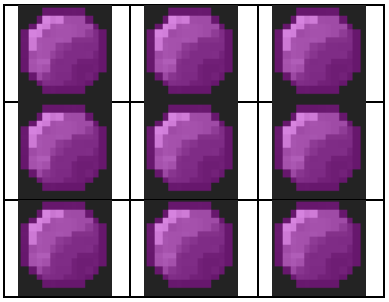
*Raw Diamond [coming soon]

Recovery Pearl



*Credit: Minecraft Wiki

Chorine Block



Smoker cooking recipes

Product	Ingredients	Smelt time in seconds
Salted beef	Salt, beef	Default
Salted porkchop	Salt, porkchop	|

Salted chicken	Salt, raw chicken	|
Salted mutton	Salt, raw mutton	|
Salted cod	Salt, cod	|
Salted salmon	Salt, salmon	|
Salted rabbit	Salt, raw rabbit	V
Salted bass	Salt, raw bass	
Roasted mushrooms	Red mushroom, brown mushroom	Default +0.5
Roasted carrot	carrot	|
Roasted beetroot	beetroot	|
Simmered fruit	Apples or melons	|
Roasted berries	Sweet berries or glowberries	| |
Caramelized onions and glass bottle	Honey bottle, coconut oil, onion	| |
Sweet berry jam bottle	sweet berry, honey bottle	V

Furnace smelting recipes

Product	Ingredient(s)	Smelt time in seconds
Cracked exposed stone	Exposed stone	Default
Exposed stone	Cracked exposed stone, Mossy exposed stone	| |
Limestone block	Dead coral block (any type)	|
Crumbled limestone	Dead coral fan (any type)	|
Packed mud	Mud, wheat or plant fiber[realism mode only] or rope or tumbleweed or hanging roots or mangrove roots	| | |
Chorine	Chorite	|
Charred sand	Sand	|
Charred dirt	Dirt or soil or cracked dirt or coarse dirt or rooted dirt	|
Cracked dirt	Mud	V

Stone cutter recipes

Product	Ingredient(s)	Cut time
Nether brick x2	Nether bricks (any variant)	Default
Bricks x2	Bricks (any variant)	|
Netherwart brick x2	Nether bricks	|
Resin x4	Resin bricks	|
Stone slab	Exposed stone	|

Prismarine shard	Prismarine block (any variant or type)	| |
Mud brick stairs	Packed mud	|
Mud brick walls x2	Packed mud	|
Mud brick slabs x2	Packed mud	|
Mud bricks	Packed mud	V

Gemcutter recipes

Product	Ingredient(s)	Cut time
Diamond	Raw diamond, coconut oil	Default
Lapis lazuli	Raw lazurite, coconut oil	|
Blue dye, glass bottle	Lapis lazuli, coconut oil, bottle of water	| |
Emerald	Green beryl, coconut oil	|
Amythest shard x3	Amythest block	|
Sand	Glass, Slag[realism mode only]	|
Radonite	Raw radonite	|
Lumen mytanum	Raw lumen mytanum	V

Blast furnace exclusive recipes

Product	Ingredient(s)	Smelt time
Chorine	Chorite	150

Brewing recipes

Herb potion

	*Four leaf clover	
	| | V	
	*Awkward potion	

Appendix E: Commands

Changes to current

Weather

`/weather` now no longer includes snow or rain allowing for local weather only effecting the current superbiome[see also: superbiomes, changes to generation]

Command blocks

Command blocks have been replaced by console blocks, although the block ID is the same, allowing current command block builds to still be compatible.[see also: comscript]

Event

Syntax

```
/event <directory:event>
```

Usage

`/event` can be used to toggle in-game events such as eclipses.

Possible inputs

Solar eclipse

```
/event strawberrymc:solar_eclipse
```

```
//Only works during the day during clear weather
```

Lunar eclipse/blood moon

```
/event strawberrymc:lunar_eclipse
```

```
//Only works at night during clear weather
```

Snow

```
/event strawberrymc:snow
```

```
//Only works during overcast weather
```

Parabola

```
/parabola - I feel like this is a bad idea
```

Syntax

```
/parabola abc <a b c>
/parabola akh <a k h>
```

Usage

Turns on parabola mode if a is negative, turns off if a is positive. Displays the parabola on a graph in standard form (abc) or vertex form (akh).

Appendix F: Raw changelog (draft) [original ideas]

The following is the raw nbt data that will be used:
The update:
Update section [End update]
Subsection blocks:
Added Endillary Plant > Endillary Plant Family > nbt:<hasNectar?>,<didNaturalGenerate?>[needsSilkTouch=true]
Added Chorite ore //"Raw unrefined state is Chorite, Refined state is Chorine"
Added Chorine Block > Chorine Ore Block Family >
nbt:<isChiseled?>,<isPolished?>,<isCracked>,<isStairs?>{<east?>,<west?>,<south?>,<north?>,<cornerIn?>,<cornerOut?>},<isSlab>{<topSlab?>,<bottomSlab?>},<verticalSlab?>
//"Crafted from 9 Chorite shards, prob someone will use for decor"
Added Sensegrass Tuber > Sensegrass Tuber Family > nbt:<isCorrupted?>,<isDecor?>,<inHotBiome?>[Change palette] //"Will sense player and retract into tuber, outputs redstone signal on how far away player is"
Added End Vine > End Vine Family > nbt:<onTop?>,<onMiddle?>,<onBottom?>
Added Void Anchor > Void Anchor family > nbt:<charged1?>,<charged2?>,<charged3?>,<charged4?>
Subsection items:
Added Endillary nectar bottle [/Effect @s End_Immunity] //"Will be useful for levitation effect in end cities"
Added Recovery template //"Crafted with Chorite Shard and 7 diamonds"
Added Chorite Shard //"Obtained from mining Chorite Ore with
Added Potion of Sense //"Obtained by brewing with Sensegrass Tuber, still working out what will happen when drunk"
Added Void Rod //"Made by plating fishing rod; works in the void and will fish lost items out, low durability"
Subsection structures:
Added structure "\structure:end_anchor" //"Place for players to set respawn point in end, uses jigsaw generation"
Subsection generation:
Modified dim 1 terrain generation //"3d terrain generation for outer End Islands"
Modified dim 1 terrain generation //"<isDecor?> == false atributed Endillary plants having a 50% chance of generating on "\structure:end_anchor"
Modified dim 1 terrain generation //"End caves; 2 types"
Subsection entities:
Added "strawberrymc:groundling" //"Octopus like creature that will tunnel through endstone and ambush players (simmilar to evoker fangs)"
Subsection boolean events:
Now checking for blocks:
"strawberrymc:sensegrass_tuber" or "strawberrymc:sensegrass" ~~-1~ of "strawberrymc:sensegrass" during block updates
"minecraft:endstone" or "strawberrymc:chorus_hummus" or "minecraft:obsidian" ~~-1 of "strawberrymc:endillary_flower" or "strawberrymc:endillary_stem" during block updates
Now checking for entities:
"\natural_entity{}" on ~-1~-1~-1 to ~1~1~1 on "strawberrymc:sensegrass_tuber" containing attribute condition <isCorrupted?> == true, for running [/effect <target> blindness until //"Not fulfilling previous condition(s)]",[/ damage <target>//"Take damage"]
Subsection existing mechanics/textures
Changed texture of block: "minecraft:string": //"Lowered opacity of string texture"
Subsection nbt data:
Added attribute to block: "minecraft:endstone" : > Endstone Family > <isMited?>
Added attribute to block: "minecraft:sand": > Sand Family > <inEndDim?> //"Attribute for sand in areas near Groundling spawns"
Added attribute to item: "minecraft:**<var>_shovel": > Shovel Family > <isPlated?>
Added attribute to item: "minecraft:**<var>_sword": > Sword Family > <isPlated?>
Added attribute to item: "minecraft:**<var>_hoe": > Hoe Family > <isPlated?>
Added attribute to item: "minecraft:**<var>_axe": > Axe Family > <isPlated?>
Added attribute to item: "minecraft:**<var>_pickaxe": > Pickaxe Family > <isPlated?>
Added attribute to item: "minecraft:**<var>_spear": > Spear Family > <isPlated?>
Added attribute to item: "minecraft:**<var>_mace": > Mace Family > <isPlated?>
Added attribute to item: "minecraft:**<var>_bow": > Bow Family > <isPlated?>
Added attribute to item: "minecraft:**<var>_crossbow": > Crossbow Family > <isPlated?>
Added attribute to item: "minecraft:**<var>_trident": > Trident Family > <isPlated?>
Added attribute to item: "minecraft:**<var>_trident": > Flint Family > <isPlated?>
Subsection entity effects:
Added "strawberrymc:fritliosis" //"Revamped purple nausea"
Added "strawberrymc:effect_immunity" //"Self explanatory"
Subsection UI and commands:
For BE:
Added attribute to "/gamerule" with "maxMinecartSpeed" with states == true, false
Added command "/tick" //"increase speed of everything, eg entity movement, etc."
Update section [Terra]:
Subsection blocks:

Added Salt Block
Added Hollow Termite Mound Block
Added Occupied Termite Mound Block
Added “strawberrymc:cracked_sand”
Added “strawberrymc:biocrust” //”Sand/dirt biocrust”
Added “strawberrymc:poor_dirt” //”Note is different from pre-existing “minecraft:dirt””
Subsection items:
Added “strawberrymc:salt”
Added “strawberrymc:hollowed_*<var>_log
Added “strawberrymc:cracked_*<var>_planks
Added “strawberrymc:four_leaved_clover” //”1% drop rate from mining clover mat, used to brew potion of luck”
Added “strawberrymc:three_leaved_clover” //”Found in clover glades”
Added “strawberrymc:crab_claw” //”Used to reach further away”
Added “strawberrymc:wool_hood”
Added Wool Vest
Added Wool Pants
Added Wool Boots
Added Leaf Hat
Added Leaf Vest
Added Leaf Skirt
Added Leaf Pads
Subsection biomes:
Added “strawberrymc:clover_glades” //”
Subsection structures:
Added “\structure:gyeser” //”Generates on to of sulfur caves, has gyeser block”
Subsection entities:
Added “strawberrymc:termite”
Added “strawberrymc:crab”
Added “strawberrymc:athensissuperior”
Subsection boolean events:
“\natural_entities:sulfur_cube” with var [contain] == barrel //”portable moving storage”
Subsection nbt data:
Modified nbt data for previously “minecraft:dirt” to “minecraft:soil”
Added attribute to “minecraft:soil”: <isMud?>
Added attribute to “minecraft:soil”: <nextWaterR?>
Added attribute to “minecraft:farmland”: <isFertile?>,<isOkay?>,<isDegraded?>
Added attribute to “minecraft:noteblock” : <isSticky?>
Added subattribute to all variants of “minecraft:<var>_log: <isHollow?>,<hasTermites>,<hasBlock?>{//”All small hitbox items”}

Appendix G: Technical debugging

Flora and fauna rendering not rendering:

1. Use `/give @p <item name> [item_model=strawberrymc:<path>]`
2. Make sure the Fabric UV is in `assets/minecraft/texture` [java]
3. Try command in step 1 again
4. Crash out and cry for help

Appendix H: Procedures

GenerateParticleSphereProcedure [Raw]

```
import net.minecraft.core.particles.ParticleOptions;
import net.minecraft.world.level.Level;
Import net.minecraft.cor.BlockPos;

public static void GenerateParticleSphereProcedure(
        Level level,
        ParticleOptions particle,
        BlockPos pos,
        double radius,
        int count
```

```
) {
    double goldenAngle = Math.PI * (3.0 - Math.sqrt(5.0));

    for (int i = 0; i < count; i++) {
        double y = 1.0 - 2.0 * (i + 0.5) / count;
        double r = Math.sqrt(1.0 - y * y);
        double theta = goldenAngle * i;

        double x = Math.cos(theta) * r;
        double z = Math.sin(theta) * r;

        level.addParticle(
            particle,
            pos.getX() + radius * x,
            pos.getY() + radius * y,
            pos.getZ() + radius * z,
            0, 0, 0
        );
    }
}
```

CoconutImpact

Procedure that generates `coconutParticle` particles triggered when a Coconut projectile impacts a solid block.

CalcGemcutterOutput

GUI procedure that calculates output(s) based from the 3 input slots.

AddArrowNBT

GUI procedure that adds arrow NBT based off a given arrow, and 5 input slots corresponding to coating, potion coating, tip, shaft, and fletching.

CoconutPlant

Block procedure that is responsible for planting a `coconutSeedling` block when the player right clicks sand, red sand, or gravel.

```
package io.github.strawberrymc.procedures; import net.minecraft.world.level.LevelAccessor;
import net.minecraft.world.item.ItemStack; import net.minecraft.world.entity.player.Player;
import net.minecraft.world.entity.Entity; import net.minecraft.core.BlockPos; import
io.github.strawberrymc.init.StrawberrymcModItems; import
io.github.strawberrymc.init.StrawberrymcModBlocks; public class CoconutPlantProcedure { public
static void execute(LevelAccessor world, double x, double y, double z, Entity sourceentity)
{ if (sourceentity == null) return; world.setBlock(BlockPos.containing(x, y + 1, z),
```

```
StrawberrymcModBlocks.PALM_SEEDLING.get().defaultBlockState(), 3); if (sourceentity instanceof
Player player & ) { ItemStack stktoremove = new
ItemStack(StrawberrymcModItems.PALMSEEDLING.get());
_player.getInventory().clearOrCountMatchingItems(p -> _stktoremove.getItem() == p.getItem(),
1, _player.inventoryMenu.getCraftSlots()); } } }
```

RecoveryPearlFetch

Procedure responsible for teleporting any entity impacted by a `recoveryPearl` to teleport to the thrower.

Raw: UnminableReigonData

```
import net.minecraft.core.BlockPos;
import net.minecraft.core.HolderLookup;
import net.minecraft.nbt.CompoundTag;
import net.minecraft.nbt.ListTag;
import net.minecraft.nbt.NbtUtils;
import net.minecraft.nbt.Tag;
import net.minecraft.world.level.Level;
import net.minecraft.world.level.saveddata.SavedData;
import net.minecraft.server.level.ServerLevel;

import java.util.HashSet;
import java.util.Set;

public class UnminableRegionData extends SavedData {
    private final Set<BlockPos> protectedPositions = new HashSet<>();

    // Standard factory setup for NeoForge/Minecraft
    public static final SavedData.Factory<UnminableRegionData> FACTORY = new
SavedData.Factory<>(
            UnminableRegionData::new,
            UnminableRegionData::load,
            net.minecraft.util.datafix.DataFixTypes.LEVEL
    );

    public UnminableRegionData() {}

    public void protect(BlockPos pos) {
        if (protectedPositions.add(pos)) {
            this.setDirty(); // Tells Minecraft to save the file to the disk
        }
    }

    public boolean isProtected(BlockPos pos) {
        return protectedPositions.contains(pos);
    }

    // Helper to grab the data instance from the server world
    public static UnminableRegionData get(Level level) {
        if (level instanceof ServerLevel serverLevel) {
            return serverLevel.getDataStorage().computeIfAbsent(FACTORY,
"unminable_regions");
        }
        return new UnminableRegionData(); // Fallback for safety
```

```
    }

    // --- NBT Saving and Loading ---
    public static UnminableRegionData load(CompoundTag tag, HolderLookup.Provider
lookupProvider) {
        UnminableRegionData data = new UnminableRegionData();
        ListTag list = tag.getList("Positions", Tag.TAG_COMPOUND);
        for (int i = 0; i < list.size(); i++) {
            data.protectedPositions.add(NbtUtils.readBlockPos(list.getCompound(i),
"pos").orElse(BlockPos.ZERO));
        }
        return data;
    }

    @Override
    public CompoundTag save(CompoundTag tag, HolderLookup.Provider lookupProvider) {
        ListTag list = new ListTag();
        for (BlockPos pos : protectedPositions) {
            CompoundTag posTag = new CompoundTag();
            posTag.put("pos", NbtUtils.writeBlockPos(pos));
            list.add(posTag);
        }
        tag.put("Positions", list);
        return tag;
    }
}
```

Raw: `CalculateBlockProtected`[provisional]

Responsible for making Sensors work.

```
package io.github.strawberrymc;

/*Various imports here*/

@EventBusSubscriber(modid=”strawberrymc”) public class
CalculateBlockProtectedProcedure {

@SubscribeEvent public static void execute(BlockEvent.breakEvent
event) {

Player player = event.getPlayer(); Blockstate state =
event.getBlockState(); Level world = event.getLevel;

if (player.isCreative()) {return;} if (level.isServerSide()) {return;}

boolean isUnminable = UnminableRegionData.get(level).isProtected(pos);

if (isUnminable)
{ event.setCancellationResult(InteractionResult.FAIL);
```

```
player.sendSystemMessage("&sect;kAAA&sect;rYou may not break blocks here. &sect;kAAAAAA&sect;r"); event.setCanceled(true); } } }
```

Appendix I: Raw committed \elements [outdated]

File name: Commit directory:

.DS_Store	raw ores, gemcutter, misc	3 days ago
CalcGemcutterOut.mod.json	raw ores, gemcutter, misc	3 days ago
Chorine.mod.json	initial2.0	5 days ago
ChorineBlock.mod.json	initial2.0	5 days ago
ChorineBlockOreRecipe.mod.json	initial2.0	5 days ago
ChorineOre.mod.json	initial2.0	5 days ago
ChorineOreBlockRecipe.mod.json	initial2.0	5 days ago
Chorite.mod.json	initial2.0	5 days ago
ChoriteRefine.mod.json	raw ores, gemcutter, misc	3 days ago
CloverCluff.mod.json	clover, glowroot fixes	3 days ago
CloverGlades.mod.json	un corrupted	2 days ago
CloverMat.mod.json	clover, glowroot fixes	3 days ago
CloverPatch.mod.json	un corrupted	2 days ago
CloverPlant.mod.json	clover, glowroot fixes	3 days ago
Coconut.mod.json	recovery pearl, coconuts, misc	5 days ago
CoconutFragment.mod.json	recovery pearl, coconuts, misc	5 days ago

CoconutHitParticle.mod.json	recovery pearl, coconuts, misc	5 days ago
CoconutOil.mod.json	un corrupted	2 days ago
CoconutProjectile.mod.json	recovery pearl, coconuts, misc	5 days ago
CrystalHollows.mod.json	initial2.0	5 days ago
CrystalHollowsHere.mod.json	glowroot	4 days ago
CrystallineCalcite.mod.json	initial2.0	5 days ago
DendriteBlock.mod.json	initial2.0	5 days ago
DendriteCrystal.mod.json	initial2.0	5 days ago
EchowoodBoat.mod.json	initial2.0	5 days ago
EchowoodBoatRecipe.mod.json	initial2.0	5 days ago
EchowoodButton.mod.json	initial2.0	5 days ago
EchowoodButtonRecipe.mod.json	initial2.0	5 days ago
EchowoodChestBoat.mod.json	initial2.0	5 days ago
EchowoodChestBoatRecipe.mod.json	initial2.0	5 days ago
EchowoodDoor.mod.json	initial2.0	5 days ago
EchowoodDoorRecipe.mod.json	initial2.0	5 days ago
EchowoodFence.mod.json	initial2.0	5 days ago
EchowoodFenceGate.mod.json	initial2.0	5 days ago
EchowoodFenceGateRecipe.mod.json	initial2.0	5 days ago
EchowoodFenceRecipe.mod.json	initial2.0	5 days ago

EchowoodHangingSign.mod.json	initial2.0	5 days ago
EchowoodHangingSignRecipe.mod.json	initial2.0	5 days ago
EchowoodLeaves.mod.json	initial2.0	5 days ago
EchowoodLeavesComposting.mod.json	initial2.0	5 days ago
EchowoodLog.mod.json	initial2.0	5 days ago
EchowoodPlanks.mod.json	initial2.0	5 days ago
EchowoodPlanksRecipe.mod.json	initial2.0	5 days ago
EchowoodPressurePlate.mod.json	initial2.0	5 days ago
EchowoodPressurePlateRecipe.mod.json	initial2.0	5 days ago
EchowoodSign.mod.json	initial2.0	5 days ago
EchowoodSignRecipe.mod.json	initial2.0	5 days ago
EchowoodSlab.mod.json	initial2.0	5 days ago
EchowoodSlabRecipe.mod.json	initial2.0	5 days ago
EchowoodStairs.mod.json	initial2.0	5 days ago
EchowoodStairsRecipe.mod.json	initial2.0	5 days ago
EchowoodTrapdoor.mod.json	initial2.0	5 days ago
EchowoodTrapdoorRecipe.mod.json	initial2.0	5 days ago
EchowoodWood.mod.json	initial2.0	5 days ago
EchowoodWoodRecipe.mod.json	initial2.0	5 days ago
Electricution.mod.json	glowroot	4 days ago
EntityShock.mod.json	glowroot	4 days ago

Fiber.mod.json	raw ores, gemcutter, misc	3 days ago
ForgeHammer.mod.json	un corrupted	2 days ago
ForgingTable.mod.json	un corrupted	2 days ago
Garnet.mod.json	initial2.0	5 days ago
GarnetBlock.mod.json	initial2.0	5 days ago
GarnetCrystal.mod.json	initial2.0	5 days ago
GarnetNodule.mod.json	initial2.0	5 days ago
Gemcutter.mod.json	raw ores, gemcutter, misc	3 days ago
Glowroot.mod.json	clover, glowroot fixes	3 days ago
GlowrootDummyPatch.mod.json	clover, glowroot fixes	3 days ago
GlowrootDummySpawner.mod.json	clover, glowroot fixes	3 days ago
GlowrootGeneration.mod.json	clover, glowroot fixes	3 days ago
GlowrootHavest.mod.json	clover, glowroot fixes	3 days ago
GlowrootPaste.mod.json	raw ores, gemcutter, misc	3 days ago
GlowrootPasteRecipe.mod.json	raw ores, gemcutter, misc	3 days ago
GlowrootTuber.mod.json	glowroot fixes	4 days ago
GlowrootTuberGrow.mod.json	glowroot fixes	4 days ago
GlowrootTuberPlant.mod.json	glowroot	4 days ago
GlowrootTuberStone.mod.json	glowroot fixes	4 days ago

GreenBeryl.mod.json	un corrupted	2 days ago
Herb.mod.json	un corrupted	2 days ago
HerbBrewing.mod.json	un corrupted	2 days ago
IronDust.mod.json	un corrupted	2 days ago
IronDustToNugget.mod.json	un corrupted	2 days ago
Lazurite.mod.json	raw ores, gemcutter, misc	3 days ago
LuckBrewing.mod.json	un corrupted	2 days ago
Palm.mod.json	recovery pearl, coconuts, misc	5 days ago
PalmBoat.mod.json	initial2.0	5 days ago
PalmBoatRecipe.mod.json	initial2.0	5 days ago
PalmButton.mod.json	initial2.0	5 days ago
PalmButtonRecipe.mod.json	initial2.0	5 days ago
PalmChestBoat.mod.json	initial2.0	5 days ago
PalmChestBoatRecipe.mod.json	initial2.0	5 days ago
PalmDoor.mod.json	initial2.0	5 days ago
PalmDoorRecipe.mod.json	initial2.0	5 days ago
PalmFence.mod.json	initial2.0	5 days ago
PalmFenceGate.mod.json	initial2.0	5 days ago
PalmFenceGateRecipe.mod.json	initial2.0	5 days ago
PalmFenceRecipe.mod.json	initial2.0	5 days ago

PalmHangingSign.mod.json	initial2.0	5 days ago
PalmHangingSignRecipe.mod.json	initial2.0	5 days ago
PalmLeaves.mod.json	initial2.0	5 days ago
PalmLeavesComposting.mod.json	initial2.0	5 days ago
PalmLog.mod.json	initial2.0	5 days ago
PalmPlanks.mod.json	initial2.0	5 days ago
PalmPlanksRecipe.mod.json	initial2.0	5 days ago
PalmPressurePlate.mod.json	initial2.0	5 days ago
PalmPressurePlateRecipe.mod.json	initial2.0	5 days ago
PalmSeedling.mod.json	recovery pearl, coconuts, misc	5 days ago
PalmSeedlingGrow.mod.json	recovery pearl, coconuts, misc	5 days ago
PalmSign.mod.json	initial2.0	5 days ago
PalmSignRecipe.mod.json	initial2.0	5 days ago
PalmSlab.mod.json	initial2.0	5 days ago
PalmSlabRecipe.mod.json	initial2.0	5 days ago
PalmStairs.mod.json	initial2.0	5 days ago
PalmStairsRecipe.mod.json	initial2.0	5 days ago
PalmTrapdoor.mod.json	initial2.0	5 days ago
PalmTrapdoorRecipe.mod.json	initial2.0	5 days ago
PalmWood.mod.json	initial2.0	5 days ago

PalmWoodRecipe.mod.json	initial2.0	5 days ago
PlantCoconut.mod.json	glowroot	4 days ago
RadoniteArmor.mod.json	initial2.0	5 days ago
RadoniteArmorBootsRecipe.mod.json	initial2.0	5 days ago
RadoniteArmorChestplateRecipe.mod.json	initial2.0	5 days ago
RadoniteArmorHelmetRecipe.mod.json	initial2.0	5 days ago
RadoniteArmorLeggingsRecipe.mod.json	initial2.0	5 days ago
RadoniteAxe.mod.json	initial2.0	5 days ago
RadoniteAxeRecipe.mod.json	initial2.0	5 days ago
RadoniteBlock.mod.json	initial2.0	5 days ago
RadoniteBlockOreRecipe.mod.json	initial2.0	5 days ago
RadoniteHoe.mod.json	initial2.0	5 days ago
RadoniteHoeRecipe.mod.json	initial2.0	5 days ago
RadoniteIngot.mod.json	initial2.0	5 days ago
RadoniteOre.mod.json	initial2.0	5 days ago
RadoniteOreBlockRecipe.mod.json	initial2.0	5 days ago
RadoniteOreSmelting.mod.json	initial2.0	5 days ago
RadonitePickaxe.mod.json	initial2.0	5 days ago
RadonitePickaxeRecipe.mod.json	initial2.0	5 days ago
RadoniteShovel.mod.json	initial2.0	5 days ago
RadoniteShovelRecipe.mod.json	initial2.0	5 days ago

RadoniteSword.mod.json	initial2.0	5 days ago
RadoniteSwordRecipe.mod.json	initial2.0	5 days ago
RareCloverCluff.mod.json	clover, glowroot fixes	3 days ago
RawDiamond.mod.json	un corrupted	2 days ago
RawRadonite.mod.json	initial2.0	5 days ago
RecoveryPearl.mod.json	un corrupted	2 days ago
RecoveryPearlActivation.mod.json	recovery pearl, coconuts, misc	5 days ago
RecoveryPearlFail.mod.json	recovery pearl, coconuts, misc	5 days ago
RecoveryPearlParticle.mod.json	recovery pearl, coconuts, misc	5 days ago
RecoveryPearlProjectile.mod.json	un corrupted	2 days ago
RecoveryPearlRecipe.mod.json	recovery pearl, coconuts, misc	5 days ago
Shocking.mod.json	glowroot	4 days ago
StrippedEchowoodLog.mod.json	initial2.0	5 days ago
StrippedEchowoodWood.mod.json	initial2.0	5 days ago
StrippedEchowoodWoodRecipe.mod.json	initial2.0	5 days ago
StrippedPalmLog.mod.json	initial2.0	5 days ago
StrippedPalmWood.mod.json	initial2.0	5 days ago
StrippedPalmWoodRecipe.mod.json	initial2.0	5 days ago